

Signal

- * It is defined as any physical phenomena that conveys (or) carries some information
- * Generally the signals are classified into
 - * analog signal
 - * Discrete (or) digital signals

Representation of discrete time signals

- * Discrete time signals are defined as only of discrete incident at time
- * They are four (4) ways representation
 - 1) Functional form (or) functional representation
 - 2) Graphical representation
 - 3) Sequence representation
 - 4) Tabular representation

Example:-

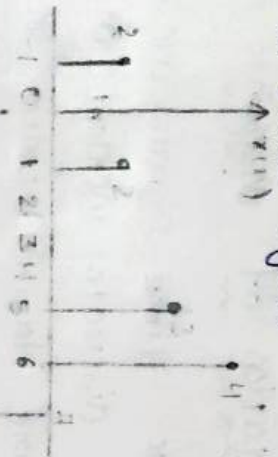
- 1) consider a discrete time signal $x(n)$ with values
 - $x(-1) = 2$
 - $x(0) = 1$
 - $x(1) = 2$
 - $x(5) = 3$

functional form for the above example

is

$$x[n] = \begin{cases} 2 & \text{for } n = -1 \\ 1 & \text{for } n = 0 \\ 2 & \text{for } n = 1 \\ 3 & \text{for } n = 5 \\ 4 & \text{for } n = 6 \\ -5 & \text{for } n = 7 \end{cases}$$

2) Graphical form (or) Graphical representation
Graphical form for the above example is shown in below figure



Sequence form

Sequence form for the above example is $x[n] = \{2, 1, 2, 0, 0, 0, 3, 4, 5\}$

NOTE:- The arrow (up arrow) indicates for $n=0$

4) Tabular form

Tabular form for the above example is shown in below

n	-1	0	1	5	6	7
x[n]	2	1	2	3	4	-5

Signal processing

It is an operation that changes the characteristics of the signal

The characteristics includes the amplitude shape, frequency (or) phase of a

Signal system

General block diagram of digital signal processing

The General block diagram of digital signal processing is shown in below figure



It consists of five blocks

1) Analog filter (low pass filter)

2) A/D converter

3) DSP

4) D/A converter

5) Reconstruction filter

Analog filter

- * It takes analog signal and gives the band limited signal

DSP converter

- * It converts the band limited signal into discrete (or) digital form

Digital signal processor

- * This is the heart of dsp
- * The digital signal processed in this block

Representation filter

- * It is used to reconstruct the original signal

Advantages of DSP

- * Better signal quality
- * Greater flexibility
- * High noise immunity
- * It enables multi dimensional transmission very easily

Applications of DSP

- * Space
- * Communication
- * Medical

- * military
- * Industrial
- * Image Processing
- * commercial

Basic elementary discrete time signals

- * The basic elementary discrete time signals are

- 1, unit step signal
- 2, unit ramp signal
- 3, unit parabolic signal
- 4, Impulse signal
- 5, sinusoidal signal
- 6, Real exponential
- 7, complex exponential signal

- * unit step signal (or) unit step sequence

- * The discrete time unit step signal is defined as

$$u(n) = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$

- * The graphical representation is shown below

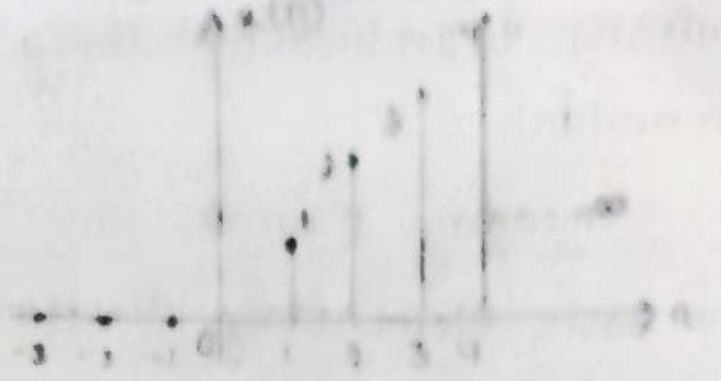


- * unit ramp signal

- * The discrete time unit ramp signal is defined as

$$r(n) = \begin{cases} n & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$

- * The graphical representation is shown below



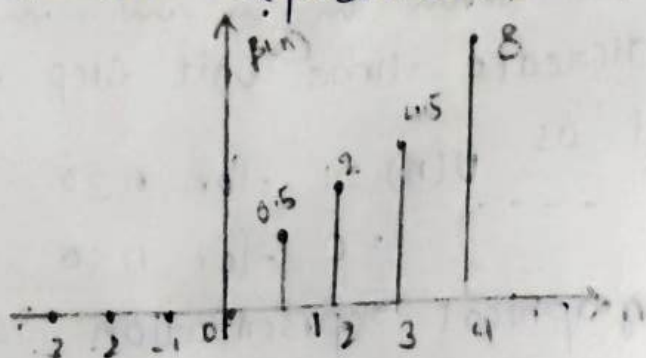
Unit Parabolic Signal

The unit parabolic signal (or) sequence is defined as

$$P(n) = \frac{n^2}{2} \quad \text{for } n \geq 0$$

$$= 0 \quad \text{for } n < 0$$

* The graphical representation show below



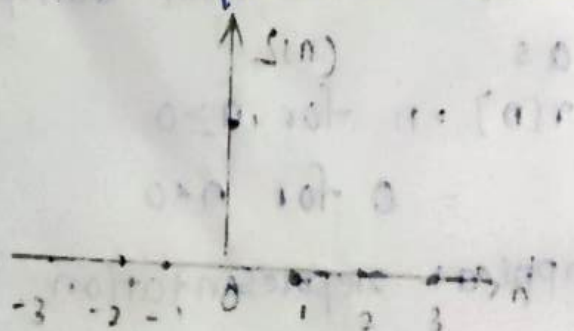
* Unit Impulse signal (or) Sequence

* The discrete-time unit impulse signal is defined as

$$s(n) = 1 \quad \text{for } n = 0$$

$$= 0 \quad \text{elsewhere}$$

* The graphical representation is show below



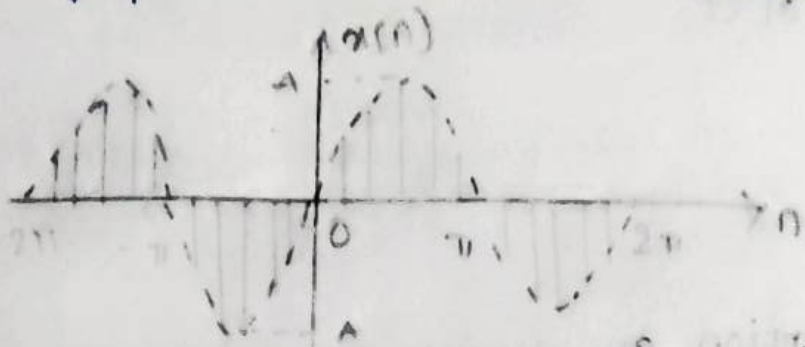
Sinusoidal Sequence

* The discrete-time sinusoidal signal (or) sequence is defined as

$$x[n] = A \sin(\omega n + \phi)$$

where A = Amplitude, ω = Angular frequency
 ϕ = phase angle

* The graphical representation is shown below



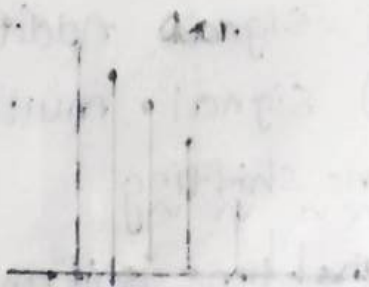
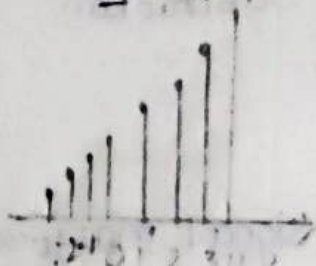
Real exponential Sequence (or) signal

* The discrete-time Real exponential signal or sequence defined as

$$x[n] = a^n \text{ for all 'n' values.}$$

* The graphical representation shown below.

TWO CASES



Complex exponential signal (or) Sequence

The discrete-time complex exponential signal (or) Sequence is defined as

$$x[n] = a^n \cdot e^{j(\omega n + \phi)}$$

$$\{ e^{j\theta} = \cos\theta + j\sin\theta \}$$

$$= a^n \cos(\omega n + \phi) + j a^n \sin(\omega n + \phi)$$

* The graphical representation is shown below
Three Cases

i) $|a| = 1$

ii) $|a| > 1$

iii) $|a| < 1$

* operation on signals (or) sequences:-

* The basic operation on signals are

- 1) Time shifting
- 2) Time Reversal
- 3) Time scaling
- 4) Amplitude scaling
- 5) Signal addition (or) Subtraction
- 6) Signal multiplication

Time shifting

* The time shifting of a signal may result in time delay (or) time advance

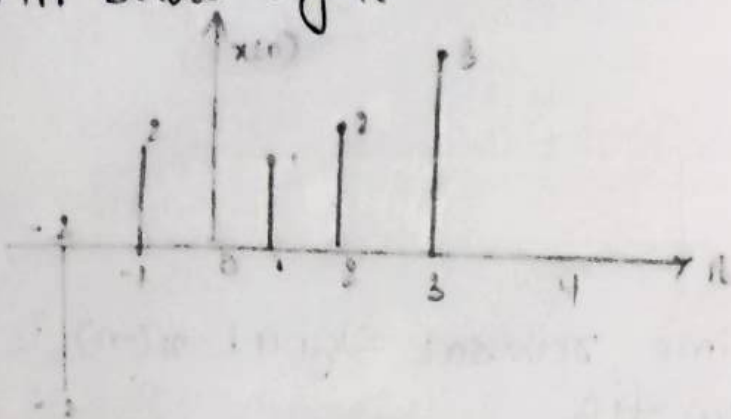
* The time shifting operation of a discrete time signal can be represented as

$$y(n) = x[n \pm k]$$

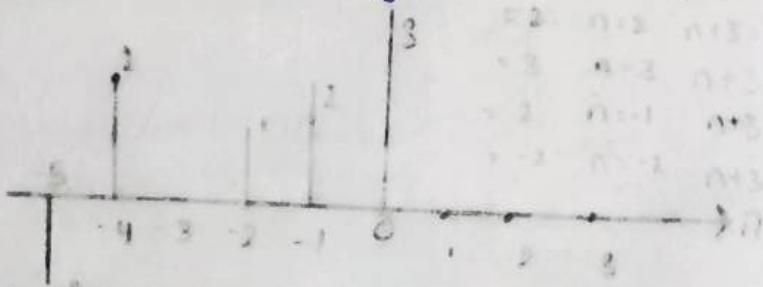
$x(n+k) \rightarrow$ Delayed

Example

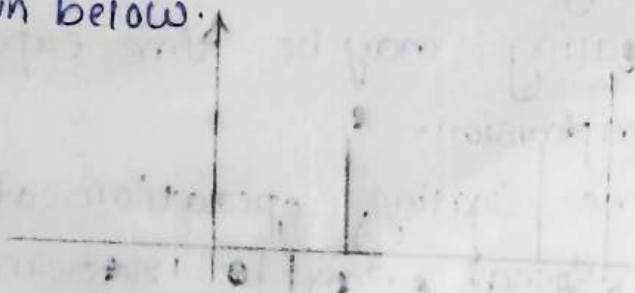
* Let us consider an arbitrary signal $x(n]$ as shown in below figure



* The delayed version of $x[n]$ that is $x[n+3]$ is shown in below fig



* The advanced version of $x[n]$ that is $x[n-3]$ is shown in below.



Time reversal

* The time reversal is also called time folding or discrete time signal $x[n]$ can be obtained by folding the sequence

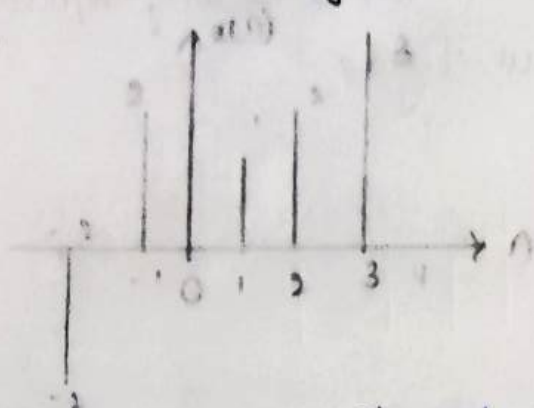
* The time reversal operation of a discrete time signal can be represented as

$$y[n] = x[-n]$$

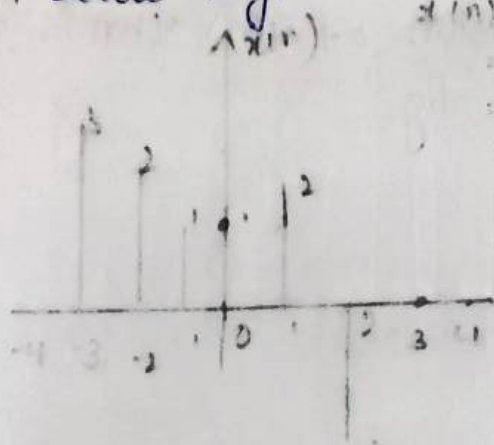
... (or) attenuation

Ex

Consider an arbitrary signal $x(n]$ is shown below.



* The time reversal signal $x[-n]$ is shown in below fig.



$x[n]$	1	2	1	2	1
$x[-n]$					
n	-3	-2	-1	0	1

* Time Scaling

* Time scaling may be time expansion or time compression.

* The time scaling operation of a discrete-time signal can be represented as

$$y[n] = x[an]$$

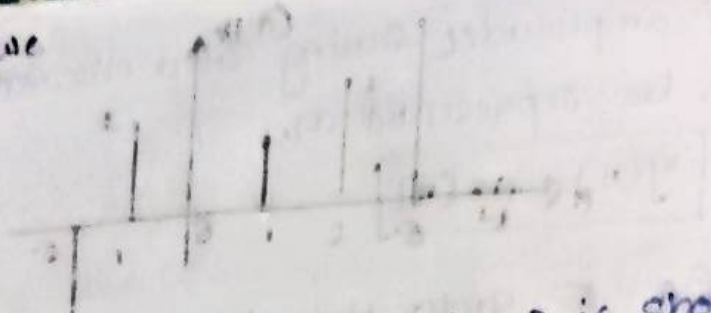
* if a is greater than one ($a > 1$) time compression

* if a is less than one ($a < 1$) time expansion

Example

Consider an arbitrary signal $x[n]$ is shown

below - figure



time compression signal $x[2n]$ is shown in figure below.

$$y[n] = x[2n]$$

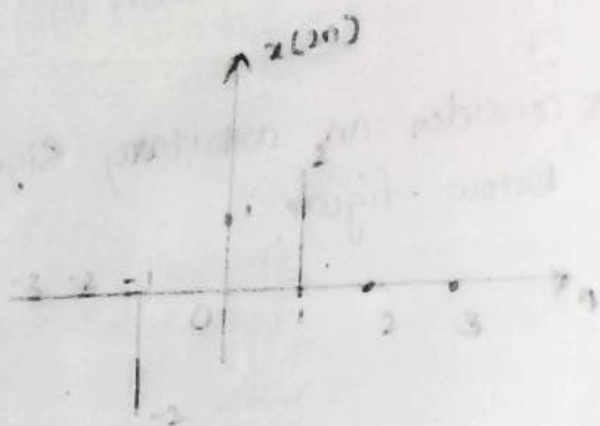
$$y[0] = x[0] = 1$$

$$y[1] = x[2] = 0$$

$$y[2] = x[4] = 0$$

$$y[-1] = x[-2] = -2$$

$$y[-2] = x[-4] = 0$$



* time expansion signal $x[n/2]$ is shown in below figure

$$y[n] = x[n/2]$$

$$y[0] = x[0] = 1$$

$$y[1] = x[0.5] = x$$

$$y[2] = x[1] = 2$$

$$y[3] = x[1.5] = x$$

$$y[4] = x[2] = 0$$

$$y[5] = x[2.5] = x$$

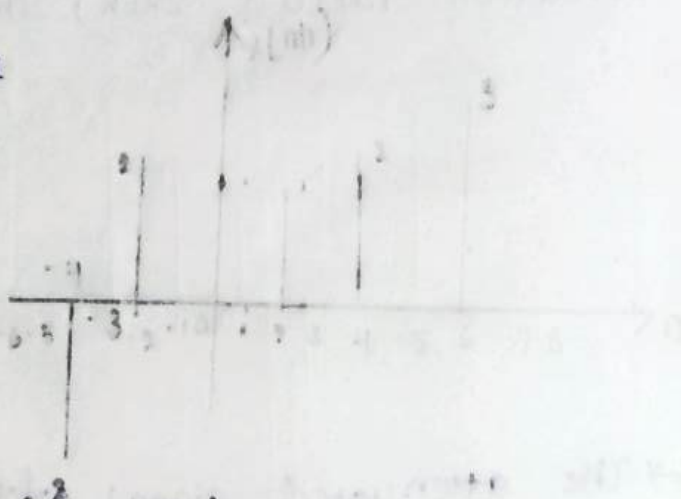
$$y[6] = x[3] = 2$$

$$y[8] = x[4] = 0$$

$$y[-2] = x[-1] = 2$$

$$y[-4] = x[-2] = -2$$

$$y[-6] = x[-3] = 0$$



* Amplitude Scaling

* The amplitude scaling of discrete time system may be amplification (or) attenuation

The amplitude scaling of a discrete-time system can be represented as

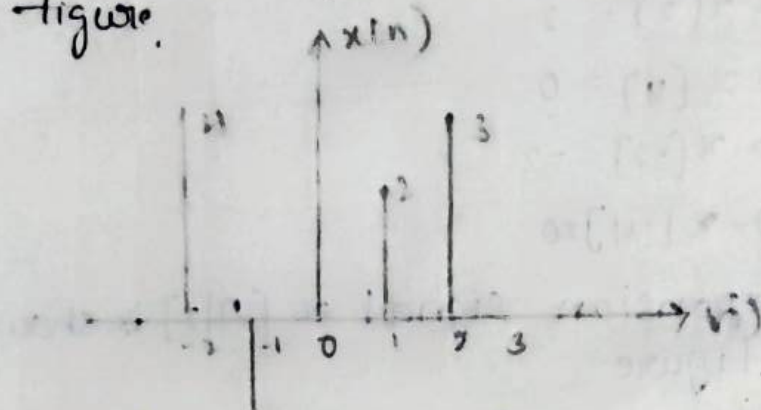
$$y[n] = ax[n]$$

* if a is greater than 1) $a > 1$ - amplification

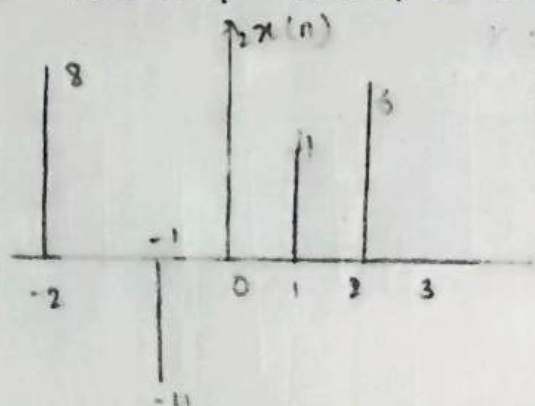
* if a is less than 1) $a < 1$ - attenuation

Ex

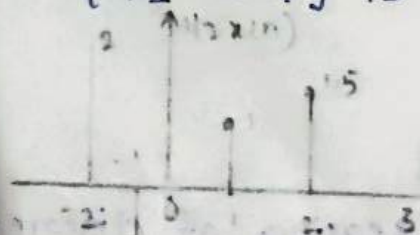
* Consider an arbitrary signal $x[n]$ is shown in below figure.



* The amplified signal of $x[n]$ with amplification factor 2 $2x[n]$ shown in below fig.



* The attenuated signal of $x[n]$ with attenuation factor $1/2$ $[1/2 x[n]]$ is shown in below fig.

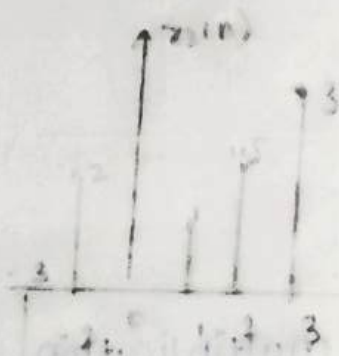
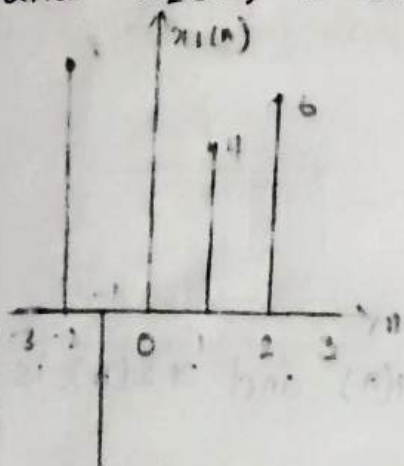


5. Signal addition (or) subtraction :-

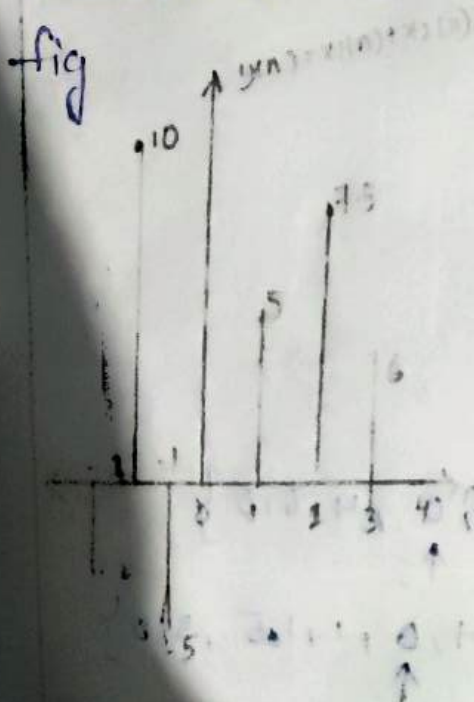
* In discrete time domain - the sum of two signals $x_1(n)$ and $x_2(n)$ can be obtained by adding - the corresponding sample values be obtained by subtracting each sample of $x_2(n)$ from - the corresponding sample of $x_1(n)$

Ex

Let us consider two arbitrary signals $x_1(n)$ and $x_2(n)$ is show in below.



* Addition of $x_1(n)$ and $x_2(n)$ is show in below



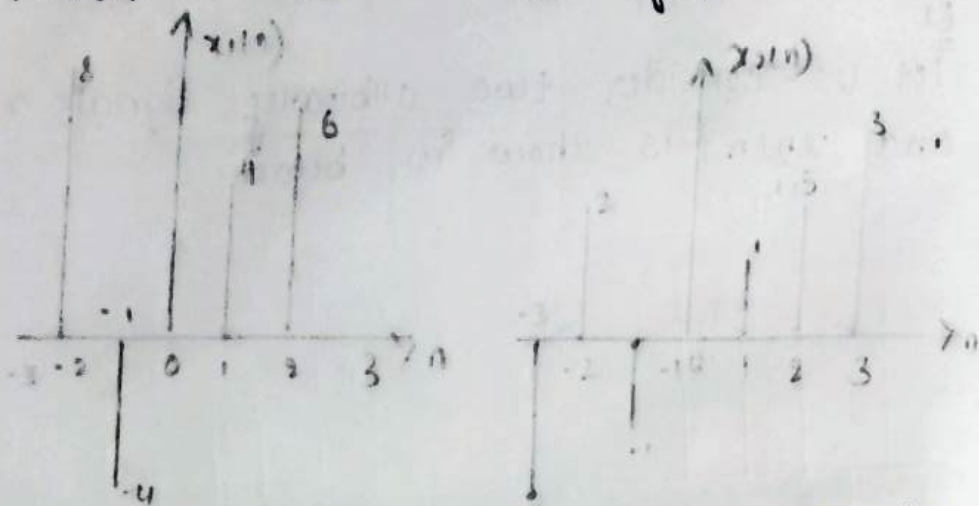
The subtraction of $x_1(n)$ and $x_2(n)$ is show in figure

* Signal multiplication

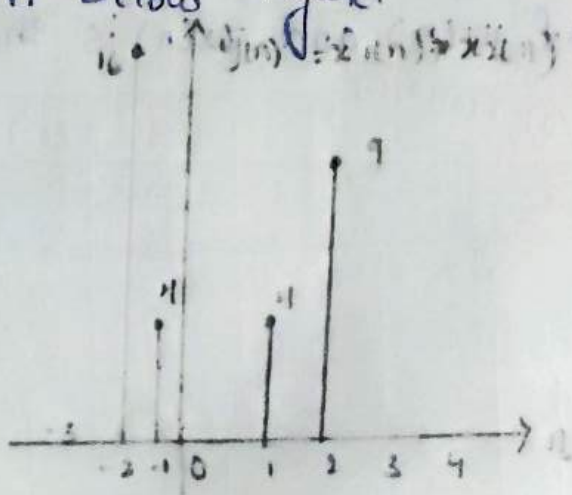
* The multiplication of two discrete time sequence (signals) $x_1(n)$ and $x_2(n)$ can be obtained by multiplying the corresponding values.

Ex

* let us consider two arbitrary signals $x_1(n)$ & $x_2(n)$ is show in below figure.



* The multiplication of $x_1(n)$ and $x_2(n)$ is show in below figure.



$$x_1(n) = \{0, 8, -4, 0, 4, 6, 0\}$$

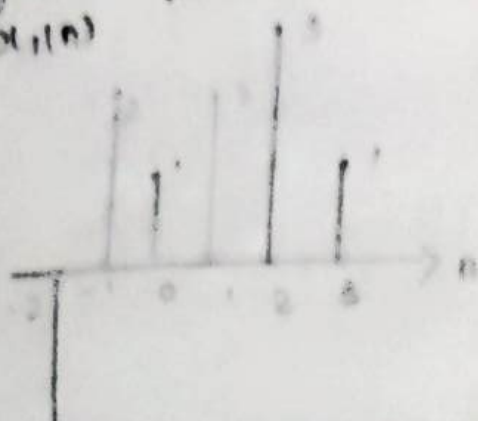
↑

$$x_2(n) = \{0, -2, 2, -1, 0, 1, 5, 3, 0\}$$

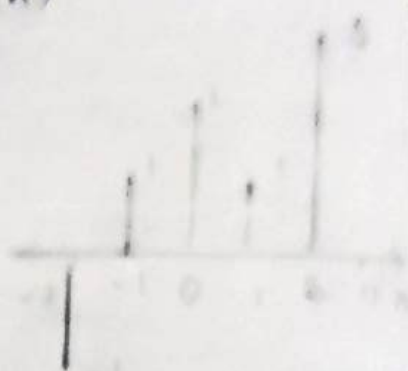
↑

* Perform the following operations on the given signals.

$x_1(n)$



$x_2(n)$

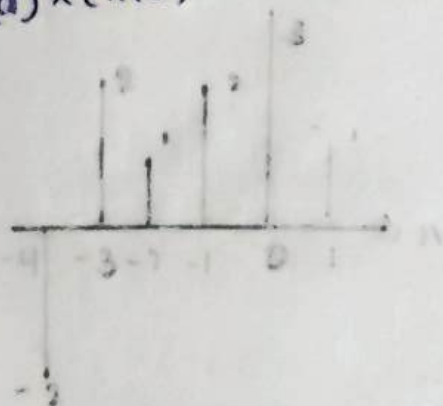


(a) $x_1(n+2)$, (b) $x_2(n-3)$, (c) $x_1(2n)$, (d) $x_2(n)$

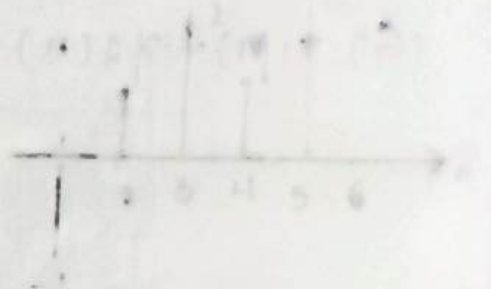
(e) $3x_1(n)$, (f) $\frac{1}{4}x_2(n)$, (g) $x_1(n) + x_2(n)$

(h) $x_1(n) * x_2(n)$, (i) $x_1(-n)$

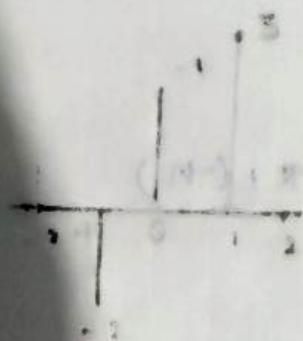
(a) $x(n+2)$



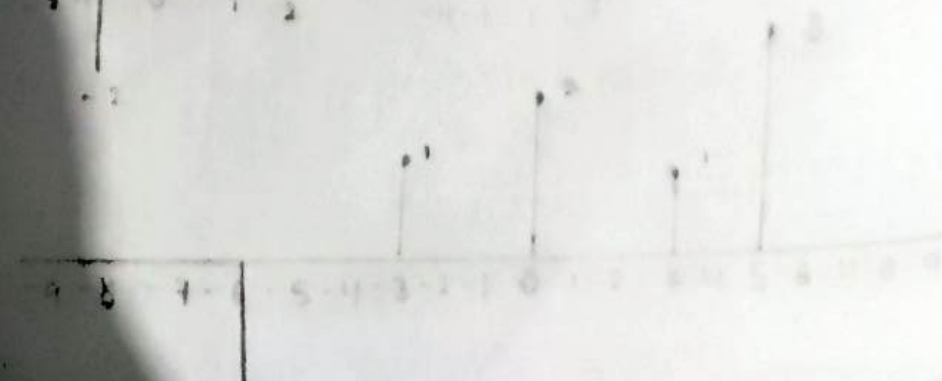
(b) $x_2(n-3)$



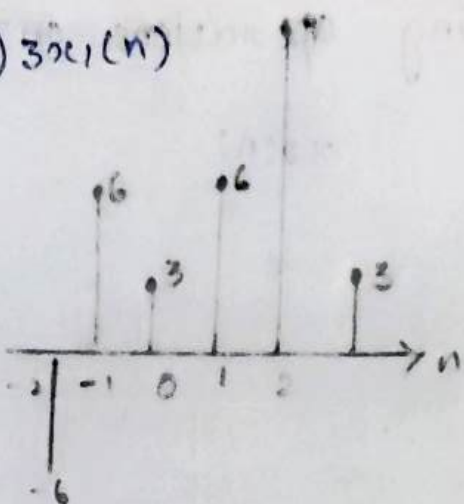
(c) $x_1(2n)$



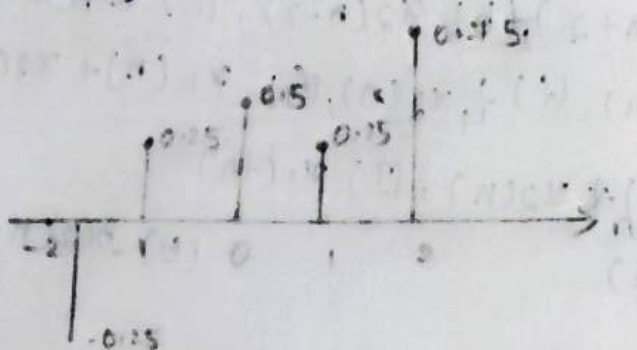
(d) $x_2(n/3)$



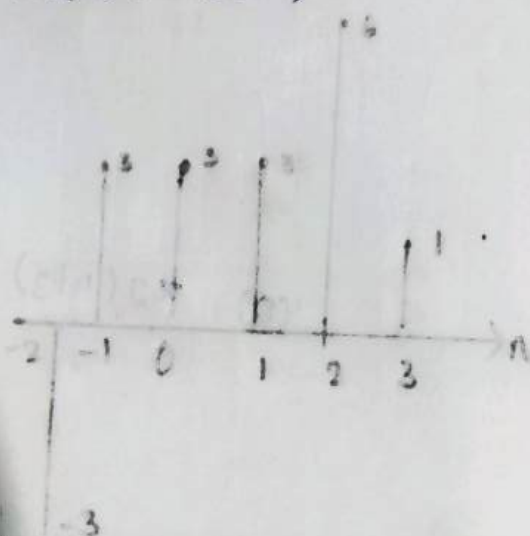
$$(e) 3x_1(n)$$



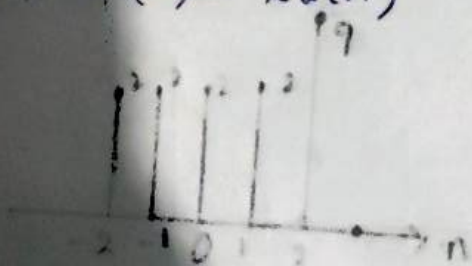
$$(f) \frac{1}{4} x_2(n)$$



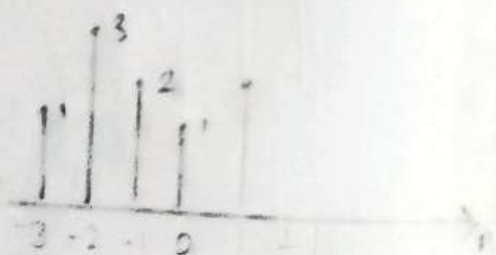
$$(g) x_1(n) + x_2(n)$$



$$(h) x_1(n) * x_2(n)$$



$$(i) x_1(-n)$$



Types of signals :- $\rightarrow$ signals can be classified as

* They are five categories

* Deterministic & Non Deterministic Signals

* periodic and aperiodic

* casual & non casual

* energy and power

* even and odd

* deterministic & random signals

A signal exhibiting no uncertainty at its magnitude and phase at any given instant of time is called deterministic signals or sequence's

Ex: $x(n)$ - unit step Sequence

$x(n] = 1$ -

$x(n) = n$ - for n is ≥ 0

Non-Deterministic

* A signal characterised by uncertainty above occurrence is called Non Deterministic or random signals

Ex

Thermal noise

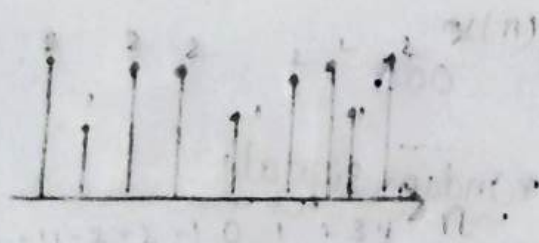
periodic and aperiodic signals

A signal which has a definite pattern and repeat it self at regular intervals of time is called periodic signals.

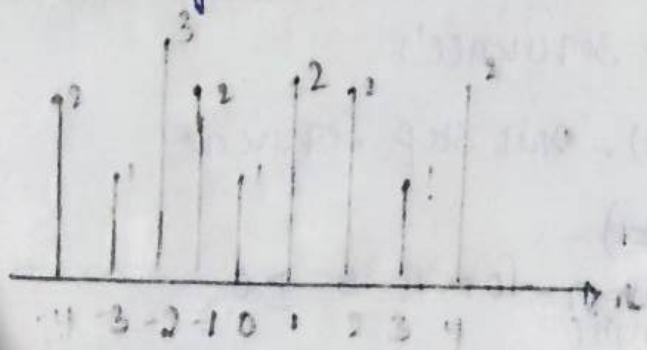
A discrete time signal $x(n]$ is set to be periodic if it satisfy's the condition $x(n) = x(n+N]$ for all n

where N is fundamental period $= \frac{2\pi}{\omega}$

Ex:-



* A signal which doesnot repeat at regular interval of time is called non periodic of A periodic signals.



Causal & Non Causal

A discrete-time signals $x(n]$ is set to be causal is $x(n)=0$ for n is less than 0 ($n < 0$) otherwise the signals is Non causal

Ex

$u(n)$ is causal

$u(-n)$ is non causal

energy and power signals:

* $0 < E < \infty \rightarrow$ is energy signal

* The total energy of a discrete time signal $x(n)$ is defined as

$$E = \sum_{n=-\infty}^{\infty} [x(n)]^2 = \lim_{N \rightarrow \infty} \sum_{n=-N}^N [x(n)]^2$$

* if the energy is $\neq$ of the signal is finite

$$\therefore 0 < E < \infty$$

then the signal is set to be energy signal

* The average power of a discrete time signal $x(n)$ is defined as

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N [x(n)]^2$$

* if the power of a signal is finite that is $0 < P < \infty$ then the signal is set to be power signal

NOTE: a) for the energy signal power is 0

b) for power signal energy is ∞

Even and Odd Signals:

* A discrete time signal $x(n)$ is set to be even if it satisfies the condition $x(n) = x(-n)$

for all n

* A discrete time signal $x(n)$ is set to be an

an odd if it satisfies the condition
 $x(n) = -x(-n)$

NOTE

Any arbitrary signal $x(n)$ can be converted into even and odd signals as follows
 - the even part of $x(n)$ is $x_e(n)$

$$x_e(n) = \frac{1}{2} [x(n) + x(-n)]$$

- the odd part of $x(n)$ is $x_o(n)$

$$x_o(n) = \frac{1}{2} [x(n) - x(-n)]$$

* verify the following signals are periodic or not and also find fundamental period

(a) $\cos x(n) = \cos 4n$ (b) $x(n) = \sin 5\pi n$

(c) $x(n) = \sin \frac{2\pi n}{3} + \cos \frac{2\pi n}{5}$

(d) $x(n) = \cos 4n$

We have $x(n) = \cos 2\pi n$ compare with the given $x(n) = \sin 4n$ therefore $f = 2/n$ it is not a rational number

the given signal $x(n)$ is set to be non periodic

(b) $x(n) = \sin 5\pi n$

comparing with $x(n) = \sin 2\pi n$ we have

if $f = 5/n$ then $f = 5/n$ it is a rational number
 - the given signal $x(n)$ is set to be periodic - the fundamental period $N=2$

if $f = 5/n$
 $f = 5/2 = k/N$

$N=2$

Comparing with $x(n) = \sin 2\pi f n + \cos 2\pi f n$

then $f_1 = 1/3 = \frac{k_1}{N_1}$

$f_2 = 1/5 = \frac{k_2}{N_2}$

$N_1 = 3$ $N_2 = 5$

Since $\frac{N_1}{N_2} = 3/5$

as a ratio of 2 integers the given sequence

$x(n)$ is periodic $N=15$

* The fundamental period of $x(n)$ is the LCM of N_1 & N_2 that is shown above equation

⇒ energy and power signals

if infinite

$1+r+r^2+\dots+r^\infty = \frac{1}{1-r}$ $r < 1$

$1+r+r^2+\dots+r^\infty = \infty$ $r > 1$

finite $1+r+r^2+\dots+r^N = \frac{1-r^{N+1}}{1-r}$ $r < 1$

$$\begin{aligned}
 * \quad 1 + r + r^2 + \dots + r^{N-1} &= \frac{1 - r^N}{1 - r} \quad r < 1 \\
 * \quad 1 + r + r^2 + \dots + r^{N-1} &= \frac{r^{N+1} - 1}{r - 1} \quad r > 1 \\
 * \quad 1 + r + r^2 + \dots + r^{N-1} &= \frac{r^{N+1} - 1}{r - 1} \quad r > 1
 \end{aligned}$$

$$\begin{aligned}
 * \quad E &= \sum_{n=-\infty}^{\infty} |x[n]|^2 \\
 * \quad P &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2
 \end{aligned}$$

* check whether the following signals are energy & power signals.

$$\begin{aligned}
 (a) \quad x[n] &= \left(\frac{1}{2}\right)^n u[n] \quad \bullet \quad x[n] = \sin\left(\frac{\pi}{3}n\right) \\
 (c) \quad x[n] &= n u[n]
 \end{aligned}$$

Ans: (a) $x[n] = \left(\frac{1}{2}\right)^n u[n]$

Ans: $x[n] = \left(\frac{1}{2}\right)^n u[n]$

$$\begin{aligned}
 E &= \sum_{n=-\infty}^{\infty} |x[n]|^2 \\
 &= \sum_{n=-\infty}^{\infty} \left| \left(\frac{1}{2}\right)^n u[n] \right|^2 \\
 &= \sum_{n=-\infty}^{-1} \left| \left(\frac{1}{2}\right)^n \right|^2 u[n] + \sum_{n=0}^{\infty} \left| \left(\frac{1}{2}\right)^n \right|^2 u[n] \\
 &= 0 + \sum_{n=0}^{\infty} (1/4)^n \\
 &= 1 + 1/4 + (1/4)^2 + (1/4)^3 + \dots + (1/4)^{\infty}
 \end{aligned}$$

$$\frac{1}{1-1/4} = 4/3 \text{ joules}$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \left| \left(\frac{1}{2}\right)^n u[n] \right|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N (1/4)^n$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \left[1 + 1/4 + \dots + (1/4)^N \right]$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \left[\frac{1 - (1/4)^{N+1}}{1 - 1/4} \right]$$

$$= \frac{4}{3} \lim_{N \rightarrow \infty} \frac{1}{2N+1} \left[1 - \frac{1}{4^{N+1}} \right]$$

$$= 0$$

Ans: (b) $x[n] = \sin\left(\frac{\pi}{3}n\right)$

Ans: $E = \sum_{n=-\infty}^{\infty} |x[n]|^2$

$$\begin{aligned}
 &= \sum_{n=-\infty}^{\infty} \left| \sin\left(\frac{\pi}{3}n\right) \right|^2 \\
 &= \sum_{n=-\infty}^{\infty} \sin^2\left(\frac{\pi}{3}n\right)
 \end{aligned}$$

$$= \frac{1}{2} \left[\sum_{n=-\infty}^{\infty} \frac{1 - \cos\left(\frac{2\pi}{3}n\right)}{2} \right]$$

$$\frac{1}{2} \left[\sum_{n=-\infty}^{\infty} 1 - \sum_{n=-\infty}^{\infty} \cos \frac{2\pi}{3} n \right]$$

∴ no energy signal

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \sin^2 \left(\frac{\pi}{3} n \right)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \left[\frac{1 - \cos \frac{2\pi}{3} n}{2} \right]$$

$$= \frac{1}{2} \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N 1$$

$$= \frac{1}{2} \lim_{N \rightarrow \infty} \frac{1}{2N+1} \times 2N+1 = \frac{1}{2} \text{ watts}$$

∴ Power signal

(c) $x(n) = n u(n)$

$$E = \sum_{n=-\infty}^{\infty} |n u(n)|^2$$

$$= \sum_{n=-\infty}^{\infty} n^2 u(n)$$

$$= \sum_{n=-\infty}^{-1} n^2 u(n) + \sum_{n=0}^{\infty} n^2 u(n)$$

$$= \sum_{n=0}^{\infty} n^2$$

$$= 0 + 1 + 4 + 9 + \dots + \infty$$

$$= \infty$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N n^2 u(n)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \left[\sum_{n=-N}^{-1} n^2 u(n) + \sum_{n=0}^N n^2 u(n) \right]$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N n^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} [0 + 1^2 + 2^2 + 3^2 + \dots + N^2]$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \left[\frac{N(N+1)(2N+1)}{6} \right]$$

$$= \infty$$

∴ It is neither energy & power signal

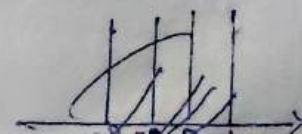
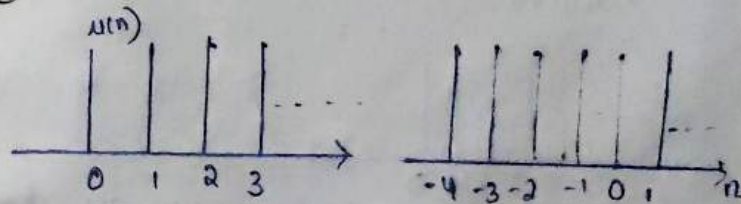
* Check whether the following signals are causal & non causal & not

(a) $x(n] = u(n+4) - u(n-2)$

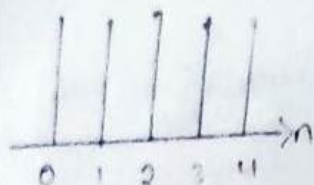
(b) $x(n] = \left(\frac{1}{4}\right)^n u(n+9) - \left(\frac{1}{2}\right)^n u(n-4)$

(c) $x(n] = u(-n)$

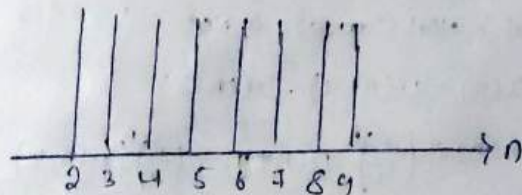
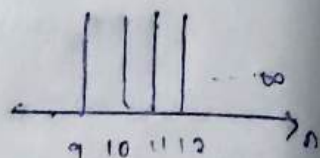
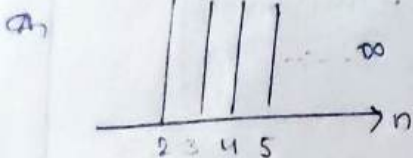
(d) $x(n] = u(n+4) - u(n-2) \quad u(n+4)$



$$x(n+2)$$

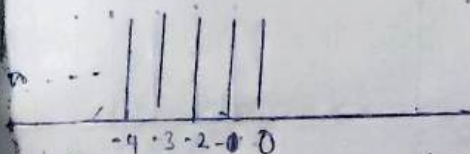


$$x(n) = \left(\frac{1}{4}\right)^n u(n-2) + \left(\frac{1}{2}\right)^n u(n-9)$$



$$x(n) = x(-n) \quad x(n) = 0 \text{ for } n < 0$$

$$x(n) \neq 0 \quad n < 0$$



therefore the given signal is non causal

convert the following signal into even and odd parts $x(n) = \{-3, 1, 2, -4, -2\}$

NOTE:

arrow indicates the value of $n=0$ if arrow not given the $n=0$ value in the sequence will be treated as at $n=0$ (right sided sequence)

$$\Rightarrow x_e(n) = \frac{1}{2} [x(n) + x(-n)] \quad \text{formula for odd}$$

$$x_o(n) = \frac{1}{2} [x(n) - x(-n)] \quad \text{formula for even}$$

$$x_e(n) = \{-0.5, -1.5, 2, -1.5, -0.5\}$$

$$x_o(n) = \{-2.5, 2.5, 0, -2.5, 2.5\}$$

* classification of discrete time systems:-

* a system is defined as an entity (device) that acts on input signal that transform it into an output signal the systems may be classified into continuous time and discrete time systems

* A continuous time system is one which transform continuous time input into continuous time output

(c) If $x_1(n)$ is the input the o/p $y_1(n) = x_1(n^2) + x_{11}(n)$
 if $x_2(n)$ is the input the o/p $y_2(n) = x_2(n^2) + x_{21}(n)$
 if $ax_1(n) + bx_2(n)$ is the input and the o/p of

$$y(n) = ax_1(n^2) + bx_2(n^2) + ax_1(n) + bx_2(n)$$

$$= a[x_1(n^2) + x_1(n)] + b[x_2(n^2) + x_2(n)]$$

$$y(n) = ay_1(n) + by_2(n)$$

It is linear.

* Time invariant and variant (1) shift invariant and shift variant

* A system is set to be time invariant or shift invariant if its input and output characteristics does not change with time mathematically $Tx(n) = y(n)$ then

$$T[x(n-k)] = y(n-k)$$

* determine whether the following system is time invariant or not

(a) $y(n) = x^2(n-2)$

(b) $y(n) = x(n) + x_2(n-2)$

(c) $y(n) = x(n/2)$

Ans:- $T[x(n) + nx(n-2)] = y(n)$

if $x(n-k)$ is the input then o/p is $y(n-k)$
 $= x(n-k) = x(n-k) + nx(n-2-k)$

if $y(n)$ is shifted by k then the o/p

$$y(n-k) = x(n-k) + n-k \cdot x(n-2-k)$$

$$y(n+k) \neq y(n-k)$$

Time invariant

b) $T[x(n)] = y(n)$ then $T[x(n-k)] = y(n-k)$

$$y(n) = x^2(n-2)$$

$$T[x^2(n-2)] = y(n)$$

if $x(n-k)$ is the i/p then o/p is $y(n,k) = x^2(n-2-k)$

If $y(n)$ is shifted by k then the o/p is $y(n-k) = x^2(n-k-2)$

$$y(n,k) = y(n-k)$$

time invariant

(c)

$$y(n) = x(n/2)$$

$$T[x(n/2)] = y(n)$$

if $x(n-k)$ is the i/p then o/p is

$$y(n,k) = x\left(\frac{n-k}{2}\right)$$

$y(n)$ is shifted by k then the o/p is

$$y(n-k) = x\left(\frac{n-k}{2}\right)$$

$$y(n,k) = y(n-k)$$

the given system is time invariant.

Causal & Non causal

- * A system is said to be causal if the output of the system at any instant of n depends only on the present and past values of the input but not depends on future inputs.
- * A system is said to be non causal if the output of the system at any instant of n depends on future values of input.
- * Check whether the following systems are causal or not

(a) $y(n) = x(n) + x(n-2)$

(b) $y(n) = x(2n)$

(c) $y(n) = \sin[x(n)]$

A. a) $y(0) = x(0) + x(-2)$ b) $y(0) = x(0)$

$y(1) = x(1) + x(-1)$

$y(1) = x(2)$

$y(2) = x(2) + x(0)$

$y(-1) = x(-2)$

$y(-1) = x(-1) + x(-3)$

$y(-2) = x(-2) + x(-4)$

Non-causal

Causal

(11)

c) $y(0) = \sin[x(0)]$

$y(1) = \sin[x(1)]$

$y(-1) = \sin[x(-1)]$

Causal

static [memory less] and dynamic [memory]
systems

* A system is set to be static & memory less if the response depends on present pip only

* A system is set to be dynamic & memory if the response depends on future pip's are past pip's are past o/p's

* find whether the following systems are dynamic & not

(a) $y(n) = x(n+2)$

(b) $y(n) = x^2(n)$

(c) $y(n) = x(n-2) + x(n)$

(d) $y(n) + 4y(n-1) + 4y(n-2) = x(n)$

(a) $y(0) = x(2)$

$y(1) = x(3)$

$y(-1) = x(1)$

$y(-2) = x(0)$

$y(-3) = x(-1)$

Dynamic

(b) $y(0) = x^2(0)$

$y(1) = x^2(1)$

$y(2) = x^2(1)$

$y(-2) = x^2(-2)$

Static

(c) $y(0) = x(-2) + x(0)$

$y(1) = x(-1) + x(1)$

$y(-1) = x(-3) + x(-1)$

Dynamic

(d) $y(n) + 4y(n-1) + 4y(n-2) = x(n)$

$y(n) = x(n) - 4y(n-1) - 4y(n-2)$

$y(0) = x(0) - 4y(-1) - 4y(-2)$

∴ It is Dynamic System

Stable and Unstable System

* A bounded signal is a signal whose amplitude and magnitude is always

a finite value that is $|x(n)| \leq m$

* where m is positive real-finite number.

* System is set to be BIBO [Bounded input & Bounded output] stable if and only if every bounded input produces a bounded output.

* let signal is bounded $x(n)$ is bounded (finite)

$$|x(n)| \leq m < \infty \quad \text{then} \quad |y(n)| \leq m_y < \infty$$

then the system is set to be stable otherwise unstable.

* check the stability of the system's define by (a) $y(n) = ax(n-1)$

$$(b) y(n) = x(n) + \frac{1}{2}x(n-1) + \frac{1}{2}x(n-2)$$

$$(c) h(n) = a^n \quad 0 < n < \infty \quad (d) h(n) = 2^n u(n)$$

A a) let $x(n) = \delta(n)$

$$y(n) = h(n)$$

$$h(n) = a\delta(n-1)$$

$$\delta(n) = 1 \quad n=0$$

$$\delta(n-1) = 1 \quad n=1$$

$$\sum_{n=-\infty}^{\infty} h(n) = \sum_{n=-\infty}^{\infty} a\delta(n-1)$$

$$= 0$$

b)

$$y(n) = x(n) + \frac{1}{2}x(n-1) + \frac{1}{2}x(n-2)$$

$$\text{let } x(n) = \delta(n)$$

$$y(n) = h(n)$$

$$h(n) = \delta(n) + \frac{1}{2}\delta(n-1) + \frac{1}{2}\delta(n-2)$$

$$|h(n)| < \infty$$

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty \Rightarrow \sum_{n=-\infty}^{\infty} \delta(n) + \frac{1}{2} \delta(n-2) + \frac{1}{4} \delta(n-4) = \frac{7}{4} < \infty$$

- the summation is

$$n=0 \Rightarrow \delta(0) + \frac{1}{2} \underbrace{\delta(-1)}_0 + \frac{1}{4} \underbrace{\delta(-2)}_0 = 1$$

$$n=1 \Rightarrow \delta(1) + \frac{1}{2} \delta(0) + \frac{1}{4} \delta(-1) = \frac{1}{2}$$

$$n=2 \Rightarrow \delta(2) + \frac{1}{2} \delta(0) + \frac{1}{4} \delta(0) = \frac{1}{4}$$

$\therefore$ The system is stable.

c) $h(n) = a^n \quad 0 < n < \infty$

$$|h(n)| < \infty$$

$$\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=0}^{\infty} a^n = a^0 + a^1 + \dots + a^{\infty} = \frac{1-a^{\infty}}{1-a}$$

if 'a' is finite and < 1 then the system is set to be stable

d) $\underline{h(n)} = \underline{2^n u(n)}$

q.m: $|h(n)| < \infty$

$$\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=-\infty}^{\infty} 2^n u(n) = \sum_{n=0}^{\infty} 2^n u(n) + \sum_{n=0}^{\infty}$$

$$2^n u(n) = \sum_{n=0}^{\infty} 2^n = \infty$$

$$h(n) < \infty$$

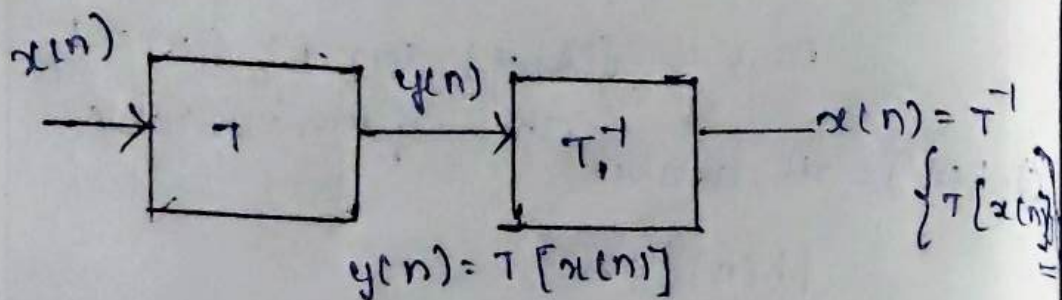
$\therefore$ The system is unstable

invertible and non invertible systems

A system is known as invertible only if an inverse system exist which

when cascaded with the original system produces an op equal to the input of the first system

* The graphical representation of an invertible system is shown in below figure



* A system does not satisfy the above criteria is known as the non invertible system that is $T^{-1}\{T[x(n)]\} \neq x(n)$

Ex: $y(n) = 3x(n) \rightarrow$ invertible system

$y(n) = x^2(n) \rightarrow$ noninvertible system

* finite impulse response [FIR] and infinite impulse response ^{discrete time}

* linear time invariant system can be classified according to the type of impulse response.

* if the impulse response sequence of a system is finite duration the system

is called the finite impulse response (FIR) system

* In the impulse response of a system is of infinite duration then the system is set to be infinite impulse response (IIR) system

* Check the following system LTI or not
 $y(n) = x(n/2)$

Ans.

Overview of z-transforms:

* The z-transform of any arbitrary sequence, $x(n)$ is defined as $z[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$
 $\Rightarrow$ [Two sided]

$$\Rightarrow X(z) = \sum_{n=0}^{\infty} x(n)z^{-n} \text{ [one-sided]}$$

* find the transform of the following sequence (a) $x(n) = a^n u(n)$ (b) $x(n) = -a^n u(-n-1)$

a) $x(n) = a^n u(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n} = a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

$$= \frac{1}{1 - az^{-1}} \quad |az^{-1}| < 1$$

$$= \frac{z}{z-a} \quad z > a$$

$$\begin{cases} |z^{-1}| < |a| \\ ||z| < |a| \\ z > a \end{cases}$$

b) $x(n) = -a^n u(-n-1)$

$$X(z) = - \sum_{n=-\infty}^{\infty} a^n (-n-1) z^{-n}$$

$$= - \sum_{n=-\infty}^{-1} a^n z^{-n}$$

$$= - \sum_{n=1}^{\infty} (a^{-1}z)^n$$

$$= - \left[\sum_{n=0}^{\infty} (a^{-1}z)^n - 1 \right]$$

$$\left[\frac{1}{1-a|z|^{-1}} \right]$$

$$= \frac{z}{z-a} \quad (|z| < a)$$

* properties of z-transforms

Statement

if $x_1(n)$

if $x_1(n) \xrightarrow{+} x_1(z)$ and $x_2(n) \xrightarrow{+} x_2(z)$

then $ax_1(n) + bx_2(n) \xrightarrow{+} ax_1(z) + bx_2(z)$

Proof

$$z[x(n)] = x(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$x(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$z[x_1(n)] = x_1(z) = \sum_{n=-\infty}^{\infty} x_1(n) z^{-n}$$

$$z[x_2(n)] = x_2(z) = \sum_{n=-\infty}^{\infty} x_2(n) z^{-n}$$

$$z[x_2(n)] = x_2(z) = \sum_{n=-\infty}^{\infty} x_2(n) z^{-n}$$

$$\text{then } z[ax_1(n) + bx_2(n)] = \sum_{n=-\infty}^{\infty} [ax_1(n) + bx_2(n)] z^{-n}$$

$$= a \sum_{n=-\infty}^{\infty} x_1(n) z^{-n} + b \sum_{n=-\infty}^{\infty} x_2(n) z^{-n}$$

$\underbrace{\hspace{10em}}_{x_1(z)} \qquad \underbrace{\hspace{10em}}_{x_2(z)}$

$$= ax_1(z) + bx_2(z)$$

the ROC of $x_1(z)$ is R_1 the ROC of $x_2(z)$ is R_2 then the ROC of $ax_1(z) + bx_2(z)$ then $R_1 \cap R_2$

* find z transform of and it's ROC of $x(n) = 2(5/6)^n u(-n-1) + 3(1/4)^n u(n)$

Ans:

$$\begin{aligned}
 X(z) &= \sum_{n=-\infty}^{\infty} [2(5/6)^n u(-n-1) + 3(1/4)^n u(n)] z^{-n} \\
 &= 2 \left[\sum_{n=-\infty}^{\infty} (5/6)^n u(-n-1) z^{-n} \right] + 3 \sum_{n=-\infty}^{\infty} (1/4)^n u(n) z^{-n} \\
 &= 2 \sum_{n=-\infty}^{-1} (5/6)^{-n-1} z^{-n} + 3 \sum_{n=0}^{\infty} (1/4)^n z^{-n} \\
 &= 2 \sum_{n=1}^{\infty} \left[\left(\frac{5}{6} \right)^{-1} z \right]^n + 3 \sum_{n=0}^{\infty} \left[\frac{1}{4} z^{-1} \right]^n \\
 &= 2 \left[\frac{(5/6)^{-1} z}{1 - (5/6)^{-1} z} \right] + 3 \left[\frac{1}{1 - \frac{1}{4} z^{-1}} \right] \begin{cases} (5/6)^{-1} z < 1 \Rightarrow z < 5/6 \\ 1/4 z^{-1} < 1 \Rightarrow z > 1/4 \end{cases} \\
 &= 1/4 < z < 5/6
 \end{aligned}$$

* time shifting property.

Statement:

* If $x(n) \xrightarrow{Z} X(z)$ z-transform pair then $x(n-m) \xrightarrow{Z} z^{-m} X(z)$ [if initial conditions are equal to zero]

$$x(n-m) \xrightarrow{Z} z^{-m} X(z) + z^{-m} \sum_{n=1}^{-m} x(-n) z^n$$

Proof 1

$$Z[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$Z[x(n-m)] = \sum_{n=-\infty}^{\infty} [x(n-m)]z^{-n}$$

$$\text{let } n-m = p \quad p \rightarrow \infty \Rightarrow n = m+p$$

$$= \sum_{p=-\infty}^{\infty} x(p)z^{-(m+p)} \quad p \rightarrow \infty$$

$$= \sum_{p=-\infty}^{\infty} x(p)z^{-m}z^{-p}$$

$$p = -\infty$$

$$= z^{-m} \sum_{p=-\infty}^{\infty} x(p)z^{-p}$$

$$= z^{-m} X(z)$$

$$* Z[x(n-m)] = \sum_{n=0}^{\infty} x(n-m)z^{-n}$$

$$n-m = p$$

$$p \rightarrow -m$$

$$p \rightarrow \infty$$

$$= \sum_{p=-m}^{\infty} x(p)z^{-m}z^{-p}$$

$$z = -m$$

$$= z^{-m} \sum_{p=-m}^{\infty} x(p)z^{-p} + z^{-m} \sum_{p=0}^{\infty} x(p)z^{-p}$$

$$p = -m$$

$$Z[x(n-m)] = z^{-m} \sum_{p=1}^m x(-p)z^p + z^{-m} X(z)$$

* Similarly $z[x(n+m)] = z^m x(z) - z^m \sum_{p=0}^{m-1} x(p) z^{-p}$

$\Rightarrow z^m [x(z) - n+m=p \Rightarrow n=p-m]$

$p \rightarrow m \text{ to } \infty$

$$= \sum_{p=m}^{\infty} x(p) z^{-(p-m)}$$

$$= z^m \sum_{p=m}^{\infty} x(p) z^{-p}$$

$$= z^m \left[\sum_{p=0}^{\infty} x(p) z^{-p} \right] - \sum_{p=0}^{m-1} x(p) z^{-p}$$

$$z[x(n+m)] = z^m x(z) - \sum_{p=0}^{m-1} x(p) z^{-p}$$

$$= z^m x(z) - z^m x(0) - x(1) z^{m-1} - x(2) z^{m-2} \dots$$

$$= z^m \left[x(z) - x(0) - \frac{x(1)}{z} - \frac{x(2)}{z^2} - \dots - \frac{x(m-1)}{z^{m-1}} \right]$$

$$z[x(n-m)] = z^{-m} x(z) + x(-1) z^{-(m-1)} + x(-2) z^{-(m-2)} + \dots + x(-m)$$

$$z[x(n)]$$

$$z[x(n+m)] = z^m x(z) - z^m x(0) - z^{m-1} x(1) - \dots - \frac{x(m-1)}{z^{m-1}}$$

$$\Rightarrow Z[x(n-1)] = Z^{-1}x(z) + x(-1)$$

$$\Rightarrow Z[x(n+1)] = Zx(z) - Zx(0)$$

$$\Rightarrow Z[x(n-2)] = Z^{-2}x(z) + Z^{-1}x(-1) + x(-2)$$

$$\Rightarrow Z[x(n+2)] = Z^2x(z) - Z^2x(0) - x(1)Z$$

$$\Rightarrow Z[x(n+3)] = Z^3x(z) + Z^{-2}x(-1) + Z^{-3}x(-2) + x(-3)$$

* Time Scaling property

if $x(n) \xrightarrow{Z} x(z)$ then $a^n x(n) \xrightarrow{Z} x(z/a)$

$$Z[x(n)] = \sum_{n=-\infty}^{\infty} x(n) Z^{-n} = x(z)$$

$$Z[a^n x(n)] = \sum_{n=-\infty}^{\infty} [a^n x(n)] Z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) (Z/a)^{-n} = x(z/a)$$

Time reversal property

if $x(n) \xrightarrow{Z} x(z)$ form a Z-transform then $x(-n) \xrightarrow{Z} x(1/z)$

$$Z[x(n)] = \sum_{n=-\infty}^{\infty} x(n) Z^{-n} = x(z)$$

$$Z[x(-n)] = \sum_{n=-\infty}^{\infty} x(-n) Z^{-n}$$

let $-n = p$

$$\sum_{p=-\infty}^{\infty} x(p) Z^p$$

$$= \sum_{p=-\infty}^{\infty} x(p) (z^{-1})^p = x(z)$$

differentiation in z-domain

* if $x(n) \xleftrightarrow{z} x(z)$ then $n x(n) \xleftrightarrow{z} -z \frac{d}{dz} x(z)$

$$Z[x(n)] = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = x(z)$$

$$\frac{d x(z)}{dz} = \frac{d}{dz} \left[\sum_{n=-\infty}^{\infty} x(n) z^{-n} \right]$$

$$= \sum_{n=-\infty}^{\infty} x(n) \frac{d}{dz} z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) (-n) z^{-n-1}$$

$$-z \frac{d}{dz} x(z) = \sum_{n=-\infty}^{\infty} n x(n) z^{-n}$$

$$= z \cdot [n x(n)]$$

solution of difference equations by using z-transform

~~* determine $y(n)$ for $z \geq 0$ if $y(n) = \frac{1}{2} y(n-1) + z(n)$~~

* determine $y(n)$ for $z \geq 0$ if $y(n) = \frac{1}{2} y(n-1) + z(n)$
 for $x(n) = (1/3)^n u(n)$ the initial condition $y(-1) = 0$

$$y(n) = \frac{1}{2} y(n-1) + x(n)$$

$$x(n) = (1/3)^n u(n)$$

$$y(z) = \frac{1}{2} [z^{-1} y(z) + 1] + \frac{z}{z-1/3}$$

$$= \frac{1}{2} [z^{-1} y(z) + 1] + \frac{z}{z-1/3}$$

$$y(z) [1 - 0.5z^{-1}] = 0.5z^{-1} y(z) + 0.5 + \frac{z}{z-1/3}$$

$$y(z) [1 - 0.5z^{-1}] = 0.5 + \frac{z}{z-1/3}$$

$$y(z) = \frac{0.57}{z-0.5} + \frac{z^2}{(z-1/3)(z-0.5)}$$

$$\frac{y(z)}{z} = \frac{0.5}{z-0.5} + \frac{z}{(z-0.5)(z-1/3)}$$

$$y(z) = \frac{0.57}{z-0.5} + \frac{3z}{z-0.5} - \frac{2z}{z-1}$$

$$y(n) = -\left[\frac{1}{2} (0.5)^n + 3 (0.5)^{n-2} (1/3)^n \right]$$

• determine unit step response of the system whose difference equation is,

$$y(n) = 0.7 y(n-1) + 0.12 y(n-2) + x(n-1) + x(n-2)$$

$$\text{if } y(1) = y(2) = 1$$

$$\begin{aligned} \text{or } y(z) &= 0.7 [z^{-1} y(z) + y(0)] + 0.12 \left\{ \begin{array}{l} x(z) = \frac{z}{z-1} \\ x(0) = \frac{z}{z-1} \end{array} \right\} \\ &\quad [z^{-2} y(z) + z^{-1} y(0) + y(0)] \\ &= [z^{-1} x(z) + x(0)] + [z^{-2} x(z) + z^{-1} x(0) + x(0)] \end{aligned}$$

$$\begin{aligned} y(z) [1 - 0.7z^{-1} + 0.12z^{-2}] &= 0.58 + 0.12z^{-1} x(z) [z^{-1} + z^{-2}] \\ &= \frac{z}{z-1} [z^{-1} + z^{-2}] = \frac{1+z^{-1}}{z-1} \end{aligned}$$

$$y(z) = \frac{1+z^{-1}}{(z-1)(1-0.7z^{-1}+0.12z^{-1})} + \frac{0.58z - 0.12z^{-1}}{1-0.7z^{-1}+0.12z^{-2}}$$

$$\begin{aligned} &= \frac{z+1}{(z-1)(z^2-0.7z+0.12)} + \frac{0.58z - 0.12}{z^2-0.7z+0.12} \\ &= \frac{z+1}{(z-1)(z^2-0.7z+0.12)} + \frac{0.58z - 0.12}{z^2-0.7z+0.12} \end{aligned}$$

$$\begin{aligned} \frac{y(z)}{z} &= \frac{z+1}{(z-1)(z-0.3)(z-0.4)} + \frac{0.58z - 0.12}{(z-0.3)(z-0.4)} \\ &= \left[\frac{A}{z-1} + \frac{B}{z-0.3} + \frac{C}{z-0.4} \right] + \left[\frac{D}{(z-0.3)} + \frac{E}{(z-0.4)} \right] \end{aligned}$$

$$A = \frac{z+1}{(z-1)(z-0.3)(z-0.4)} \times (z-1) \Big|_{z=1} = 4.76$$

$$B = \frac{z+1}{(z-1)(z-0.3)(z-0.4)} \times (z-0.3) \Big|_{z=0.3} = 18.57$$

$$C = \frac{z+1}{(z-1)(z-0.3)(z-0.4)} \times (z-0.4) \Big|_{z=0.4} = 23.33$$

$$D = \frac{0.58z - 0.7z}{(z-0.3)(z-0.4)} \times (z-0.3) \Big|_{z=0.3} = -0.54$$

$$E = \frac{0.58z - 0.7z}{(z-0.3)(z-0.4)} \times (z-0.4) \Big|_{z=0.4} = 1.12$$

$$\frac{Y(z)}{z} = \frac{4.76}{z-1} + \frac{18.57}{z-0.3} + \frac{23.33}{z-0.4} + \frac{0.54}{z-0.3} + \frac{1.12}{z-0.4}$$

$$Y(z) = 4.76 \frac{z}{z-1} + 18.57 \frac{z}{z-0.3} - 23.33 \frac{z}{z-0.4} - 0.54 \frac{z}{z-0.3} + 1.12 \frac{z}{z-0.4}$$

$$\frac{z}{z-0.3} + 1.12 \frac{z}{z-0.4}$$

$$y(n) = [4.76(1)^n + 18.57(0.3)^n - 23.33(0.4)^n - 0.54(0.3)^n + 1.12(0.4)^n] u(n)$$

$$y(n) = [4.76(1)^n + 18.03(0.3)^n - 22.21(0.4)^n] u(n)$$

* find a unit step response of the system whose difference equation is

$$y(n) - (1/4)y(n-1) + (1/4)y(n-2) - 1/16 y(n-3) = 2x(n) - 3x(n-1)$$

Ans: The z transform on both sides of above equation

$$Y(z) - \frac{1}{4} [z^{-1} Y(z) + y(-1)] + \frac{1}{4} [z^{-2} Y(z) + z^{-1} y(-1) + y(-2)] - \frac{1}{16} [z^{-3} Y(z) + z^{-2} y(-1) + z^{-1} y(-2) + y(-3)]$$

$$= 2X(z) - 3 [z^{-1} X(z) + x(-1)]$$

$$Y(z) \left[1 - \frac{1}{4} z^{-1} + \frac{1}{4} z^{-2} - \frac{1}{16} z^{-3} \right] = X(z) [2 + 3z^{-1}]$$

$$Y(z) \left[\frac{z^3 - 1/4 z^2 + 1/4 z - 1/16}{z^3} \right] = \frac{2z+3}{z-1} [2 + 3z^{-1}] = \frac{2z+3}{z-1}$$

$$Y(z) = \left(\frac{2z+3}{z-1} \right) \left(\frac{z^3}{z^3 - \frac{1}{4} z^2 + \frac{1}{4} z - \frac{1}{16}} \right)$$

$$\frac{Y(z)}{z-1} = \frac{2z+3}{(z-1)^2 (z^2 + 1/4)(z-1/4)}$$

* find the impulse response of the system whose difference equation is given by

$$y(n) = \frac{3}{4} y(n-1) + \frac{1}{8} y(n-2) + x(n)$$

* The Z transform of the above equation on both sides is $Y(z) = \frac{3}{4} [z^{-1}Y(z) + y(-1)] + \frac{1}{8} [z^{-2}Y(z) + z^{-1}y(-1) + y(-2)] = X(z)$

$$Y(z) = \frac{3}{4} z^{-1} Y(z) + \frac{1}{8} z^{-2} Y(z) = X(z)$$

$$Y(z) \left[1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2} \right] = 1$$

$$\frac{Y(z)}{z^2} \left[z^2 - \frac{3}{4} z + \frac{1}{8} \right] = 1$$

impulse
 $x(n) = \delta(n)$
 $\sum_{n=-\infty}^{\infty} x(n) z^{-n}$
 $\delta(n) = 1 \text{ if } n=0$
 0 otherwise

$$\frac{Y(z)}{z^2} = \frac{1}{z^2 - \frac{3}{4} z + \frac{1}{8}}$$

$$\frac{Y(z)}{z} = \frac{z}{(z^2 - \frac{1}{4})(z - \frac{1}{2})}$$

$$\frac{A}{z-1/4} + \frac{B}{z-1/2}$$

$$A = \frac{z}{(z-1/4)(z-1/2)} \times (z-1/2) \Big|_{z=1/4} = -1$$

$$B = \frac{z}{(z-1/4)(z-1/2)} \times (z-1/4) \Big|_{z=1/2} = 2$$

$$Y(z) = \frac{-z}{z-1/4} + 2 \frac{z}{z-1/2}$$

$$y(n) = \left[-(1/4)^n + 2(1/2)^n \right] u(n)$$

Realisation of discrete time systems

To realise a discrete system of the given difference equation

To be converted into an algebraic sum of terms of the above equation can be represented by a suitable element the symbol of basic element use for constructing the block diagram of a discrete time system are

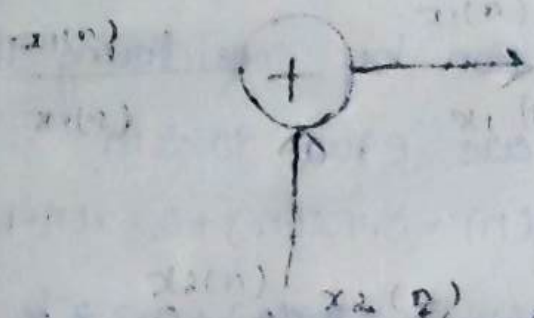
- 1) adder
- 2) multiplier
- 3) delay element

change
change
change
change
change

adder

* An adder is used to add two or more signals

* The symbol for adder is shown in below



multiplier & constant multiplier

* A multiplier is used to multiply the signal by a constant. The output of multiplier is product of input signals and the constant of the multiplier.

* The symbol for the multiplier is shown in below figure.



delay element & unit delay element

* A unit delay element is used to delay the signal passing through it.



* construct a block diagram for the discrete time system whose input and output are described by the following difference equation $y(n) = 0.7x(n) + 0.3x(n-1)$

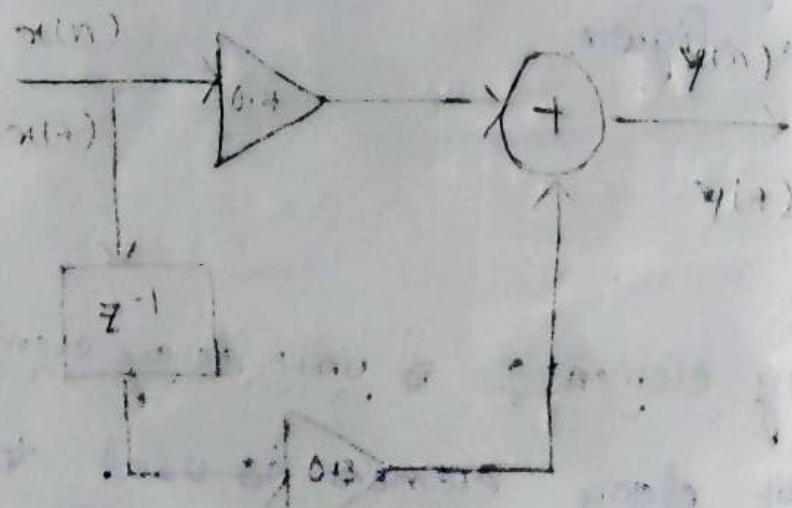
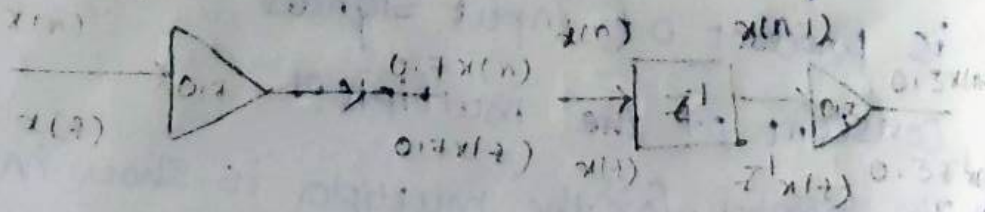
* Taking z-transform on both sides of below equation by considering the initial conditions are equal to zero

$$y(n) = 0.7x(n) + 0.3x(n-1)$$

~~$$y(z) = 0.7x(z) + 0.3z^{-1}x(z)$$~~

$$y(z) = 0.7x(z) + 0.3z^{-1}x(z)$$

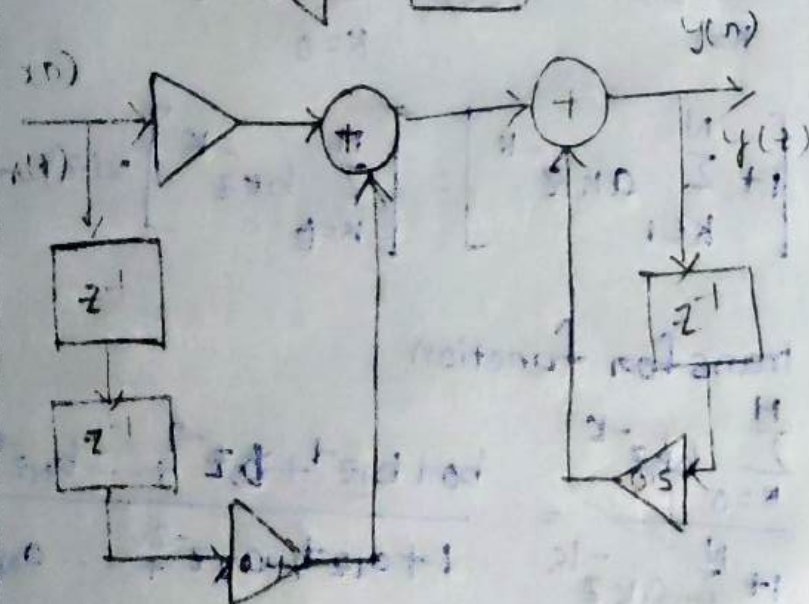
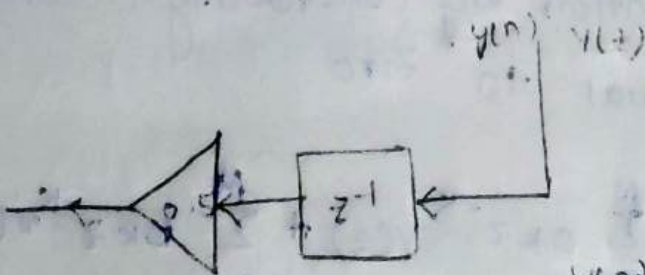
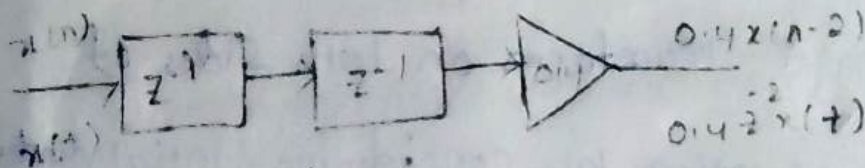
The above equation can be realised as follows



* construct a block diagram whose i/p and o/p relations are described by the following difference equation.

$$y(n) = 0.5 y(n-1) + 0.8 x(n) + 0.4 x(n-1)$$

Ans- $y(z) = 0.5 z^{-1} y(z) + 0.8 x(z) + 0.4 z^{-1} x(z)$



Realisation of discrete time systems from

Transfer function :-

Transfer function of IIR system :-

* IIR is defined as $y(n) = \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$

$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

The transform function of DTR in general an DTR system can be described by the following difference equation.

* Taking z -transform on both sides of above equation by considering initial conditions are equal to zero

$$Y(z) = - \sum_{k=1}^N a_k z^{-k} Y(z) + \sum_{k=0}^M b_k z^{-k} X(z)$$

$$Y(z) \left[1 + \sum_{k=1}^N a_k z^{-k} \right] = \left[\sum_{k=0}^M b_k z^{-k} \right] X(z)$$

$\therefore$ the transform function

$$\frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}}$$

The different types of structures for realising DTR system :-

* direct form one

* direct form two

* direct form two

- * Transposed form
- * cascaded form
- * Parallel form
- * lattice structure
- * ladder structure

* direct form ONE

* direct form one realisation of an IR system is the direct implementation of difference equation of transfer function. It is simplified and most straight forward realisation.

* The difference equation of IR system is

$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

Taking z transform on b.s of above equation by considering the initial conditions are equal to zero.

$$Y(z) = - \sum_{k=1}^N a_k z^{-k} Y(z) + \sum_{k=0}^M b_k z^{-k} X(z)$$

$$\text{let } W(z) = \sum_{k=0}^M b_k z^{-k} X(z)$$

$$Y(z) = -a_1 z^{-1} Y(z) - a_2 z^{-2} Y(z) - \dots - a_N z^{-N} Y(z) + W(z)$$

$$W(z) = b_0 X(z) + b_1 z^{-1} X(z) + \dots + b_M z^{-M} X(z)$$

NOTE

IF IR system is more complex that is

higher order - then introduce an intermediate variable $w(z)$

* realise the following IIR system using direct form one $y(n) + 2y(n-1) + 3y(n-2) = 4x(n) + 5x(n-1) + 6x(n-2)$

Ans:

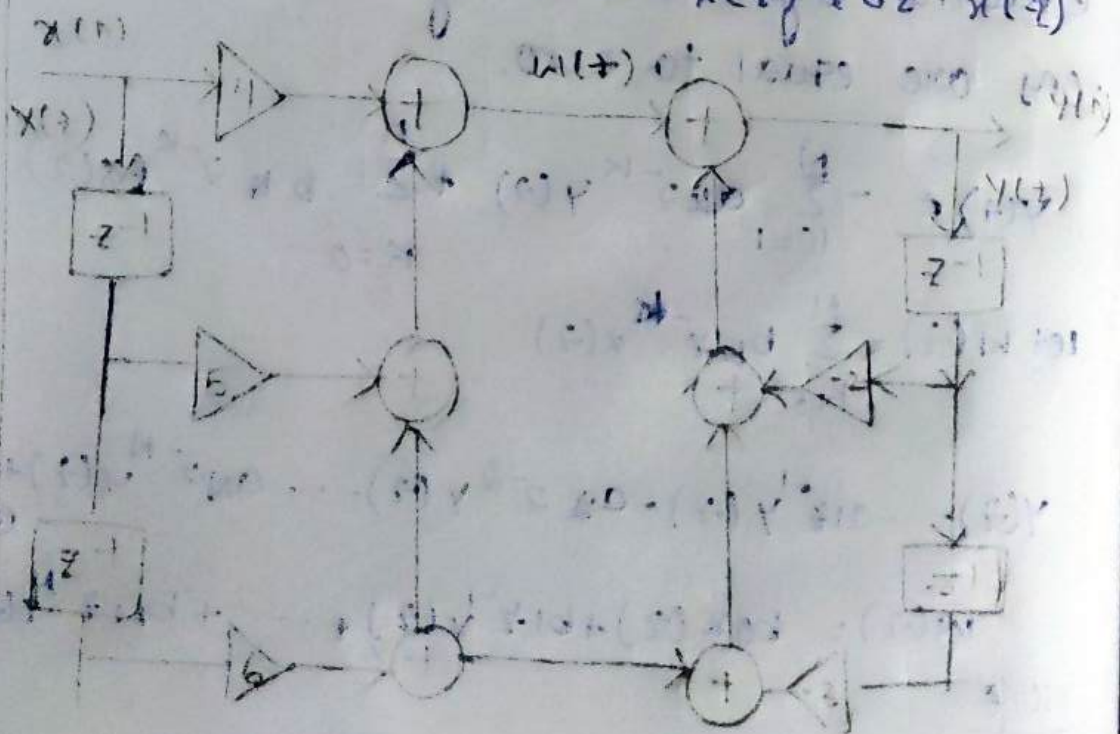
$$y(n) + 2y(n-1) + 3y(n-2) = 4x(n) + 5x(n-1) + 6x(n-2)$$

$$y(z) + 2z^{-1}y(z) + 3z^{-2}y(z) = 4x(z) + 5z^{-1}x(z) + 6z^{-2}x(z)$$

$$y(z) = -2z^{-1}y(z) - 3z^{-2}y(z) + 4x(z) + 5z^{-1}x(z) + 6z^{-2}x(z)$$

$$\text{Let } w(z) = -2z^{-1}y(z) - 3z^{-2}y(z) + w(z)$$

$$w(z) = 4x(z) + 5z^{-1}x(z) + 6z^{-2}x(z)$$



Find the direct form one realisation
 a discrete time ~~signal~~ system represented
 by the transfer function $H(z) = \frac{3z^3 - 5z^2 + 9z + 3}{(z-1)(z^2 - z + 1)}$

$$H(z) = \frac{3z^3 - 5z^2 + 9z + 3}{(z-1)(z^2 - z + 1)}$$

$$= \frac{3z^3 - 5z^2 + 9z + 3}{z^3 - \frac{3}{2}z^2 + \frac{5}{6}z - \frac{1}{6}}$$

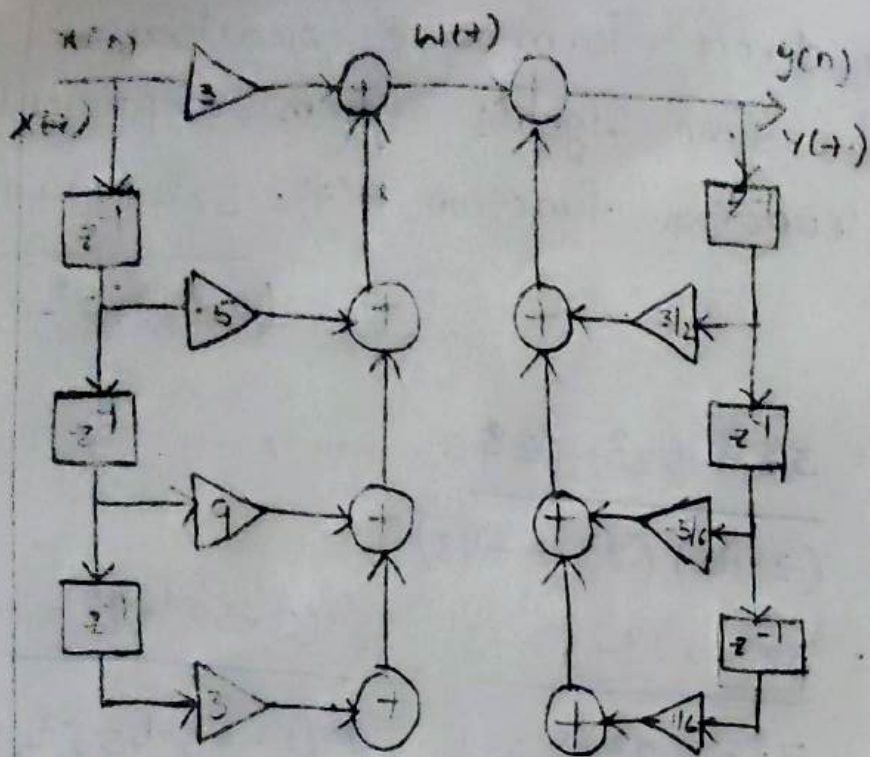
$$= \frac{3(3 - 5z^{-1} + 9z^{-2} + 3z^{-3})}{1 - \frac{3}{2}z^{-1} + \frac{5}{6}z^{-2} - \frac{1}{6}z^{-3}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3 - 5z^{-1} + 9z^{-2} + 3z^{-3}}{1 - \frac{3}{2}z^{-1} + \frac{5}{6}z^{-2} - \frac{1}{6}z^{-3}}$$

$$Y(z) = \frac{3}{2}z^{-1}Y(z) + \frac{5}{6}z^{-2}Y(z) + \frac{1}{6}z^{-3}Y(z) + 3X(z) - 5z^{-1}X(z) + 9z^{-2}X(z) + 3z^{-3}X(z)$$

$$Y(z) = \frac{3}{2}z^{-1}Y(z) + \frac{5}{6}z^{-2}Y(z) + \frac{1}{6}z^{-3}Y(z) + W(z)$$

$$W(z) = 3X(z) - 5z^{-1}X(z) + 9z^{-2}X(z) + 3z^{-3}X(z)$$



* Direct-form Two

* The direct form Two Structure is an alternative to direct form one structure. It is more advantageous than direct form one because it uses less no. of delay units than direct form one structure.

* In direct form Two the given transfer function is splitted into the 2 parts as given below

Consider the transfer function $\frac{y(z)}{x(z)}$

$$\frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_m z^{-m}}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}}$$

$$\frac{y(z)}{x(z)} = \frac{w(z)}{x(z)} \times \frac{y(z)}{w(z)}$$

let

$$\frac{w(z)}{x(z)} = \frac{1}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}}$$

$$x(z) = w(z) + a_1 z^{-1} w(z) + a_2 z^{-2} w(z) + \dots + a_N z^{-N} w(z)$$

by cross multiplication

$$\therefore w(z) = x(z) - a_1 z^{-1} w(z) - a_2 z^{-2} w(z) - \dots - a_N z^{-N} w(z)$$

$$\frac{y(z)}{w(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_m z^{-m}}{1}$$

by cross multiplication

$$y(z) = b_0 w(z) + b_1 z^{-1} w(z) + \dots + b_m z^{-m} w(z)$$

→ Realise the following DTF system using direct form II $y(n) + 2y(n-1) + 3y(n-2) = 4x(n) + 5x(n-1) + 6x(n-2)$

→ find a transform

Find a transform on b.s on considering initial conditional are equal to zero

$$y(z) + 2z^{-1}y(z) + 3z^{-2}y(z) = 4x(z) + 5z^{-1}x(z) + 6z^{-2}x(z)$$

$$y(z) [1 + 2z^{-1} + 3z^{-2}] = x(z) [4 + 5z^{-1} + 6z^{-2}]$$

then the transfer function

$$\frac{y(z)}{x(z)} = \frac{4 + 5z^{-1} + 6z^{-2}}{1 + 2z^{-1} + 3z^{-2}}$$

$$\frac{y(z)}{x(z)} = \frac{w(z)}{x(z)} \times \frac{y(z)}{w(z)}$$

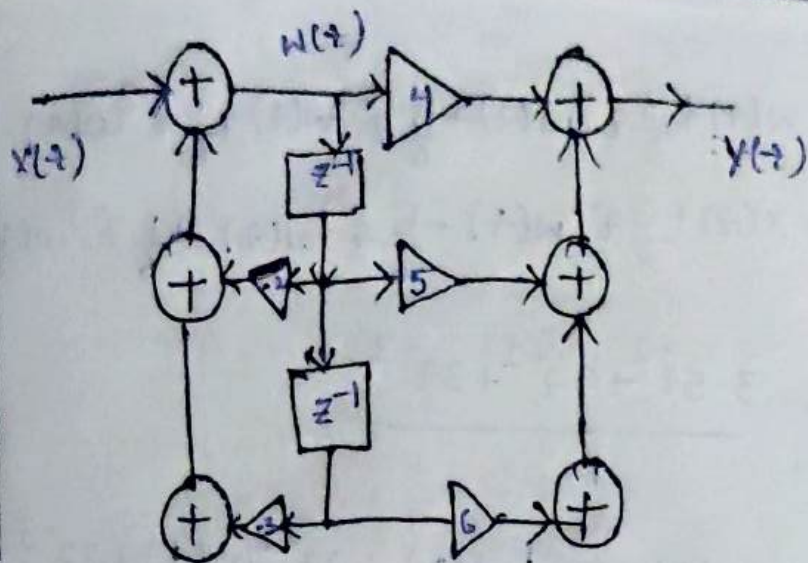
$$\frac{w(z)}{x(z)} = \frac{1}{1 + 2z^{-1} + 3z^{-2}}$$

$$x(z) = w(z) + 2z^{-1}w(z) + 3z^{-2}w(z)$$

$$w(z) = x(z) - 2z^{-1}w(z) - 3z^{-2}w(z)$$

$$\frac{y(z)}{w(z)} = \frac{4 + 5z^{-1} + 6z^{-2}}{1}$$

$$y(z) = 4w(z) + 5z^{-1}w(z) + 6z^{-2}w(z)$$



* find direct form to realization of a system in the system whose transfer function is given by $H(z) = \frac{3z^3 - 5z^2 + 9z + 3}{(z-1/2)(z^2 + 1/3)}$

$$\alpha: H(z) = \frac{3z^3 - 5z^2 + 9z + 3}{(z-1/2)(z^2 + 1/3)}$$

$$\frac{Y(z)}{X(z)} = \frac{3z^3 - 5z^2 + 9z + 3}{z^3 - \frac{3}{2}z^2 + \frac{5}{6}z - \frac{1}{6}}$$

$$\frac{Y(z)}{X(z)} = \frac{z^3 (3 - 5z^{-1} + 9z^{-2} + 3z^{-3})}{z^3 (1 - \frac{3}{2}z^{-1} + \frac{5}{6}z^{-2} - \frac{1}{6}z^{-3})}$$

$$\frac{Y(z)}{X(z)} = \frac{W(z)}{X(z)} \cdot \frac{Y(z)}{W(z)}$$

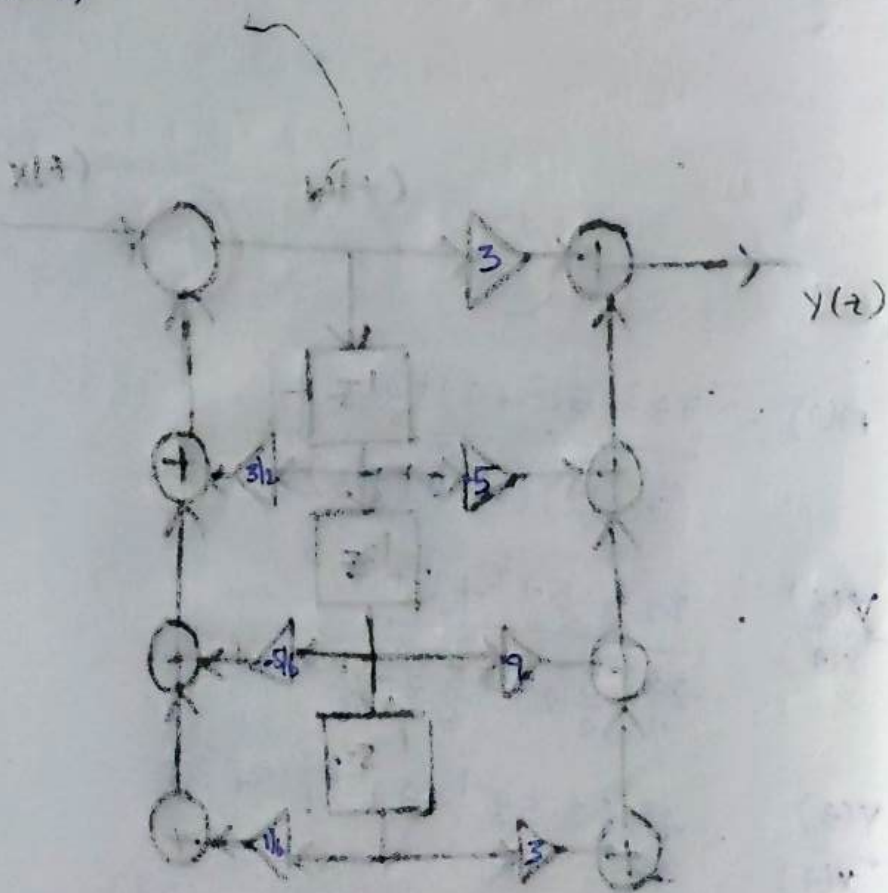
$$\frac{W(z)}{X(z)} = \frac{1}{1 - 3/2z^{-1} + 5/6z^{-2} - 1/6z^{-3}}$$

$$x(z) = w(z) + \frac{3}{2}z^{-1}w(z) + \frac{5}{6}z^{-2}w(z) + \frac{1}{6}z^{-3}w(z)$$

$$w(z) = x(z) + \frac{3}{2}z^{-1}w(z) - \frac{5}{6}z^{-2}w(z) + \frac{1}{6}z^{-3}w(z)$$

$$\frac{Y(z)}{W(z)} = \frac{3 - 5z^{-1} + 9z^{-2} + 3z^{-3}}{1}$$

$$Y(z) = 3w(z) - 5z^{-1}w(z) + 9z^{-2}w(z) + 3z^{-3}w(z)$$



* Realise the following difference equation by using direct form 1 and direct form 2.

$$y(n) = -\frac{13}{12}y(n-1) - \frac{9}{24}y(n-2) - \frac{1}{24}y(n-3) + 14x(n-1) + 3x(n-2)$$

2. Draw an LTI system in direct form one and direct form 2 whose transfer function is given by $H(z) = \frac{2 + 0.3z^{-1} + 0.5z^{-2}}{1 + 0.7z^{-1} + 0.9z^{-2}}$

Direct form-1

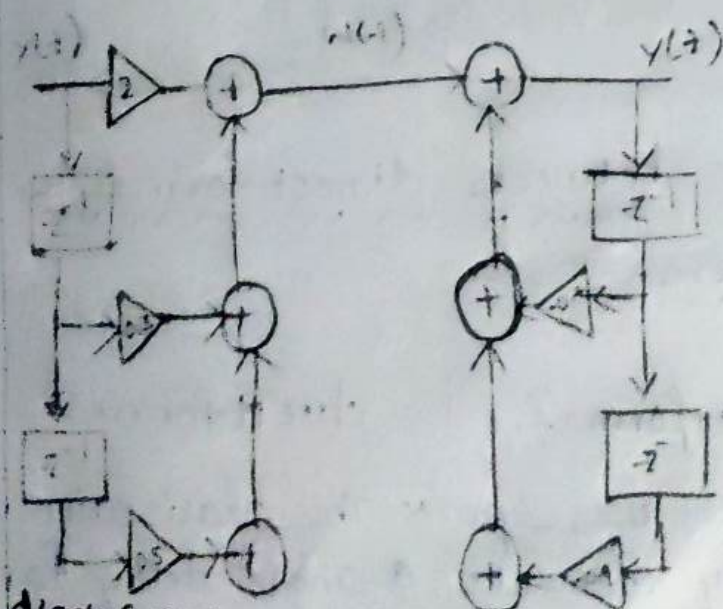
$$\text{Ans. } H(z) = \frac{Y(z)}{X(z)} = \frac{2 + 0.3z^{-1} + 0.5z^{-2}}{1 + 0.7z^{-1} + 0.9z^{-2}}$$

$$Y(z) + 0.7z^{-1}Y(z) + 0.9z^{-2}Y(z) = 2X(z) + 0.3z^{-1}X(z) + 0.5z^{-2}X(z)$$

$$Y(z) = -0.7z^{-1}Y(z) - 0.9z^{-2}Y(z) + 2X(z) + 0.3z^{-1}X(z) + 0.5z^{-2}X(z)$$

$$= -0.7z^{-1}Y(z) - 0.9z^{-2}Y(z) + W(z)$$

$$W(z) = 2X(z) + 0.3z^{-1}X(z) + 0.5z^{-2}X(z)$$



Direct form-1

$$H(z) = \frac{Y(z)}{X(z)} = \frac{W(z)}{X(z)} \cdot \frac{Y(z)}{W(z)} = \frac{2 + 0.3z^{-1} + 0.5z^{-2}}{1 + 0.7z^{-1} + 0.9z^{-2}}$$

* It is also called Non canonical form because it uses more no. of delays

* It is not efficient in terms of memory requirements

* For $(m-1)^{\text{th}}$ to $(N-1)^{\text{th}}$ Order FIR system requires $m+n-1$ multipliers $m+n+2$ adders and $m+n-2$ delays

* It is also called canonical form because it requires a minimum number of delays

* It is more efficient in terms of memory requirements.

* For the $(m-1)^{\text{th}}$ to $(N-1)^{\text{th}}$ Order IIR system requires $m+n-1$ multipliers $m+n+2$ adders and maximum of $[(N-1), (M+1)]$ delays

transpose form :-

* It is practically to that if we reverse the direction of all the branch transmittances and interchange the input and output in the structure, the system remains unchanged. The transpose structure of transpose form of reverse structure is obtained by reversing

the direction of all branch transmiss-
-ances and interchange the input and
output in the direct form structure

the p

* The procedure for transposed form
of structure of IR system is as follows

⇒ Step 1: Realise the given difference equa-
-tion of transfer function by using
direct form structure usually direct
form 2.

⇒ Step 2: Rewrite of transpose the direction
of signal flow and interchange the
i/p and o/p nodes

⇒ Step 3: Replace the Junction points by
adders and adders by Junction
Points.

⇒ Step 4: Fold the structure which is
transposed form of Rewrite form
of an IR system

* Realise an IR system $y(n) + 2y(n-1) + 3y(n-2) = 4x(n) + 5x(n-1) + 6x(n-2)$

$$y(n) + 2y(n-1) + 3y(n-2) = 4x(n) + 5x(n-1) + 6x(n-2)$$

$$\frac{Y(z)}{X(z)} = \frac{4 + 5z^{-1} + 6z^{-2}}{1 + 2z^{-1} + 3z^{-2}}$$

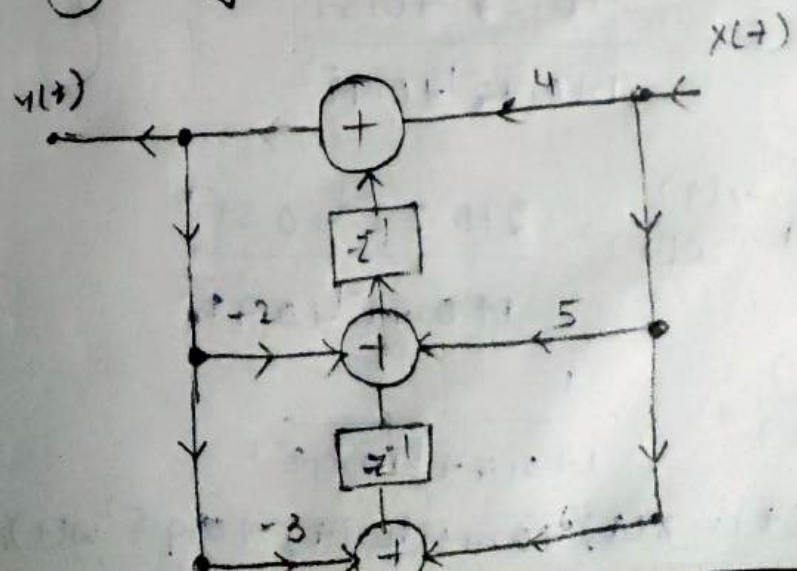
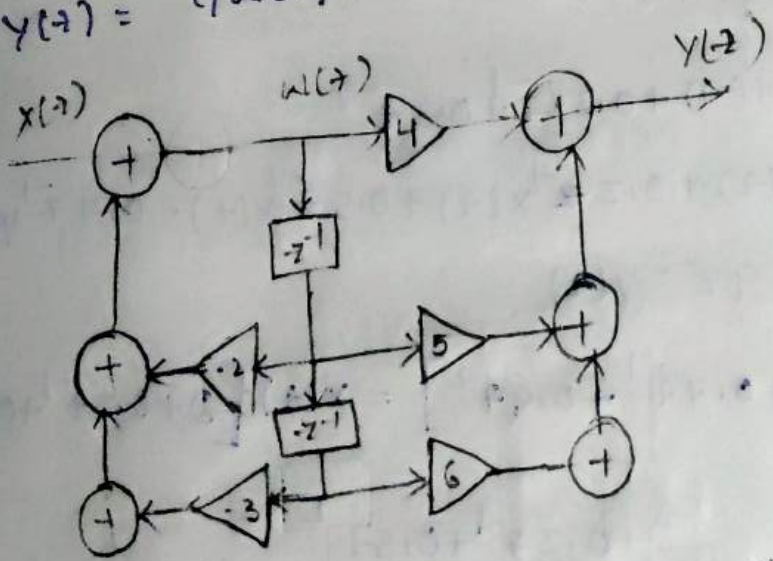
$$\frac{W(z)}{X(z)} \cdot \frac{Y(z)}{W(z)} = \frac{4 + 5z^{-1} + 6z^{-2}}{1 + 2z^{-1} + 3z^{-2}}$$

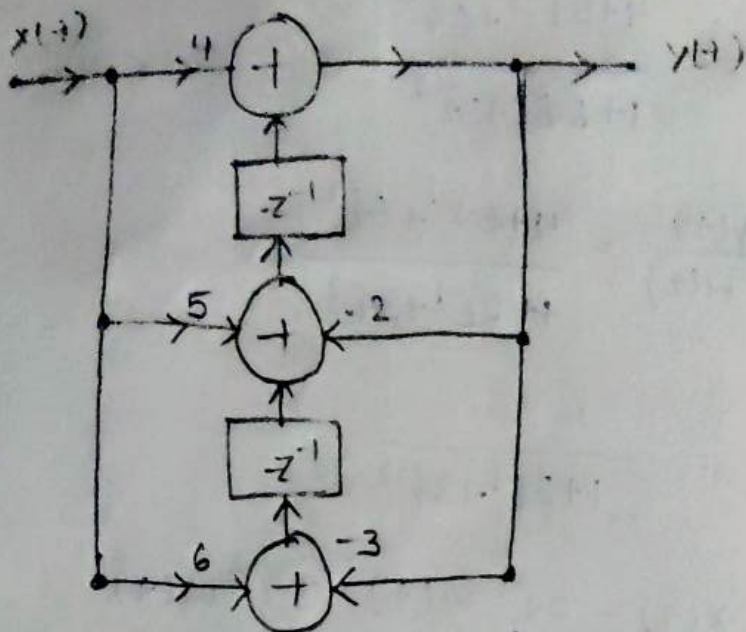
$$\frac{W(z)}{X(z)} = \frac{1}{1 + 2z^{-1} + 3z^{-2}}$$

$$W(z) = X(z) - 2z^{-1}W(z) - 3z^{-2}W(z)$$

$$\frac{Y(z)}{W(z)} = \frac{4 + 5z^{-1} + 6z^{-2}}{1}$$

$$Y(z) = 1W(z) + 5z^{-1}W(z) + 6z^{-2}W(z)$$





* find the transposed form for the system described by the difference equation

$$y(n) = 2x(n) + 0.3x(n-1) + 0.5x(n-2) - 0.7y(n-1) - 0.9y(n-2)$$

$$Y(z) = 2X(z) + 0.3z^{-1}X(z) + 0.5z^{-2}X(z) - 0.7z^{-1}Y(z) - 0.9z^{-2}Y(z)$$

$$Y(z) = 2X(z) + 0.3z^{-1}X(z) + 0.5z^{-2}X(z) - 0.7z^{-1}Y(z) - 0.9z^{-2}Y(z)$$

$$Y(z) = [1 + 0.7z^{-1} + 0.9z^{-2}] = X(z) [2 + 0.3z^{-1} + 0.5z^{-2}]$$

$$\frac{Y(z)}{X(z)} = \frac{2 + 0.3z^{-1} + 0.5z^{-2}}{1 + 0.7z^{-1} + 0.9z^{-2}}$$

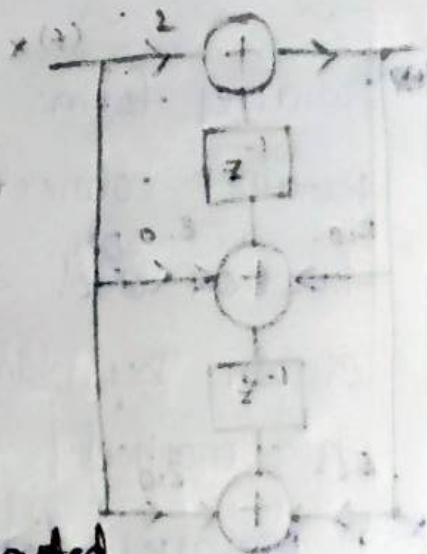
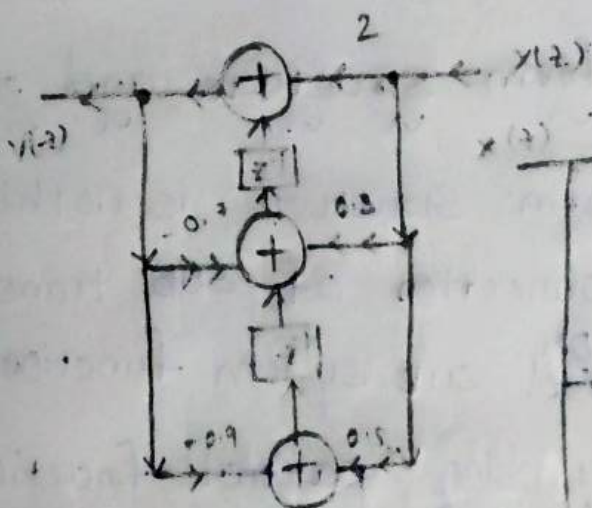
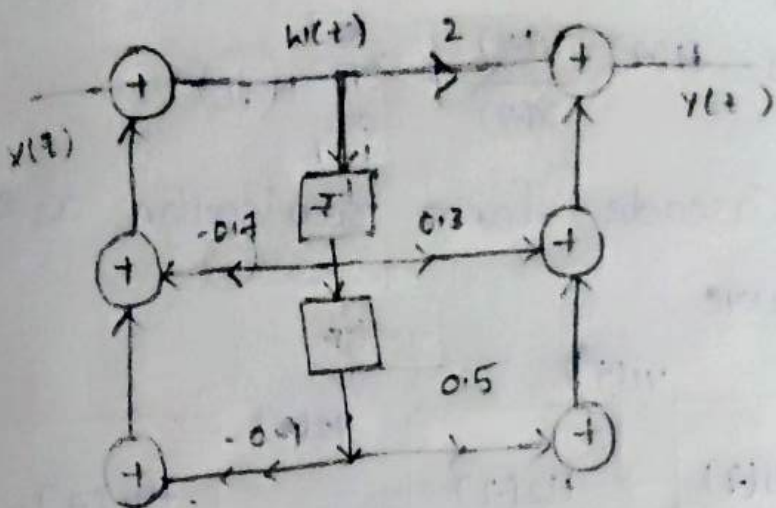
$$\frac{W(z)}{X(z)} \cdot \frac{Y(z)}{W(z)} = \frac{2 + 0.3z^{-1} + 0.5z^{-2}}{1 + 0.7z^{-1} + 0.9z^{-2}}$$

$$\frac{W(z)}{X(z)} = \frac{1}{1 + 0.7z^{-1} + 0.9z^{-2}}$$

$$W(z) = X(z) + 0.7z^{-1}W(z) + 0.9z^{-2}W(z)$$

$$\frac{Y(z)}{W(z)} = \frac{2 + 0.3z^{-1} + 0.5z^{-2}}{1}$$

$$Y(z) = 2W(z) + 0.3z^{-1}W(z) + 0.5z^{-2}W(z)$$



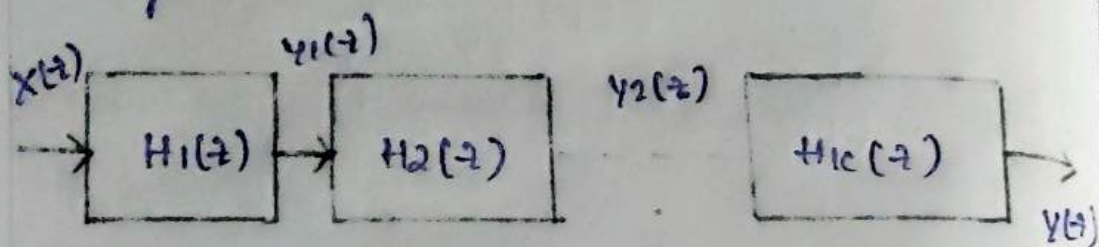
cascaded form or Cascaded
Structure

* The cascaded form structure is nothing but a cascade or series of interconnection of the sub transfer functions or sub system functions which are realised by using

either direct-form 1 or direct-form 2

* The cascade form realisation of the given transfer function $H(z)$ is expressed as $H(z) = \frac{Y(z)}{X(z)} = \prod_{i=1}^K H_i(z)$

* The cascade form realisation as shown in figure.



Parallel form structure and realisation

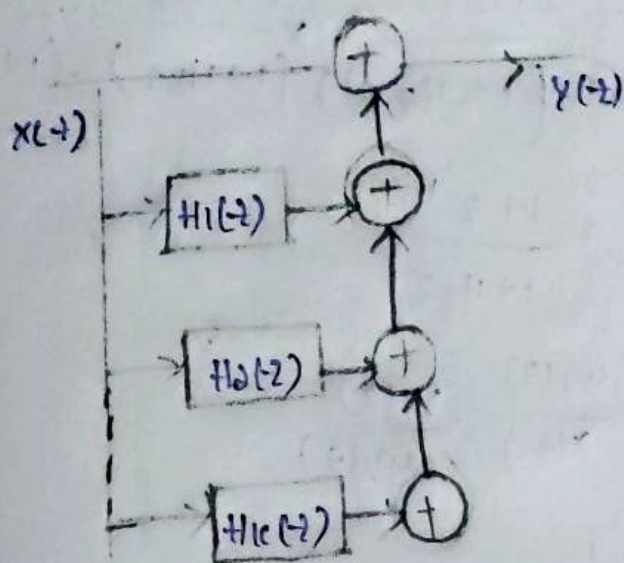
* Parallel form structure is nothing but parallel connection of sub transfer function ^{or} sub system function which are obtained by using partial fraction expansion method

* A parallel form realisation by partial fraction expansion the transfer function $H(z)$ is expressed as $H(z) =$

$$H(z) = \frac{Y(z)}{X(z)} = C + \sum_{i=1}^K H_i(z)$$

* Each of the sub transfer function can be realised by using either direct form 1 or direct form 2.

* Parallel form realisation is show in below figure.



* Obtain the cascade and parallel form realisation of LTI system is governed by the following difference equation

$$y(n) = -\frac{13}{12} y(n-1) - \frac{9}{24} y(n-2) - \frac{1}{24} y(n-3) + x(n) + 4x(n-1) + 3x(n-2)$$

$$y(z) = -\frac{13}{12} z^{-1} y(z) - \frac{9}{24} z^{-2} y(z) - \frac{1}{24} z^{-3} y(z) + x(z) + 4z^{-1} x(z) + 3z^{-2} x(z)$$

$$y(z) \left[-\frac{13}{12} z^{-1} - \frac{9}{24} z^{-2} + \frac{1}{24} z^{-3} \right] = x(z) \left[1 + 4z^{-1} + 3z^{-2} \right]$$

$$H(z) = \frac{y(z)}{x(z)} = \frac{1 + 4z^{-1} + 3z^{-2}}{1 + \frac{13}{12} z^{-1} - \frac{9}{24} z^{-2} - \frac{1}{24} z^{-3}}$$

$$= \frac{(1+z^{-1})(1+3z^{-1})}{(1+\frac{13}{12} z^{-1})(1+\frac{1}{3} z^{-1})(1+\frac{1}{4} z^{-1})}$$

Cascade form :

$$H(z) = \frac{Y(z)}{X(z)} = \frac{(1+z^{-1})}{(1+1/2z^{-1})} \times \frac{(1+3z^{-1})}{(1+1/3z^{-1})} \times \frac{1}{(1+1/4z^{-1})}$$

$$H_1(z) = \frac{Y_1(z)}{X(z)} = \frac{1+z^{-1}}{1+1/2z^{-1}}$$

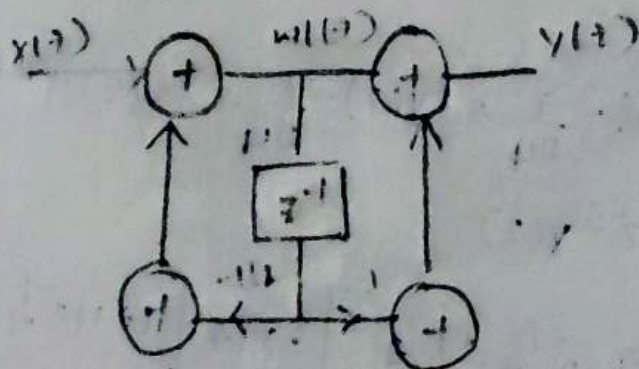
$$= \frac{W_1(z)}{X(z)} \cdot \frac{Y_1(z)}{W_1(z)}$$

$$\frac{W_1(z)}{X(z)} = \frac{1}{1+1/2z^{-1}}$$

$$W_1(z) = X(z) - 1/2z^{-1}W_1(z)$$

$$\frac{Y_1(z)}{W_1(z)} = 1+z^{-1}$$

$$Y_1(z) = W_1(z) + z^{-1}W_1(z)$$



$$H_2(z) = \frac{1+3z^{-1}}{1+1/3z^{-1}}$$

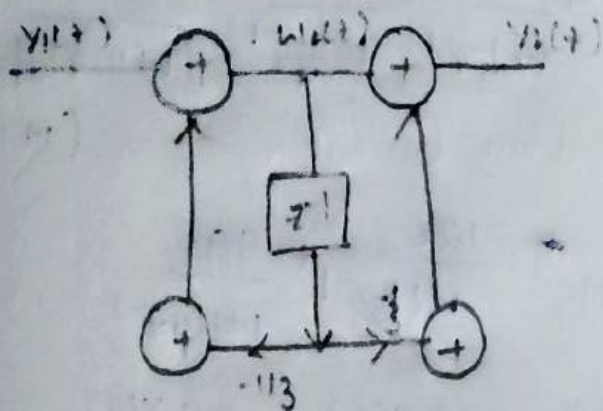
$$H_2(z) = \frac{Y_2(z)}{Y_1(z)} = \frac{W_2(z)}{Y_1(z)} \cdot \frac{Y_2(z)}{W_2(z)}$$

$$\frac{W_2(z)}{Y_1(z)} = \frac{1}{1+1/3z^{-1}}$$

$$w_2(z) = y_1(z) - \frac{1}{3}z^{-1}w_2(z)$$

$$\frac{y_2(z)}{w_2(z)} = 1 + \frac{1}{3}z^{-1}$$

$$y_2(z) = w_2(z) + \frac{1}{3}z^{-1}w_2(z)$$



$$H_3(z) = \frac{1}{1 + \frac{1}{4}z^{-1}}$$

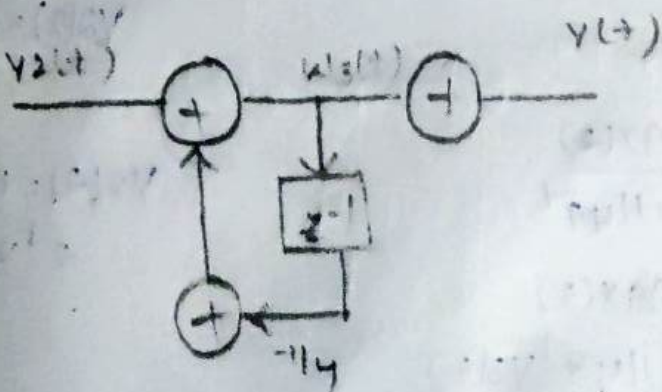
$$H_3(z) = \frac{y(z)}{y_2(z)} = \frac{w_3(z)}{y_2(z)} \cdot \frac{y_2(z)}{w_3(z)}$$

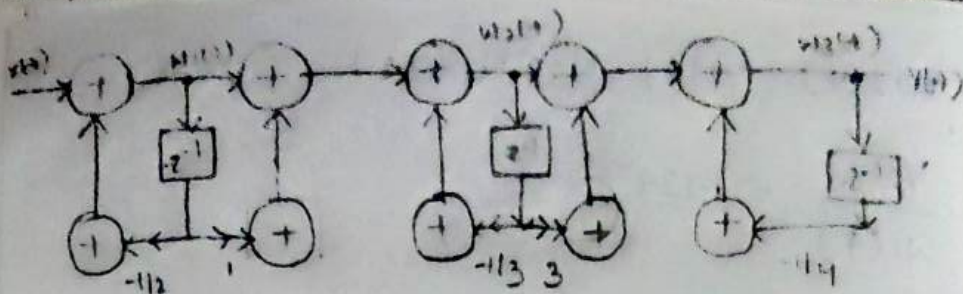
$$\frac{w_3(z)}{y_2(z)} = \frac{1}{1 + \frac{1}{4}z^{-1}}$$

$$w_3(z) = y_2(z) - \frac{1}{4}z^{-1}w_3(z)$$

$$\frac{y(z)}{w_3(z)} = 1$$

$$y(z) = w_3(z)$$





$$H(z) = \frac{A}{1 + \frac{1}{2}z^{-1}} + \frac{B}{1 + \frac{1}{3}z^{-1}} + \frac{C}{1 + \frac{1}{4}z^{-1}} = \frac{(1+z^{-1})(1+3z^{-1})}{(1+1/2z^{-1})(1+1/3z^{-1})}$$

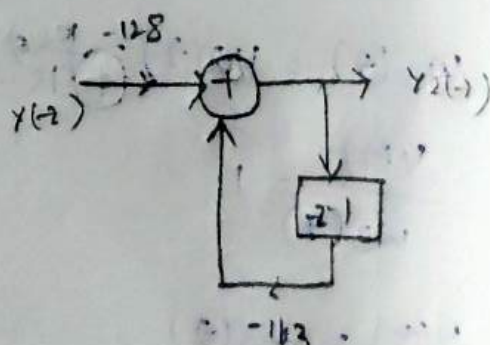
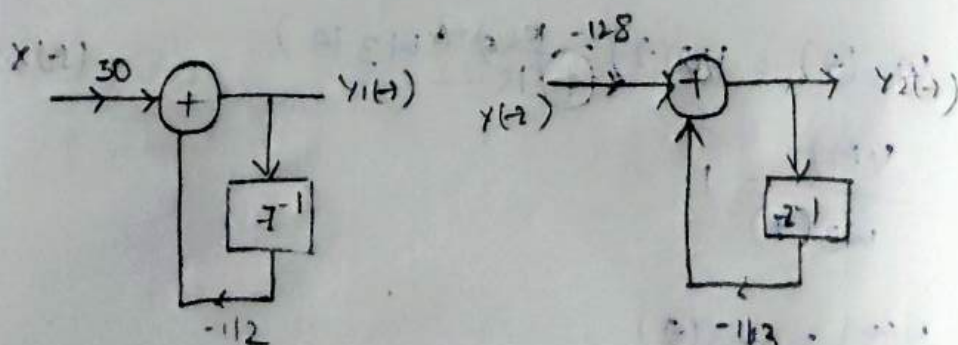
find out partial fraction $(1+1/4z^{-1})$

$$H(z) = \frac{30}{1+1/2z^{-1}} + \frac{128}{1+1/3z^{-1}} + \frac{99}{1+1/4z^{-1}} \quad \begin{matrix} A = 30 \\ B = 128 \\ C = 99 \end{matrix}$$

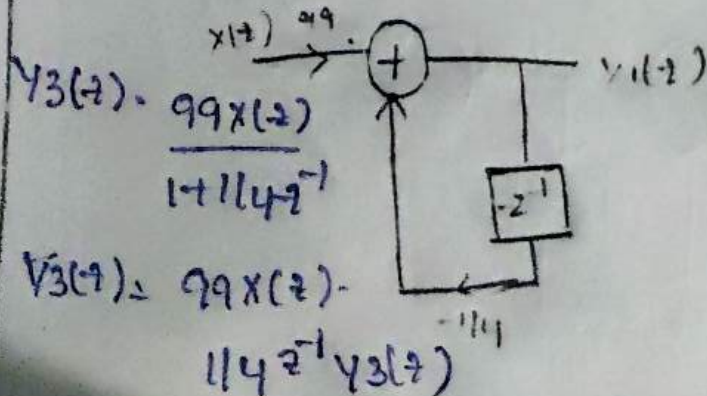
$$Y(z) = \frac{30X(z)}{1+1/2z^{-1}} - \frac{128X(z)}{1+1/3z^{-1}} + \frac{99X(z)}{1+1/4z^{-1}}$$

$$Y_1(z) = \frac{30X(z)}{1+1/2z^{-1}}$$

$$Y_1(z) = 30X(z) - \frac{1}{2}z^{-1}Y_1(z)$$



$$Y_2(z) = \frac{128X(z)}{1+1/3z^{-1}}$$



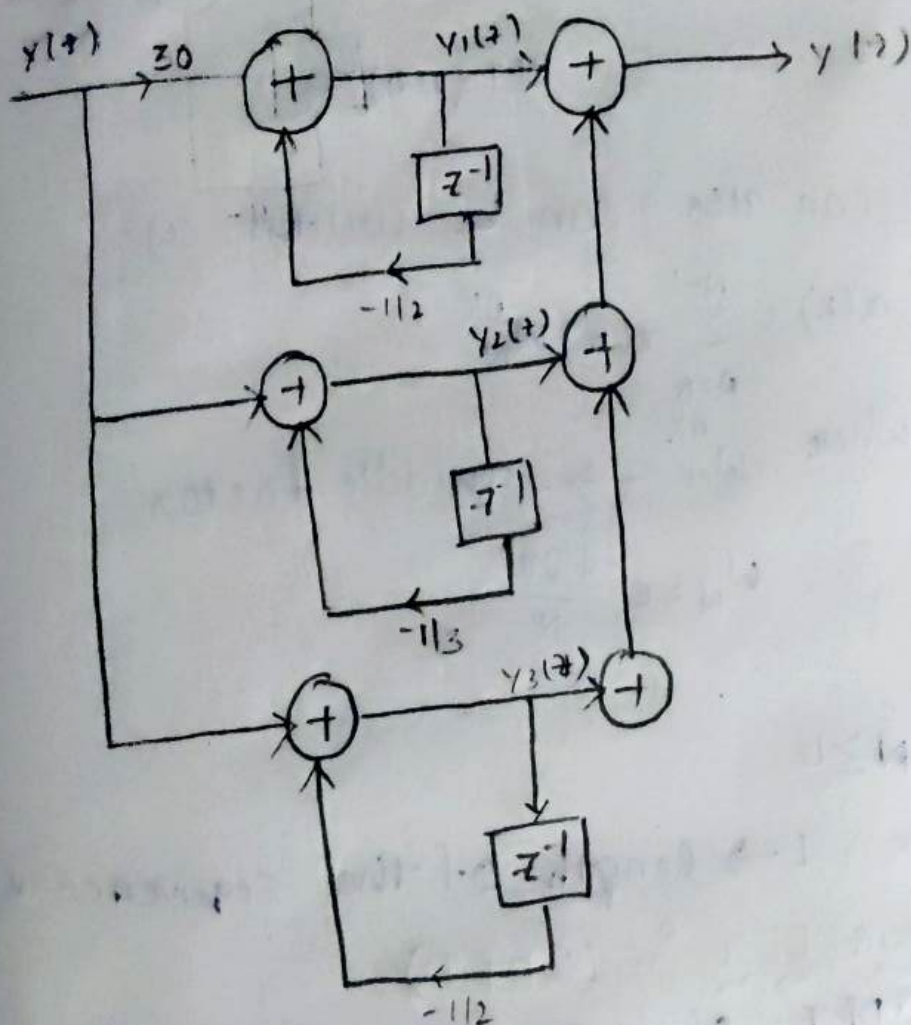
$$Y_2(z) = 128X(z) - z^{-1}Y_2(z)$$

$$Y_3(z) = 99X(z) - 1/4z^{-1}Y_3(z)$$

$$y_2(z) = \frac{128x(z)}{171132}$$

$$y_2(z) = 128x(z) - 113z^{-1}y_2(z)$$

Parallel
Parallel form :-



UNIT - II
Part-A Discrete Fourier Transform

The N-point discrete Fourier Transform (DFT) of finite duration sequence $x(n)$ of length L is defined as $X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} nk}$
 $DFT \{x(n)\} =$

This can also be written as

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk}$$

where $W_N^{nk} \rightarrow$ Twiddle factor

$$W_N = e^{-j \frac{2\pi}{N}}$$

NOTE

$$N \geq L$$

where $L \rightarrow$ length of the sequence $x(n)$

INVERSE DFT (IDFT)

The IDFT of sequence $X[k]$ of length 'N' is defined as

$$IDFT \{X(k)\} = x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi}{N} nk}$$

This can also be written as:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-nk}$$

where $W_N \rightarrow$ Twiddle factor

$$W_N = e^{-j\frac{2\pi}{N}}$$

* compute the DFT of the following finite length sequence, consider the length of sequence 'N'

(i) $x(n) = \delta(n)$ (ii) $x(n) = \delta(n-n_0)$

(iii) $x(n) = a^n$

$$\begin{aligned} \text{(i) DFT } \{x(n)\} &= \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \\ &= \sum_{n=0}^{N-1} \delta(n) e^{-j2\pi kn/N} \end{aligned}$$

$$= 1 \quad (0 \leq k \leq N-1)$$

(ii) $x(n) = \delta(n-n_0)$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

$$= \sum_{n=0}^{N-1} \delta(n-n_0) e^{-j2\pi kn/N}$$

$$= e^{-j2\pi kn_0/N} \quad (0 \leq k \leq N-1)$$

(iii) $x(n) = a^n$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

$$\sum_{n=0}^{N-1} a^n e^{-j2\pi kn/N}$$

$$= \sum_{n=0}^{N-1} (a e^{-j2\pi k/N})^n$$

formula:

$$\sum_{n=N_1}^{N_2} a^n = \frac{a^{N_1} - a^{N_2+1}}{1-a}$$

$$= \frac{(a e^{-j2\pi k/N})^0 - (a e^{-j2\pi k/N})^{N-k+1}}{1 - a e^{-j2\pi k/N}}$$

$$= \frac{1 - a e^{-j2\pi k}}{1 - a e^{-j2\pi k/N}} \quad (0 \leq k \leq N-1)$$

* find 4-point DFT of $x(n) = \{1, -1, 2, -2\}$ directly:-

* Given that

$$x(n) = \{1, -1, 2, -2\}$$

$$\text{DFT} \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

$$X(k) = \sum_{n=0}^3 x(n) e^{-j2\pi kn/4}$$

$$\begin{aligned} X(0) &= x(0) e^0 + x(1) e^0 + x(2) e^0 + x(3) e^0 \\ &= 1 - 1 + 2 - 2 = 0 \end{aligned}$$

$$\begin{aligned} X(1) &= x(0) e^0 + x(1) e^{-j2\pi/4} + x(2) e^{-j2\pi \cdot 2/4} \\ &\quad + x(3) e^{-j2\pi \cdot 3/4} \end{aligned}$$

$$= -1-j$$

$$x(2) = x(0)e^0 + x(1)e^{-j\pi} + x(2)e^{-j2\pi} + x(3)e^{-j3\pi}$$

$$= 6$$

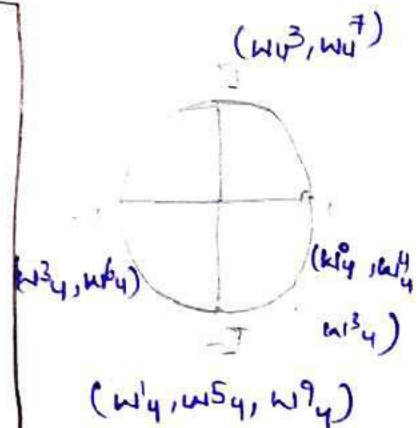
$$x(3) = x(0)e^0 + x(1)e^{-j6\pi/4} + x(2)e^{-j3\pi} + x(3)e^{-j2\pi/4}$$

$$= -1+j$$

$$x(k) = \{0, -1-j, 6, -1+j\}$$

$$W_N = W_4 = (e^{-j2\pi/4})^{nk}$$

0x0	1x0	2x0	3x0
0x1	1x1	2x1	3x1
0x2	1x2	2x2	3x2
0x3	1x3	2x3	3x3



* find 4-point DFT of the sequence

$$x(n) = \{1, -2, 3, 2\}$$

$$x(n) = \{1, -2, 3, 2\}$$

$$\text{DFT } \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

$$X(k) = \sum_{n=0}^3 x(n) e^{-j2\pi kn/4}$$

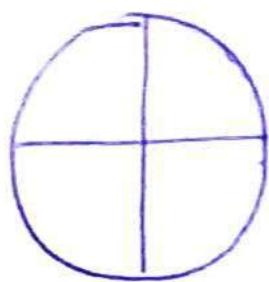
$$X(0) = x(0)e^0 + x(1)e^0 + x(2)e^0 + x(3)e^0$$

$$= 1 - 2 + 3 + 2 = 4$$

* Compute 4 point DFT of the following sequence using matrix method $x(n) = \{1, 2, -1, -2\}$

$$W^{3/4} = j = W^{7/4}$$

$$W^{6/4} = W^{2/4} = 1$$



$$W^{0/4} = 1 = W^{4/4} = W^{8/4}$$

$$W^{1/4} = -j = W^{5/4} = W^{9/4}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} \begin{bmatrix} 0 & 1 & 2 & 3 \\ W^{0/4} & W^{0/4} & W^{0/4} & W^{0/4} \\ W^{1/4} & W^{1/4} & W^{9/4} & W^{3/4} \\ W^{2/4} & W^{2/4} & W^{4/4} & W^{8/4} \\ W^{3/4} & W^{3/4} & W^{5/4} & W^{9/4} \end{bmatrix} = \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 \\ -j & j & -j & j \\ 1 & -1 & 1 & -1 \\ j & -j & j & -j \end{bmatrix}$$

$$x(0) = 1 + 2 - 1 - 2 = 0$$

$$x(1) = 1 - 2j + 1 - 2j = 2 - 4j$$

$$x(2) = 1 - 2 - 1 + 2 = 0$$

$$x(3) = 1 + 2j + 1 + 2j = 2 + 4j$$

$$x(k) = \{x(0), x(1), x(2), x(3)\}$$

$$x(k) = \{0, 2 - 4j, 0, 2 + 4j\}$$

$$y(n) = \frac{-13}{12} y(n-1) - \frac{9}{24} y(n-2) - \frac{1}{24} y(n-3)$$

$$+ x(n) + 4x(n-1) + 3x(n-2) \quad \text{Applying}$$

z.T on b.s of equation

$$y(z) = \frac{-13}{12} z^{-1} y(z) - \frac{9}{24} z^{-2} y(z) - \frac{1}{24} z^{-3} y(z)$$

$$+ x(z) + 4z^{-1} x(z) + 3z^{-2} x(z)$$

$$y(z) = \frac{-13}{12} z^{-1} y(z) - \frac{9}{24} z^{-2} y(z) - \frac{1}{24} z^{-3} y(z)$$

$$+ w(z)$$

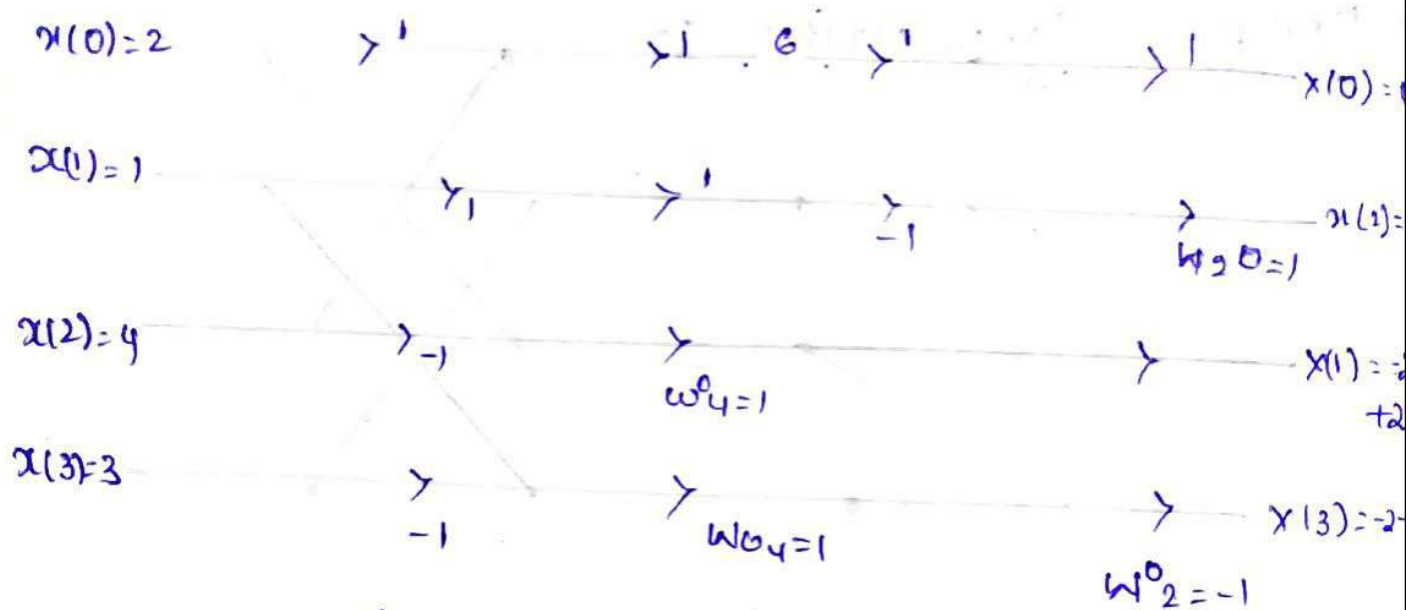
where

$$w(z) = x(z) + 4z^{-1} x(z) + 3z^{-2} x(z)$$

* Destination in frequency fast ~~Fourier~~ transform. (DIT-FFT)

* Find ^{4-point} DFT of the following sequence
by using DIT-FFT $x(n) = \{2, 1, 4, 3\}$

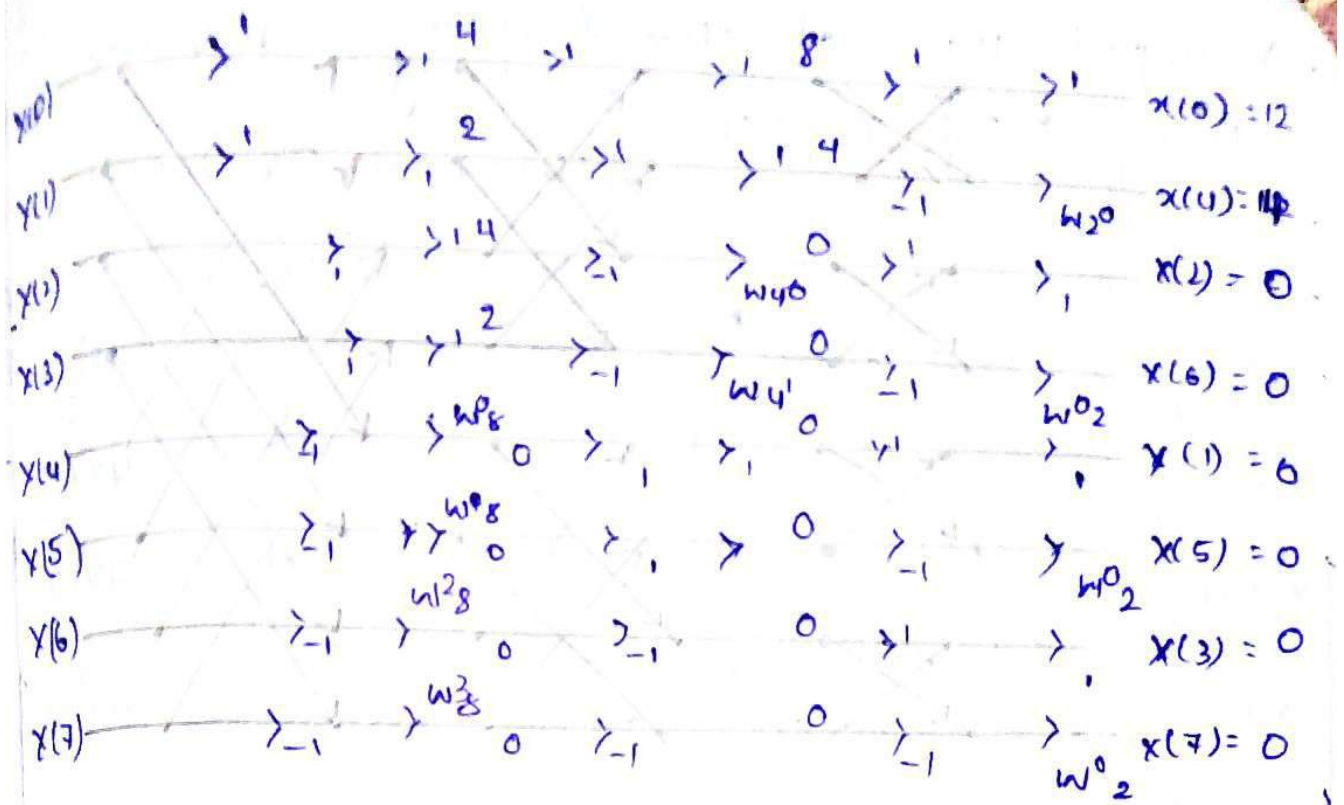
Ans:-



$$X(k) = \{10, -2+2j, 2, 2-2j\}$$

* find 8-point DFT of the following sequence by using radix 2 DIT-FFT

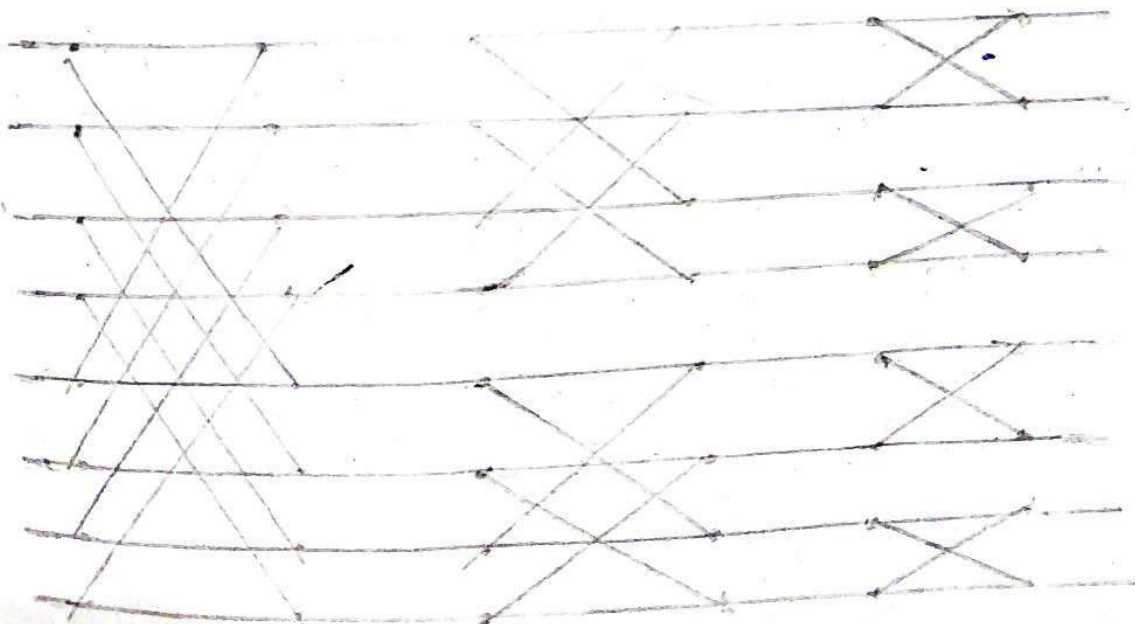
algorithm $x(n) = \{2, 1, 2, 1, 2, 1, 2, 1\}$



Find 8-point DFT of the following sequence

$$x(n) = \{2, 2, 2, 2, 1, 1, 1, 1\} \text{ by using}$$

- a) DITFFT b) DIFFFT



* Compute IDFT of the following sequence
using DIF FFT $x(k) = \{10, -2+2j, 2, -2-2j\}$



$$x^*(k) = \{10, -2-2j, 2, -2+2j\}$$

$$4x^*(n) = \{8, 4, 16, 12\}$$

$$x(n) = \{2, 1, 4, 3\}$$

HW

IIR FILTER design (digital filter)

* digital filter

Basically digital filters are linear and time invariant discrete systems

* They are 2 types

Impulse Invariant ^{response} filter (IIR)

Finite Impulse response (FIR)

frequency selective filter

* A filter is one which rejects unwanted frequency from the input signal and allows desired frequencies

* The range of frequency of the signal that are cross-through the filter is called pass band and those frequencies that are block is called stop band

* The filters are

low pass filter

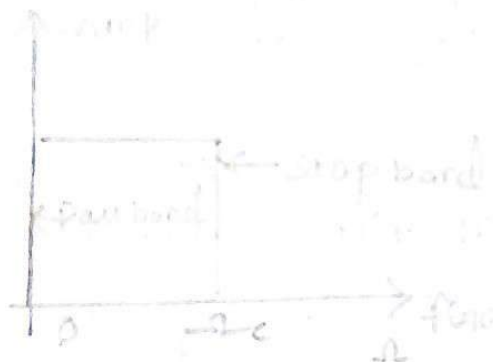
high pass filter

band pass filter

band reject filter

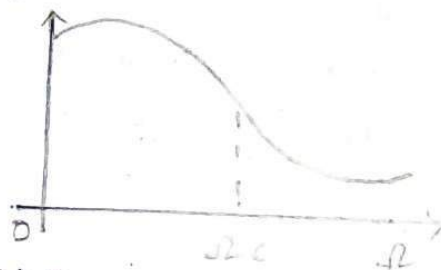
low pass filter :-

* The magnitude response of an ideal low pass filter is shown below figure.



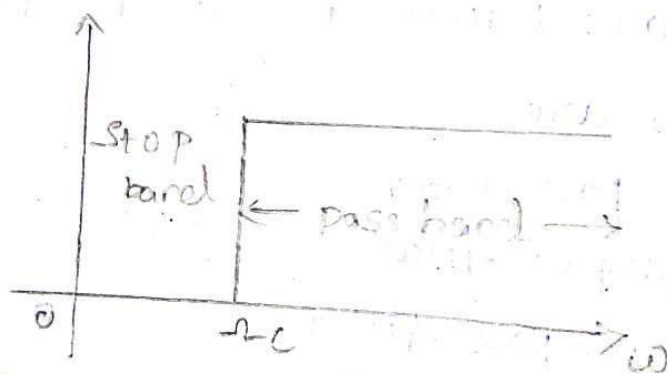
* The pass band -frequencies are $0 < \omega < \omega_c$
 Stop band -frequencies are $\omega > \omega_c$

* The practical characteristics of low pass filter is shown in below figure.



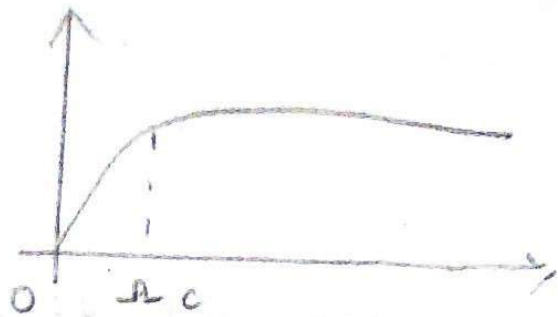
high pass filter

The ideal response of high pass filter is shown in below figure.



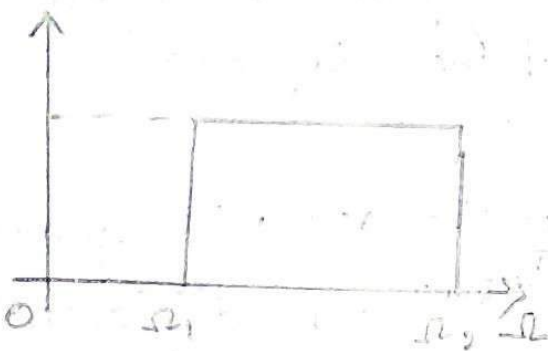
stop band frequencies are $0 < \omega < \omega_c$
 pass band frequencies are $\omega > \omega_c$

* The practical response of high pass filter is shown in below figure.



band pass filter :-

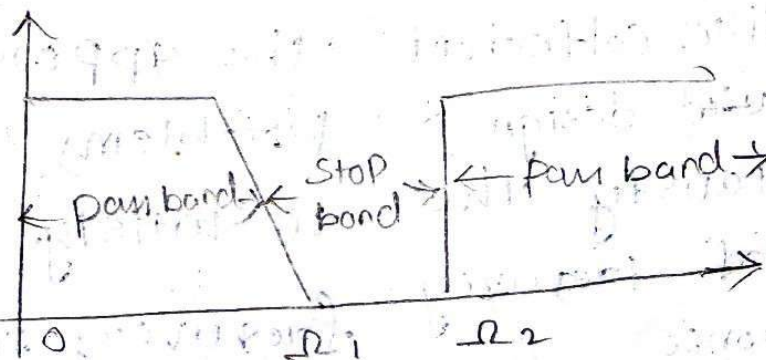
The Ideal Characteristics of band pass filter is show in below figure

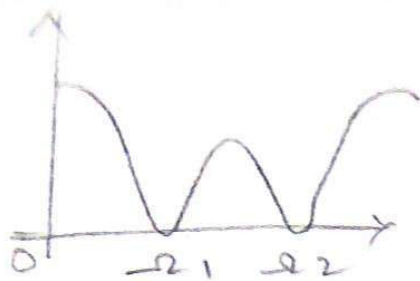


band reject filter :-

- * It rejects frequencies b/w ω_1 and ω_2 and allows all other frequencies (21)

The Ideal and practical response of band reject filter is show in below figure





difference b/w analog and digital filter

digital	analog
* It operates on digital samples of the signal	* It operates on analog signals
* It is governed by linear difference equations	* It is governed by linear differential equations
* It consists of adders, multipliers, delay units are used to implement the digital logic	* It consists of electrical elements like, resistor, capacitor, inductor.
* In a digital filter the filter coefficients are used in design to satisfy the desired frequency response	* In analog filters the approximation problem is solved to satisfy the desired frequency response

* design of IP filter by approximation of derivative :-

* In this method the digital filter system function $H(z)$ can be obtained from the analog filter transfer function $H(s)$

as follows $H(z) = H(s) \Big|_{s = \frac{1-z^{-1}}{T}}$

problem

* Convert the analog low pass filter specified by $H(s) = \frac{2}{s+3}$ a digital

filter making use of the backward difference for the derivative

as we know that mapping formula for transformation is $s = \frac{1-z^{-1}}{T}$ then

$$\text{Hence } H(z) = H(s) \Big|_{s = \frac{1-z^{-1}}{T}} = \frac{2}{\frac{1-z^{-1}}{T} + 3} = \frac{2T}{1-z^{-1} + 3}$$

for $T=1$

$$H(z) = \frac{2}{4-z^{-1}}$$

* making use of the backward difference for the derivative convert the analog filter function given below into a digital filter

$$H(s) = \frac{3}{(s+0.3)^2 + 16}$$

We know that the mapping formula is

$$s = \frac{1-z^{-1}}{T}$$

$$* H(z) = H_a(s) \bigg|_{s = \frac{1-z^{-1}}{T}} = \frac{3}{\left[\frac{1-z^{-1}}{T} + 0.3 \right]^2 + 16} = \frac{3T^2}{[1-z^{-1} + 0.3T]^2 + 16T^2}$$

for $T=1$

$$H(z) = \frac{3}{17.69 - 2.6z^{-1} + z^{-2}}$$

$$\frac{(1.3 - z^{-1})^2}{(1.3)^2 + 2^2 - 2z^{-1}} = \frac{(1.3 - z^{-1})^2}{(1.5 - z^{-1})^2}$$

design of filter by impulse invariant transformation

* The Impulse Invariant transformation for different poles as follows

$$\frac{1}{(s+p)^m} \rightarrow \frac{(-1)^{m-1}}{(m-1)!} \frac{d^{m-1}}{ds^{m-1}} \left(\frac{1}{1 - e^{-sT} z^{-1}} \right)_{s=p}$$

$$\frac{s+a}{(s+a)^2 + b^2} \rightarrow \frac{1 - e^{-aT} (\cos bT) z^{-1}}{(1 - e^{-aT} (\cos bT) z^{-1})^2 + e^{-2aT} z^{-2}}$$

$$\frac{b}{(s+a)^2 + b^2} \rightarrow \frac{e^{-at} (\sin bt) z^{-1}}{1 - ze^{-at} (\cos bt) z^{-1} + e^{-2at} z^{-2}}$$

* for analog trans for function

$$H_a(s) = \frac{2}{(s+1)(s+3)} \quad \text{determin}$$

$$H(z) \text{ if } (a) T=1S \quad (b) T=0.55 \text{ using}$$

Impulse time invariant method

$$\text{Ans:- } H_a(s) = \frac{2}{(s+1)(s+3)}$$

{simple
complex}

$$\frac{A}{s+1} + \frac{B}{s+3} \rightarrow \text{simple poles}$$

$$(a) T=1S \quad \text{and } b=1.55$$

$$H_a(s) = \frac{2}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

$$A = \frac{2}{(s+1)(s+3)} \bigg|_{s=-1} = 1 \quad B = \frac{2}{(s+1)(s+3)} \bigg|_{s=-3} = -1$$

$$H_a(s) = \frac{1}{s+1} - \frac{1}{s+3}$$

$$\frac{1}{s+1} \rightarrow \frac{1}{1 - e^{-T} z^{-1}} \quad \frac{1}{s+3} \rightarrow \frac{1}{1 - e^{-3T} z^{-1}}$$

$$H(z) = \frac{1}{1 - e^{-T} z^{-1}} - \frac{1}{1 - e^{-3T} z^{-1}}$$

$$H(z) = \frac{1}{1 - e^{-1} z^{-1}} - \frac{1}{1 - e^{-3} z^{-1}} \\ = \frac{1}{1 - 0.367 z^{-1}} - \frac{1}{1 - 0.049 z^{-1}}$$

$$H(z) = \frac{1 - 0.049z^{-1} + 0.367z^{-1}}{(1 - 0.367z^{-1})(1 - 0.049z^{-1})}$$

$$H(z) = \frac{0.318z^{-1}}{1 - 0.416z^{-1} + 0.018z^{-1}}$$

for $T = 0.5$

* Convert the analog filter $H_a(s) = \frac{s+0.1}{(s+0.1)^2 + 9}$ into a digital filter using impulse invariant transformation or method.

$$H_a(s) = \frac{s+0.1}{(s+0.1)^2 + 9} \quad a=0.1, b=3$$

$$\frac{s+a}{(s+a)^2 + b^2} \rightarrow \frac{1e^{-aT} (\cos bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}}$$

$$H(z) = \frac{1 - e^{-0.1T} (\cos 3T) z^{-1}}{1 - 2e^{-0.1T} (\cos 3T) z^{-1} + e^{-2 \cdot 0.1T} z^{-2}}$$

$$= \frac{1 - e^{-0.1} (\cos 3) z^{-1}}{1 - 2e^{-0.1} (\cos 3) z^{-1} + e^{-2 \cdot 0.1} z^{-2}}$$

$$H(z) = \frac{1 + 0.8956 z^{-1}}{1 + 1.7912 z^{-1} + 0.8187 z^{-2}}$$

* Convert analog filter with system transfer function $H_a(s) = \frac{2}{(s+0.4)^2 + 4}$ into digital filter using impulse invariant transformation.

* It is observed that the given system transfer function of the analog filter is in the standard form $H_a(s) = \frac{b}{(s+a)^2 + b^2}$
 $\therefore a=0.4, b=2$

the transformation of the above form

$$\frac{b}{(s+a)^2 + b^2} = \frac{e^{-aT} (\sin bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}}$$

$$\begin{aligned} H(z) &= \frac{e^{-0.4T} (\sin 2T) z^{-1}}{1 - 2e^{-0.4T} (\cos 2T) z^{-1} + e^{-2 \cdot 0.4T} z^{-2}} \\ &= \frac{e^{-0.4} (\sin 2) z^{-1}}{1 - 2e^{-0.4} (\cos 2) z^{-1} + e^{-2 \cdot 0.4} z^{-2}} \\ &= \frac{0.609 z^{-1}}{1 + 0.5579 z^{-1} + 0.4493 z^{-2}} \end{aligned}$$

* design of $\mathcal{B}_R$ filter by using bilinear transformation

$$* H_a(s) = \frac{b}{s+a} \Rightarrow \frac{Y(s)}{X(s)} = \frac{b}{s+a}$$

$$Y(s)s + Y(s)a = bX(s)$$

Inverse Laplace transform

$$\frac{dy(t)}{dt} + a \cdot y(t) = b x(t)$$

$$\int_{nT-T}^{nT} \frac{dy(t)}{dt} dt + a \int_{nT-T}^{nT} y(t) dt = b \int_{nT-T}^{nT} x(t) dt$$

by taking the information with respect to $nT-T$ to nT of the above equation.

$$y(nT) - y(nT-T) + \frac{aT}{2} [y(nT) + y(nT-T)] = \frac{bT}{2} [x(nT) + x(nT-T)]$$

$$\left\{ \int_{nT-T}^{nT} a(t) dt = \frac{T}{2} [a(nT) + a(nT-T)] \right\} \text{ formula}$$

$$\text{let } y(nT) = y(n) \text{ and } x(nT) = x(n)$$

$$\Rightarrow y(n) + y(n-1) + \frac{aT}{2} [y(n) + y(n-1)] = \frac{bT}{2} [x(n) + x(n-1)]$$

equating z-transform b.s. of above equation

$$\Rightarrow Y(z) - z^{-1} Y(z) + \frac{aT}{2} [Y(z) + z^{-1} Y(z)] = \frac{bT}{2} [X(z) + z^{-1} X(z)]$$

$$\Rightarrow (1 - z^{-1}) Y(z) + \frac{aT}{2} [(1 + z^{-1}) Y(z)] = \frac{bT}{2} [(1 + z^{-1}) X(z)]$$

$$\frac{Y(z)}{X(z)} = \frac{\frac{bT}{2} (1 + z^{-1})}{(1 - z^{-1}) + \frac{aT}{2} (1 + z^{-1})}$$

$$= \frac{bT/2 (1 + z^{-1}) \times \frac{z}{z} (1 + z^{-1})}{(1 - z^{-1}) \cdot \frac{z}{z} (\frac{1}{1+z^{-1}}) + \frac{aT}{2} (1 + z^{-1}) \cdot \frac{z}{z} (\frac{1}{1+z^{-1}})}$$

$$= \frac{b}{\frac{2}{T} (\frac{1 - z^{-1}}{1 + z^{-1}})}$$

Bilinear transformation
 * In this transformation mapping from

an domain to a domain

* The bilinear transformation can be obtained by using the trapezoidal formula for the numerical integration.

* Let the system function of an analog filter is $H(s) = \frac{b}{s+a}$ $\Rightarrow$ (linear sampling delay)

* by comparing $H(z)$ and $H(s)$ and

$$b = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$$

* The relation b/w analog frequency ω_a and digital frequency $\omega_d = \frac{\omega_a}{T} \tan \frac{\omega_d T}{2}$

* convert the following analog filter system function $H(s) = \frac{s+0.1}{(s+0.1)^2 + 9}$ into digital IIR filter using bilinear transformation method. digital filter having resonant frequency $\omega_r = \pi/2$

$$H(s) = \frac{s+0.1}{(s+0.1)^2 + 9}$$

$$\omega_c = \frac{2}{T} \tan \frac{\omega_r}{2} \quad \omega_c = 3$$

$$T = \frac{2}{\omega_c} \tan \frac{\omega_r}{2}$$

$$= \frac{2}{3} \tan \left(\frac{\pi}{4} \right) = 0.667$$

$$H(z) = \frac{\frac{2}{0.667} \frac{z-1}{z+1} + 0.1}{\left[\frac{2}{0.667} \frac{z-1}{z+1} + 0.1 \right]^2 + 9 (z+1)^2}$$

$$= \frac{3(z-1) + 0.1(z+1)}{(z+1)^2 + 9(z+1)^2}$$

$$H(z) = \frac{3(z-1) + 0.1(z+1)}{[3(z-1) + 0.1(z+1)]^2 + 9(z+1)^2}$$

* By applying bilinear transformation to $H(s) = \frac{4}{(s+3)(s+4)}$ with $T=0.5$ sec

$$H(z) = \frac{1/2 (1+z^{-1})^2}{(z-2)(z-1)}$$

* apply the bilinear transformation to

$$H(s) = \frac{3s}{s^2 + 0.5s + 2} \quad \text{obtain } H(z) \text{ when } t=1$$

$$H(s) = \frac{3s}{s^2 + 0.5s + 2} \quad \text{bilinear transform}$$

$$\text{relation } s \rightarrow \frac{1-z^{-1}}{1+z^{-1}}$$

$$s \rightarrow \frac{1-z^{-1}}{1+z^{-1}} \Rightarrow 2 \cdot \frac{1-z^{-1}}{1+z^{-1}}$$

$$H(z) = \frac{6 \cdot \frac{1-z^{-1}}{1+z^{-1}}}{\left(2 \cdot \frac{1-z^{-1}}{1+z^{-1}}\right)^2 + \frac{1-z^{-1}}{1+z^{-1}} + 2}$$

$$= \frac{6 \cdot \frac{1-z^{-1}}{1+z^{-1}}}{(2-2z^{-1})^2 + (1-z^{-1})(1+z^{-1}) + 2(1+z^{-1})^2}$$

$$= \frac{6 \cdot \frac{1-z^{-1}}{1+z^{-1}}}{4 + 4z^{-2} - 8z^{-1} + 1 - z^{-2} + 2 + 2z^{-2} + 4z^{-1}}$$

$$H(z) = \frac{6 \cdot \frac{1-z^{-1}}{1+z^{-1}}}{7 - 4z^{-1} + 5z^{-2}}$$

* design of Butterworth filter

Design of low pass Butterworth filter

Step 1:- derive the specifications from the given data

Ex:- Pass band frequency, stop band frequency, pass band attenuation, stop band attenuation etc...

Step 2:- find the order of the filter by using the following equations

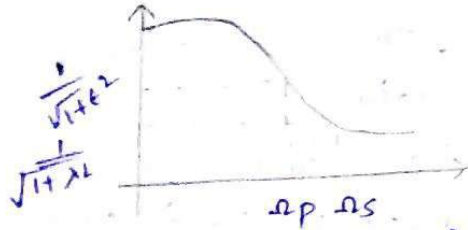
$$N \geq \frac{\log \sqrt{\frac{10^{0.55} - 1}{10^{0.15} - 1}}}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

where α_s = stop band attenuation
 α_p = pass band "
 Ω_p = passband frequency
 Ω_s = stop band frequency

if the specifications are not given directly the order of the filter is calculated by using the following equations

$$N \geq \frac{\log (1/\epsilon)}{\log (\Omega_s / \Omega_p)}$$

the λ and ϵ values are calculated by using the following diagram.



In case of bilinear transformation the Ω_s and Ω_p values are calculated as follows.

$$\Omega_s = \frac{2}{T} \tan \frac{\omega_s}{2} \quad \text{for impulse invariant transformation}$$

$$\Omega_p = \frac{2}{T} \tan \frac{\omega_p}{2} \quad \Omega_s = \frac{\omega_s}{T} \quad \Omega_p = \frac{\omega_p}{T}$$

NOTE

'N' is rounded to next highest value is the order of the filter

Example 2

Step 3

Obtain stopband filter transfer function $H(s)$ by using the following table

Order	Denominator
1	$s+1$
2	$s^2 + \sqrt{2}s + 1$
3	$(s+1)(s^2 + s + 1)$
4	$(s^2 + 0.7635s + 1)(s^2 + 1.247s + 1)$

numerator is always '1'

Step 4: The cut off frequency Ω_c is calculated as follows

i) if $\Omega_p > \Omega_s$

$$\Omega_c = \frac{\Omega_p}{\left(10^{0.1 \lambda_p} - 1\right)^{1/2N}}$$

ii) if $\Omega_p < \Omega_s$

$$\Omega_c = \frac{\Omega_s}{\left(10^{0.1 \lambda_s} - 1\right)^{1/2N}}$$

Step 5: i) for low pass filter design

$$s \rightarrow \frac{s}{\Omega_c} \quad \text{in } H(s)$$

ii) for high pass filter design

$$s \rightarrow \frac{\Omega_c}{s} \quad \text{in } H(s)$$

Step 6: Using the given transformation (bilinear or impulse invariant) the analog filter transfer function $H(s)$ is converted into the digital filter transfer function $H(z)$

Problem:-

* design a low pass butterworth filter using the bilinear transformation method for satisfying the following specifications & constraints.
pass band 0-400 Hz, stop band 2.1-4 kHz,
pass band attenuation & ripple 2db, stop band ripple 20db, sampling frequency 10 kHz ?

Step 1: derive specifications from the given data

$$\sigma_p = 2 \text{ dB}$$

$$\sigma_s = 20 \text{ dB}$$

$$f_1 = 400 \text{ Hz}$$

$$f_2 = 21 \text{ kHz}$$

$$\omega_p = 2\pi \frac{f_1}{f_s} = 2\pi \frac{400}{10000} = 0.25 \text{ rad}$$

$$\omega_s = 2\pi \frac{f_2}{f_s} = 2\pi \frac{21000}{10000} = 1.319 \text{ rad}$$

$$\sigma_p = \frac{2}{T} + \tan(\omega_p/2) = 2 \times 10000 \tan\left(\frac{0.25}{2}\right) = 2513.10$$

$$\sigma_s = \frac{2}{T} + \tan(\omega_s/2) = 2 \times 10000 \tan\left(\frac{1.319}{2}\right) = 15506.02$$

$$\frac{\sigma_s}{\sigma_p} = \frac{15506.02}{2513.102} = 6.1701$$

2) The order of the filter is evaluated by using the following formula

$$N \geq \frac{\log \sqrt{\frac{10^{\sigma_s/20} - 1}{10^{\sigma_p/20} - 1}}}{\log \left(\frac{\omega_s}{\omega_p} \right)}$$

$$N \geq \frac{\log \sqrt{\frac{10^{0.1 \times 20} - 1}{10^{0.1 \times 2} - 1}}}{\log \left(\frac{21000}{400} \right)}$$

$$\log(6.1701)$$

$$\geq 1.11$$

$$N=2$$

3) The analog transfer function $H(s)$ is

$$H(s) = \frac{1}{s^2 + 1.414s + 1}$$

4) The cutoff frequency is calculated as follows

$$\omega_c = \frac{\omega_s}{\left(\frac{10^{0.1 \times 20} - 1}{10^{0.1 \times 20} - 1} \right)^{1/2N}} = \frac{15506.02}{\left(\frac{10^{0.1 \times 20} - 1}{10^{0.1 \times 20} - 1} \right)^{1/4}} = 4915.788$$

5) For low pass filter design

$$s \rightarrow \frac{s}{\omega_c} = \frac{s}{4915.788} \text{ in } H(s)$$

$$\text{Then } H(s) = \frac{1}{\left(\frac{s}{4915.788} \right)^2 + 1.414 \left(\frac{s}{4915.788} \right) + 1}$$

$$\therefore \frac{(4915.788)^2}{s^2 + 6950.92s + (4915.788)^2}$$

$$H(s) = \frac{2.416 \times 10^7}{s^2 + 6950.92s + 2.416 \times 10^7}$$

6) By applying bilinear transformation to

$$H(s) \quad s \rightarrow \frac{2T}{1+z^{-1}} = 20000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

The digital transfer function $H(z) \approx H(z)$

$$H(z) = \frac{2.416 \times 10^{-7}}{\left[2000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 6950.92 \times 2000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$H(z) = \frac{2.416 \times 10^{-7} (1+z^{-1})^2}{4 \times 10^8 + 4 \times 10^8 z^{-2} - 8 \times 10^8 z^{-1} + 139 \times 10^6 z^{-2} + 1.39 \times 10^6 z^{-2} + 2.416 \times 10^{-7} - 2.416 \times 10^{-7} z^{-2} + 4.832 \times 10^{-7} z^{-1}}$$

$$H(z) = \frac{0.042 + 0.085z^{-1} + 0.042z^{-2}}{1 - 1.335z^{-1} + 0.506z^{-2}}$$

* design a low pass butterworth filter by using the bilinear transformation the specification of the desired low pass filter are $0.9 \leq H(\omega_p)$

$$0.9 \leq H(\omega) \leq 1 \quad 0 \leq \omega \leq \pi/2$$

$$H(\omega) \leq 0.2 \quad 3\pi/4 \leq \omega \leq \pi$$

op $\omega_p = \pi/2$ $\omega_s = 3\pi/4$

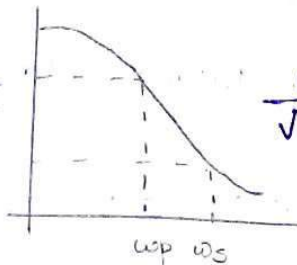
The specifications are derived from the given data is in above

The ϵ and λ values are evaluated as follows

Step 1

$$0.9 = \frac{1}{\sqrt{1+\epsilon^2}}$$

$$0.2 = \frac{1}{\sqrt{1+\lambda^2}}$$



$$\frac{1}{\sqrt{1+\epsilon^2}} = 0.9 \Rightarrow \epsilon = 0.484$$

$$\frac{1}{\sqrt{1+\lambda^2}} = 0.2 \Rightarrow \lambda = 4.899$$

Then evaluate Ω_p and Ω_s values

$$\Omega_p = \frac{2}{T} \tan(\omega_p/2) = 2 \tan(\pi/4) = 2$$

$$\Omega_s = 2/T \tan(\omega_s/2) = 2 \tan(3\pi/8) = 4.824$$

Step 2

The order of the filter is evaluated as follows

$$N \geq \frac{\log \left(\frac{10^{0.1 \lambda^2}}{10^{0.1 \epsilon^2}} \right)}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$\geq \frac{\log \left(\frac{4.899}{0.484} \right)}{\log}$$

$$N \geq 3$$

Step 3

The denominator of analog trans. function for $N=3$ is

$$(s+1)(s^2+s+1)$$

then the analog filter transfer function is $H_a(s) = \frac{1}{(s+1)(s^2+s+1)}$

step 4:

The cutoff frequency Ω_c evaluated as follows for $\omega_p < \Omega_s$

$$\Omega_c = \frac{\Omega_s}{(10^{0.1A_s} - 1)^{1/2N}} = \frac{\Omega_s}{2.11N}$$

$$\Omega_c = \frac{4.824}{(4.891)^{1/3}}$$

Answer

$$\Omega_c = 2.8403$$

step 5

for low pass filter $s \rightarrow \frac{s}{\Omega_c} = \frac{s}{2.8403}$

$$H_a(s) = \frac{1}{\left(\frac{s}{2.8403} + 1\right) \left[\left(\frac{s}{2.8403}\right)^2 + \frac{s}{2.8403} + 1\right]}$$

$$= \frac{\left(\frac{s+2.840}{2.840}\right)}{\left(\frac{s^2 + s \times 2.840 + (2.840)^2}{(2.840)^2}\right)}$$

$$= 22.9063$$

$$(s+2.840)(s^2+2.840s+8.0656)$$

step 6

For bilinear transformation the analog filter transfer function $H_a(s)$ is converted into digital filter transfer function $H(z)$ by substituting

$$s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

here $T=1$

$$s = 2 \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$22.9063$$

$H(z) =$

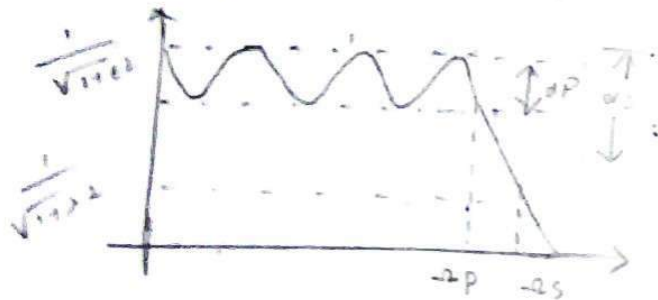
$$\frac{22.9063}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2.840 \right] \left[\left(2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right)^2 + 2.840 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 8.0656 \right]}$$

$$= \frac{22.9063 (1+z^{-1})^3}{[2-2z^{-1}+2.840+2.840z^{-1}][4+4z^{-2}-2z^{-1}+5.6-80.5680z^{-2}+8.0656+8.0656z^{-1}+56z^{-2}+16.1312z^{-1}]} \quad \text{---}$$

* design of chebyshev filter

Step 1:- derive the specifications from the given data

Step 2:- same as Butterworth filter steps in 1 and only the graph will be change



Step 3

The order of the filter is calculated by using the following equation

$$N \geq \frac{\cosh^{-1} \left(\sqrt{\frac{10^{0.1A_s} - 1}{10^{0.1A_p} - 1}} \right)}{\cosh^{-1} \left(\frac{\omega_s}{\omega_p} \right)}$$

(or)

$$\geq \frac{\cosh^{-1}(\sqrt{A_p})}{\cosh^{-1} \left(\frac{\omega_s}{\omega_p} \right)}$$

Step 3

major and minor lobe of ellipse a and b are calculated as follows

$$a = \Omega_p \left[\frac{\mu^{1/N} - \mu^{-1/N}}{2} \right]$$

$$b = \Omega_p \left[\frac{\mu^{1/N} + \mu^{-1/N}}{2} \right]$$

where $\mu = \epsilon + \sqrt{1 + \epsilon^2}$, $\epsilon = (10^{0.1A_p} - 1)^{1/2}$

Step 4

The poles of chebyshev filter on the denominator of analog transfer function is calculated as follows

$$s_k = a \cos \phi_k + j b \sin \phi_k$$

where

$$\phi_k = \frac{\pi}{2} + \left[\frac{2(k-1)}{2N} \right] \pi$$

where

$$k = 1, 2, 3, \dots, N$$

Step 5

The numerator of the transfer function is calculated as follows
Case: if 'N' is odd, put s=0 in the denominator polynomial which is

numerator

Case ii: if 'N' is even put $s=0$ in the denominator and divided by $\sqrt{1+\epsilon^2}$ which is numerator

Steps:

By using the given transformation
Convert $H_a(s)$ into $H(z)$

Problem

Design a chebyshev IIR digital low pass filter to satisfy the following constraints when $T=1$ $0.707 \leq |H(\omega)| \leq 1$

over the limits $0 \leq \omega \leq 0.2\pi$ $H(\omega) \leq 0.1$ over

the limits $0.5\pi \leq \omega \leq \pi$ Use bilinear transformation.

from problem.

$$0.707 \leq |H(\omega)| \leq 1 \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(\omega)| \leq 0.1 \quad 0.5 \leq \omega \leq \pi$$

Step 1

$$\frac{1}{\sqrt{1+\epsilon^2}} = 0.707$$

S.O.B.S

$$\frac{1}{1+\epsilon^2} = (0.707)^2$$

$$1 = (0.707)^2 (1+\epsilon^2)$$

$$1 = 0.4998 (1+\epsilon^2)$$

$$1 = 0.4998 + 0.4998 \epsilon^2$$

$$1 - 0.4998 = 0.4998 \epsilon^2$$

$$0.5002 = 0.4998 \epsilon^2$$

$$\epsilon^2 = \frac{0.5002}{0.4998}$$

$$\epsilon^2 = 1.0008$$

$$\epsilon = \sqrt{1.0008}$$

$$\epsilon = 1.0003$$

$$\epsilon = 1.0003 \approx 1$$

$$\boxed{\epsilon=1}$$

$$\frac{1}{\sqrt{1+\lambda^2}} = 0.1 \text{ (S.O.B.S)}$$

$$\frac{1}{1+\lambda^2} = 0.01$$

$$1 = 0.01 + (1+\lambda^2)$$

$$1 = 0.01 + 0.01 \lambda^2$$

$$1 - 0.01 = 0.01 \lambda^2$$

$$0.99 = 0.01 \lambda^2$$

$$\frac{0.99}{0.01} = \lambda^2$$

$$\lambda^2 = 99$$

$$\lambda = \sqrt{99}$$

$$\lambda = 9.949 \approx 9.95$$

$$\lambda = 9.95$$

$$r_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right)$$

$$r_p = 2 \tan\left(\frac{0.2\pi}{2}\right)$$

$$r_p = 0.6498$$

$$r_s = 2 \tan\left(\frac{\omega_s}{2}\right)$$

$$r_s = 2 \tan\left(\frac{0.5\pi}{2}\right)$$

$$r_s = 2$$

Step 2

$$N \geq \cosh^{-1}\left(\sqrt{\frac{\lambda}{\epsilon}}\right)$$

$$\cosh^{-1}\left(\frac{r_s}{r_p}\right)$$

$$N \geq \cosh^{-1}\sqrt{\frac{9.95}{1}}$$

$$\cosh^{-1}\sqrt{\frac{2}{0.6498}}$$

$$N \geq \frac{\cosh^{-1}(\sqrt{9.95})}{\cosh^{-1}(0.0778)} = 2$$

Step 3:

$$M = \epsilon + \sqrt{1 + \epsilon^2}$$

$$M = 1 + \sqrt{1 + 1}$$

$$M = 1 + \sqrt{2}$$

$$M = 1 + 1.414$$

$$M = 2.414$$

$$a = r_p \left[\frac{M^{\frac{1}{N}} - M^{\frac{1}{N}}}{2} \right]$$

$$a = 0.6498 \left[\frac{(2.414)^{\frac{1}{2}} - (2.414)^{\frac{1}{2}}}{2} \right]$$

$$a = 0.6498 \left[\frac{1.5337 - 0.6436}{2} \right]$$

$$a = 0.6498 \left[\frac{0.9101}{2} \right]$$

$$a = 0.6498 (0.45505)$$

$$a = 0.295$$

$$b = r_p \left[\frac{M^{\frac{1}{N}} + M^{\frac{1}{N}}}{2} \right]$$

$$b = 0.6498 \left[\frac{(2.414)^{\frac{1}{2}} + (2.414)^{\frac{1}{2}}}{2} \right]$$

$$b = 0.713$$

Step 4:

$$\phi_k = \frac{\pi}{2} + \left(\frac{2k-1}{2N} \right) \pi$$

$$\phi_1 = \frac{\pi}{2} + \left(\frac{2-1}{4} \right) \pi$$

$$\phi_1 = \frac{\pi}{2} + \frac{\pi}{4} = 90^\circ + 45^\circ = 135^\circ$$

$$\phi_0 = \frac{\pi}{2} + \left(\frac{4-1}{4}\right)\pi = \frac{\pi}{2} + \frac{3\pi}{4} = 90^\circ + 135^\circ = 225^\circ$$

$$s_k = a \cos \phi_k + j b \sin \phi_k$$

$$s_1 = a \cos \phi_1 + j b \sin \phi_1$$

$$s_1 = a \cos (135^\circ) + j b \sin (135^\circ)$$

$$s_1 = (0.295) (-0.707) + j (0.713) (0.707)$$

$$s_1 = -0.208 + j 0.504$$

$$s_2 = a \cos \phi_2 + j b \sin \phi_2$$

$$s_2 = (0.295) \cos(225^\circ) + j (0.713) \sin(225^\circ)$$

$$s_2 = (0.295) (-0.707) + j (0.713) (-0.707)$$

$$s_2 = -0.208 - j 0.504$$

$$s_2 = -0.208 - j 0.504$$

The denominator is

$$(s + 0.208 + j 0.504)(s + 0.208 - j 0.504)$$

$$(s + 0.208)^2 - (j 0.504)^2$$

$$s^2 + 0.4166 + 0.043 + 0.254$$

The denominator is

$$s^2 + 0.4166 + 0.297$$

Steps:-

The numerator in case (a) is obtained by replacing $s=0$ in denominator and divide by $\sqrt{1+e^2}$

$$\therefore \text{Numerator} = \frac{0.297}{\sqrt{1+e^2}} = \frac{0.297}{\sqrt{2}} = 0.21$$

The analog transfer function is

$$H(s) = \frac{0.21}{s^2 + 0.4166 + 0.297}$$

for bilinear transformation

$$s \rightarrow \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] \text{ in } H(s)$$

$$s \rightarrow 2 \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$H(z) = \frac{0.21}{\left[2 \left[\frac{1-z^{-1}}{1+z^{-1}} \right]^2 + 0.416 \left[2 \left[\frac{1-z^{-1}}{1+z^{-1}} \right] \right] + 0.297} \right]}$$

$$H(z) = \frac{0.21}{4 \left[\frac{1-z^{-1}}{1+z^{-1}} \right]^2 + 0.832 \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 0.297}$$

$$H(z) = \frac{0.21 (1+z^{-1})^2}{4 (1-z^{-1})^2 + 0.832 (1-z^{-2}) + 0.297 (1+z^{-2})}$$

IIR Digital Filters: Analog Filter Approximations- Butterworth and Chebyshev, Design of IIR Digital Filters from Analog Filters, Bilinear Transformation Method.

Introduction:-

- * Filters are of two types- FIR and IIR.
- * The type of filters which make use of feedback connection to get the desired filter implementation are known as recursive filters. Their impulse response is of infinite duration. So they are called IIR filters.
- * IIR filters are designed by considering all the infinite samples of the impulse response.
- * There are several techniques available for the design of digital filters having an infinite duration unit impulse response.
- * The popular methods for such filter design uses the technique of first designing the digital filter in analog domain and then transforming the analog filter into an equivalent digital filter because the analog filter design techniques are well developed.
- * IIR filters are of recursive type where the present output sample depends on present input, past input and past output.

IIR filter Properties:-

The impulse response $h(n)$ for a realizable filter is

$$h(n) = 0 \quad \text{for } n < 0$$

For stability it must satisfy the condition

$$\sum_{n=0}^{\infty} |h(n)| < \infty$$

IIR digital filter have Transfer function of the form

$$H(z) = \sum_{n=0}^{\infty} h(n) z^{-n} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

Frequency Selective Filters:-

A filter is one which rejects unwanted frequencies from the input signal and allow the desired frequencies. The range of frequencies of signal that are passed through the filter is called passband, and those frequencies that are blocked is called stopband.

The filters are

1. Lowpass filter (LPF)
2. Highpass filter (HPF)
3. Bandpass filter (BPF)
4. Bandreject filter (BRF)

1. Lowpass Filter:-

The magnitude response of an ideal low pass filter allow low frequencies in the passband $0 < \omega < \omega_c$ to pass, whereas the higher frequencies in the stopband $\omega > \omega_c$ are blocked. The frequency ω_c between the two bands is the cutoff frequency, where the magnitude $|H(j\omega)| = \frac{1}{\sqrt{2}}$

In practice it is impossible to obtain the ideal response. The practical response of a lowpass filter is shown in solid line in figure 3.1 (a)

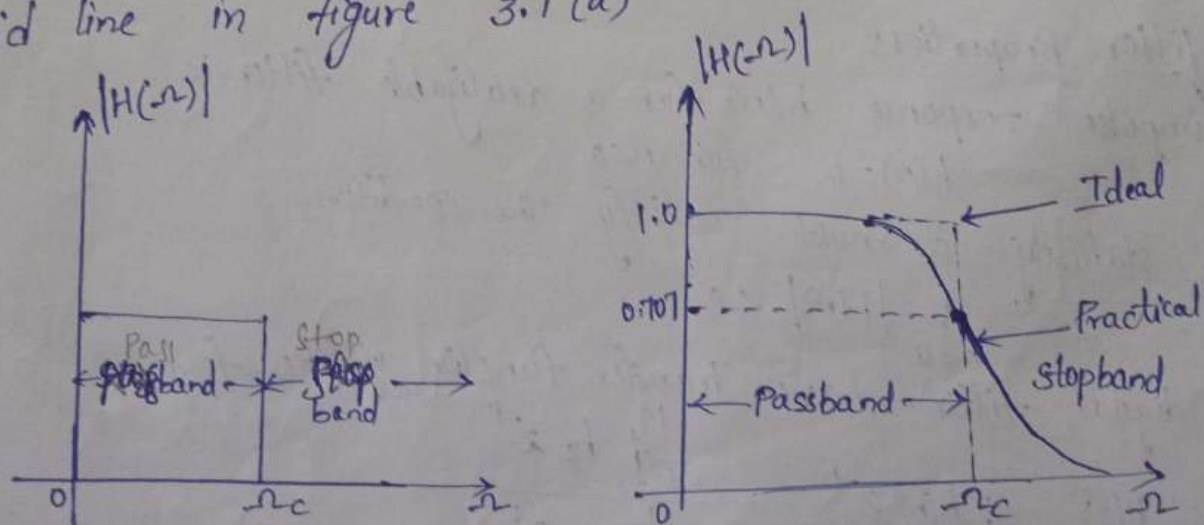


Fig 3.1 Magnitude response of filters Lowpass filter

2. Highpass Filter:-

The highpass filter allows high frequencies above $\omega > \omega_c$ and rejects the frequencies between $\omega = 0$ and $\omega = \omega_c$. The magnitude response of an ideal and practical highpass filter is shown in figure 3.2.

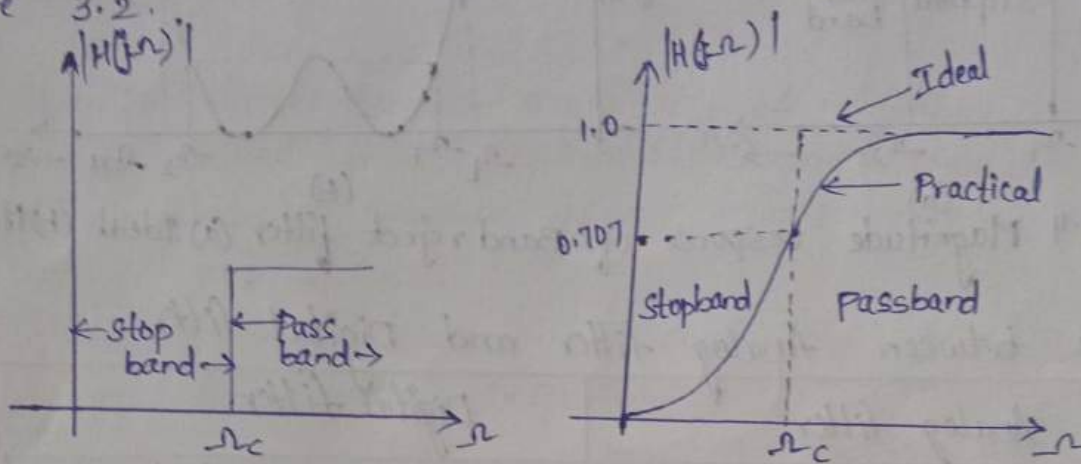


Figure 3.2: Magnitude response of Highpass Filter.

3. Bandpass Filter:-

It allows only a band of frequencies ω_1 to ω_2 to pass and stop all other frequencies. The ideal and practical response of bandpass filter are shown in figure 3.3.

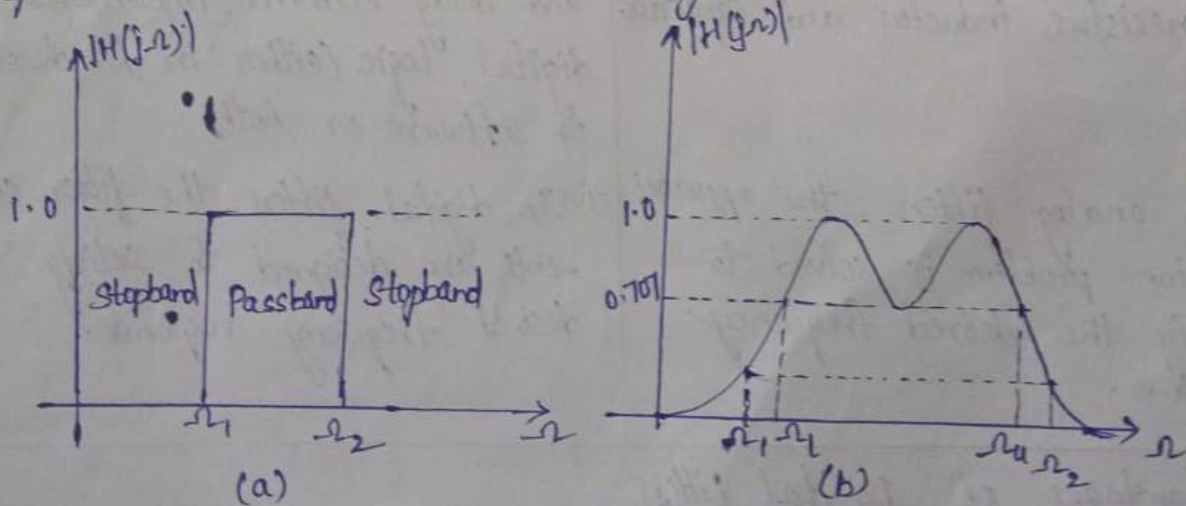


Figure 3.3: Magnitude response of Bandpass filter (a) Ideal (b) Practical

4. Band Reject Filter:-

It rejects all the frequencies between ω_1 and ω_2 and allows remaining frequencies. The magnitude response of an ideal and practical filters is shown in fig. 3.4.

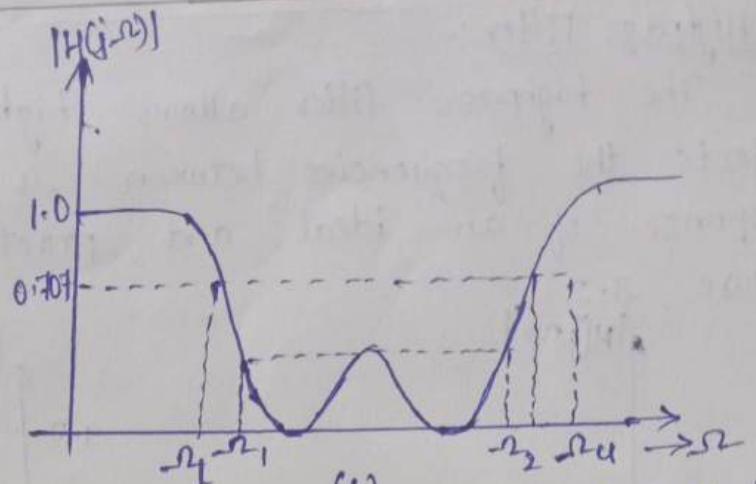
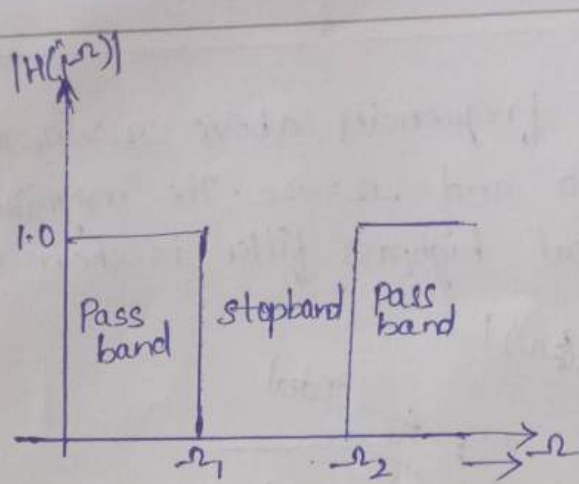


Figure: 3.4 Magnitude response of Band reject filter (a) Ideal (b) Practical

Difference between Analog filter and Digital filter

- Analog filter**
- (i) It operates on analog signals (or actual signals)
 - (ii) It is governed by the linear differential equation.
 - (iii) It consists of electrical components like resistors, inductors and capacitors.
 - (iv) In analog filters, the approximation problem is solved to satisfy the desired frequency response.

- Digital filter**
- (i) It operates on digital samples (or sampled version) of the signal.
 - (ii) It is governed by the linear difference equation.
 - (iii) It consists of adders, multipliers and delay elements implemented in digital logic (either in hardware or software or both).
 - (iv) In digital filters, the filter coefficients are designed to satisfy the desired frequency response.

Advantages of Digital Filters

1. The values of resistors, capacitors and inductors used in analog filters change with temperature. Since the digital filters do not have these components, they have high thermal stability.

2. In digital filters, the precision of the filter depends on the length (or size) of the registers used to store the filter coefficients. Hence by increasing the register bit length (in hardware) the performance characteristics of the filter like accuracy, dynamic range, stability and frequency response tolerance, can be enhanced.
- 3) The digital filters are programmable. Hence the filter coefficients can be changed any time to implement adaptive features.
- 4) A single filter can be used to process multiple signals by using the techniques of multiplexing.

Disadvantages of digital filters:

1. The bandwidth of the discrete signal is limited by the sampling frequency. The bandwidth of real discrete signal is half the sampling frequency.
- 2) The performance of the digital filter depends on the hardware (i.e., depends on the bit length of the registers in the hardware) used to implement the filter.

Important features of IIR filters:

1. The physically realizable IIR filters do not have linear phase.
2. The IIR filter specifications include the desired characteristics for the magnitude response only.

Analog Low pass Filter Design:

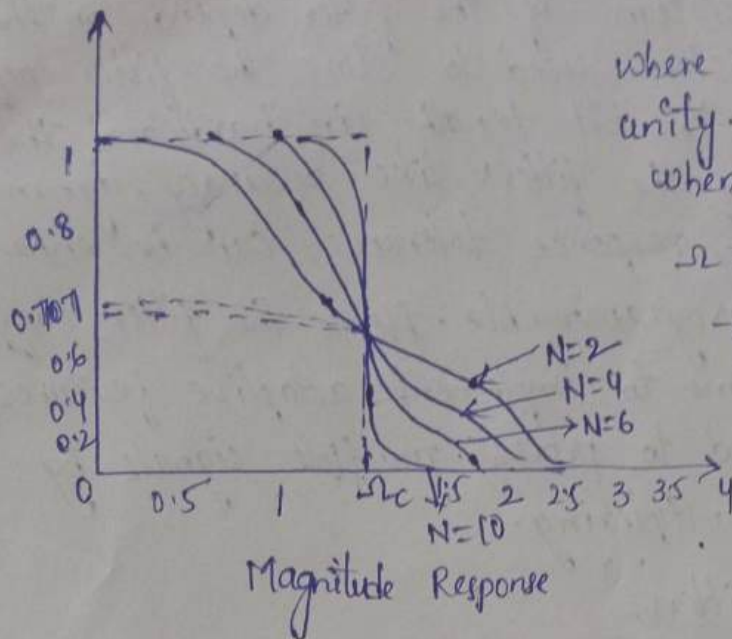
There are two types of analog filter design. They are
(i) Butterworth filter (ii) Chebyshev filter

Analog lowpass Butterworth filter:

Magnitude function of Butterworth lowpass filter is

$$|H(j\omega)| = \frac{1}{\left[1 + \left(\frac{\omega}{\omega_c}\right)^{2N}\right]^{1/2}} \quad N=1,2,3 \dots \rightarrow \textcircled{1}$$

where N is order of filter, ω_c is cutoff frequency.



where $\omega < \omega_c$ magnitude response is unity.
 when $\omega > \omega_c$ mag. response decrease
 $\omega = \omega_c$ response passes through -3dB point i.e., curve passes through 0.707.

when $\omega_c = 1$ radian/sec then

$$|H(j\omega)| = \frac{1}{[1 + (\omega)^{2N}]^{1/2}} \quad \text{--- (2)}$$

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega)^{2N}} \quad \text{--- (3)}$$

For the transfer function of a stable filter substitute $\omega = \frac{s}{j}$

$$\text{i.e., } |H(j\omega)|^2 = \frac{1}{1 + \left(\frac{s}{j}\right)^{2N}} = \frac{1}{1 + (-s^2)^N} \quad [j^2 = -1]$$

$$|H(j\omega)|^2 = H(j\omega)H(-j\omega) = \frac{1}{1 + (-s^2)^N}$$

From the above equation, the roots are obtained by equating denominator part to 0.

$$\text{For } N \text{ odd } 1 - s^{2N} = 0 \Rightarrow s^{2N} = 1 = e^{j2\pi k}$$

$$s_k = e^{j\frac{\pi k}{N}} \quad k = 1, 2, \dots, 2N$$

$$\text{For } N \text{ even } 1 + s^{2N} = 0 \Rightarrow s^{2N} = -1$$

$$\Rightarrow s^{2N} = e^{j(2k-1)\pi}$$

$$s_k = e^{j\frac{(2k-1)\pi}{2N}} \quad \text{for } k = 1, 2, \dots, 2N$$

$$\text{For } N=3 \quad s_k = e^{j\frac{\pi k}{3}}$$

$$s_6 = 1$$

$$s_1 = e^{j\frac{\pi}{3}} = \cos\frac{\pi}{3} + j\sin\frac{\pi}{3} = 0.5 + j0.866$$

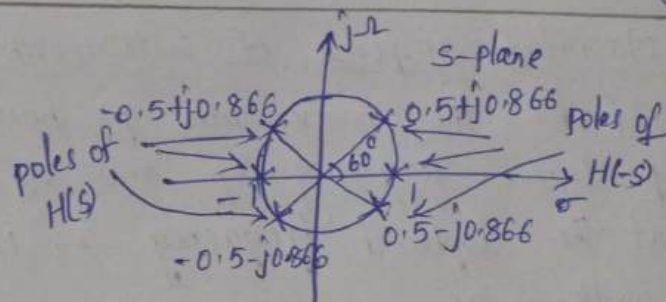
$$s_2 = e^{j\frac{2\pi}{3}} = \cos\frac{2\pi}{3} + j\sin\frac{2\pi}{3} = -0.5 + j0.866$$

$$s_3 = e^{j\pi} = \cos \pi + j \sin \pi = -1$$

$$s_4 = e^{j\frac{4\pi}{3}} = \cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3} = -0.5 - j0.866$$

$$s_5 = e^{j\frac{5\pi}{3}} = \cos \frac{5\pi}{3} + j \sin \frac{5\pi}{3} = 0.5 - j0.866$$

$$s_6 = e^{j2\pi} = \cos 2\pi + j \sin 2\pi = 1$$



The angular separation between the poles is given by $\frac{360^\circ}{2N} = \frac{360}{2 \times 3} = 60^\circ$
 All the poles are lie on unit circle.

Consider the poles at the left hand side of s-plane the denominator of the transfer function $H(s)$ is

$$(s+1) \{ (s+0.5)^2 + (0.866)^2 \} = (s+1) \{ (s^2 + 0.5s + 2 \times 0.5 \times 0.866 + 0.866^2) \}$$

$$= (s+1)(s^2 + s + 1)$$

$\therefore$ The transfer function of 3rd order butterworth filter, for cutoff frequency $\omega_c = 1$ rad/sec is

$$H(s) = \frac{1}{(s+1)(s^2 + s + 1)}$$

for N is even i.e., $N=4$

$$s_k = e^{j\phi_k} \text{ where } \phi_k = \frac{\pi}{2} + \frac{(2k-1)\pi}{2N} \quad k=1, 2, 3, \dots, N$$

for $N=4$

$$s_1 = e^{j\phi_1} = e^{j(\frac{\pi}{2} + \frac{\pi}{8})} = e^{j\frac{5\pi}{8}} = -0.3827 + j0.9239$$

$$s_2 = e^{j\frac{7\pi}{8}} = -0.9239 + j0.3827$$

$$s_3 = e^{j\frac{9\pi}{8}} = -0.9239 - j0.3827$$

$$s_4 = e^{j\frac{11\pi}{8}} = -0.3827 - j0.9239$$

The denominator of the transfer function $H(s)$ is

$$\{ (s+0.3827)^2 + (0.9239)^2 \} \{ (s+0.9239)^2 + (0.3827)^2 \} = \frac{(s^2 + 1.847s + 1)}{(s^2 + 0.765s + 1)}$$

$\therefore$ Fourth order butterworth filter transfer function for $\omega_c = 1$ rad/sec

$$H(s) = \frac{1}{(s^2 + 1.847s + 1)(s^2 + 0.765s + 1)}$$

Magnitude response of Butterworth filter:-

Consider the maximum passband attenuation in +ve dB is α_p at passband frequency ω_p and α_s is the minimum stopband attenuation at the stopband frequency ω_s . Now the magnitude function can be written as

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2N}}}$$

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2N}}$$

Apply log on both sides

$$20 \log |H(j\omega)| = 10 \log 1 - 10 \log \left(1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2N}\right) \rightarrow (1)$$

At $\omega = \omega_p$

from the figure attenuation is $-\alpha_p$

$$\therefore 20 \log |H(j\omega_p)| = -\alpha_p$$

$$-\alpha_p = -10 \log(1 + \epsilon^2)$$

$$\alpha_p = 10 \log(1 + \epsilon^2)$$

Taking antilog on both sides

$$1 + \epsilon^2 = 10^{0.1\alpha_p} \Rightarrow \epsilon = (10^{0.1\alpha_p} - 1)^{1/2}$$

At $\omega = \omega_s$ equate eq(1) to $-\alpha_s$

$$-\alpha_s = 20 \log |H(j\omega_s)| = 10 \log 1 - 10 \log \left(1 + \epsilon^2 \left(\frac{\omega_s}{\omega_p}\right)^{2N}\right)$$

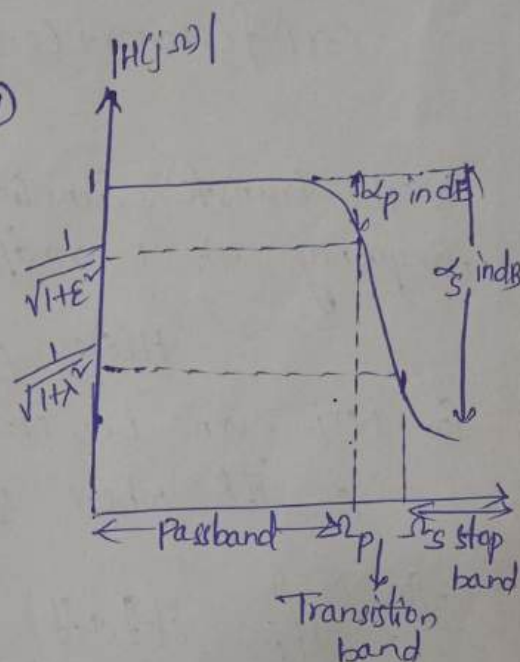
$$\alpha_s = 10 \log \left(1 + \epsilon^2 \left(\frac{\omega_s}{\omega_p}\right)^{2N}\right)$$

$$0.1\alpha_s = \log \left(1 + \epsilon^2 \left(\frac{\omega_s}{\omega_p}\right)^{2N}\right)$$

$$\left(\frac{\omega_s}{\omega_p}\right)^{2N} = \frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1}$$

Taking log on both sides

$$\log \left(\frac{\omega_s}{\omega_p}\right)^{2N} = \log \left[\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1} \right]$$



$$N = \frac{\log \sqrt{\frac{10^{0.1\alpha_S} - 1}{10^{0.1\alpha_P} - 1}}}{\log\left(\frac{\omega_S}{\omega_P}\right)}$$

To get the integer value for N , roundoff N to next higher integer.

$$N \geq \frac{\log \sqrt{\frac{10^{0.1\alpha_S} - 1}{10^{0.1\alpha_P} - 1}}}{\log\left(\frac{\omega_S}{\omega_P}\right)}$$

$$N \geq \frac{\log(\gamma_E)}{\log\left(\frac{\omega_S}{\omega_P}\right)}$$

$$\text{where } \lambda = \sqrt{10^{0.1\alpha_S} - 1}$$

$$E = \sqrt{10^{0.1\alpha_P} - 1}$$

Let $A = \gamma_E$, $k = \frac{\omega_P}{\omega_S}$ where k is transition ratio.

$$\therefore N \geq \frac{\log A}{\log(1/k)} \quad \text{where } N \text{ is the order.}$$

P. Given specifications $\alpha_P = 1\text{dB}$, $\alpha_S = 30\text{dB}$, $\omega_P = 200\text{rad/sec}$, $\omega_S = 600\text{rad/sec}$.
Determine the order of the filter.

Sol. Given $\alpha_P = 1\text{dB}$, $\alpha_S = 30\text{dB}$, $\omega_P = 200\text{rad/sec}$, $\omega_S = 600\text{rad/sec}$

$$N = \frac{\log A}{\log(1/k)} \quad A = \gamma_E = \sqrt{\frac{10^{0.1\alpha_S} - 1}{10^{0.1\alpha_P} - 1}} = \sqrt{\frac{10^{0.1 \times 30} - 1}{10^{0.1 \times 1} - 1}}$$

$$A = 62.115$$

$$k = \frac{\omega_P}{\omega_S} = \frac{200}{600} = \frac{1}{3}$$

$$N \geq \frac{\log(62.115)}{\log 3}$$

$$N \geq 3.758$$

Where ' N ' is roundoff to next higher Integer value.

$$\boxed{N = 4}$$

Relationship between ω_c , ω_p & ω_s :

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}} \rightarrow (1)$$

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2N}} \rightarrow (2)$$

Equate (1) & (2)

$$\frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}} = \frac{1}{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2N}}$$

$$1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2N} = 1 + \left(\frac{\omega}{\omega_c}\right)^{2N}$$

$$\left(\frac{\omega}{\omega_c}\right)^{2N} \left[\frac{\epsilon^2}{\left(\frac{\omega}{\omega_p}\right)^{2N}} - 1 \right] = 0$$

$$\text{where } \epsilon = (10^{0.1A_p - 1})^{1/2}$$

$$\frac{\omega_p}{\omega_c} = (10^{0.1A_p - 1})^{1/2}$$

$$\frac{\epsilon^2}{\left(\frac{\omega}{\omega_p}\right)^{2N}} = \frac{1}{\left(\frac{\omega}{\omega_c}\right)^{2N}}$$

$$\left(\frac{\omega_c}{\omega_p}\right)^{2N} \epsilon^2 = \left(\frac{\omega}{\omega_p}\right)^{2N} \Rightarrow \left(\frac{\omega_c}{\omega_p}\right)^{2N} = \frac{\left(\frac{\omega}{\omega_p}\right)^{2N}}{\epsilon^2} \Rightarrow \frac{1}{\epsilon^2} = \left(\frac{\omega_c}{\omega_p}\right)^{2N}$$

$$\epsilon^2 = \left(\frac{\omega_p}{\omega_c}\right)^{2N} = 10^{0.1A_p - 1}$$

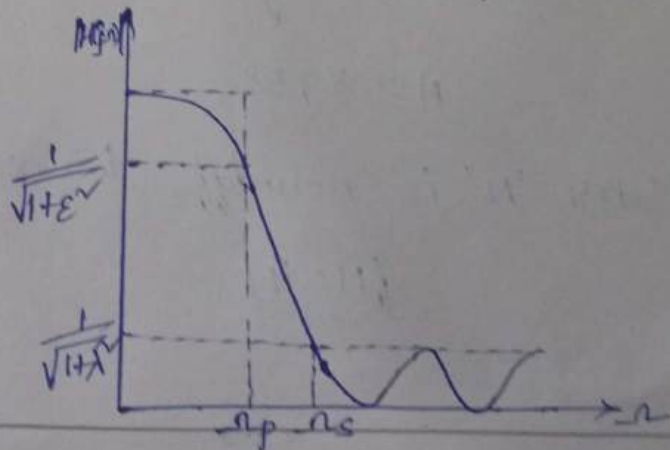
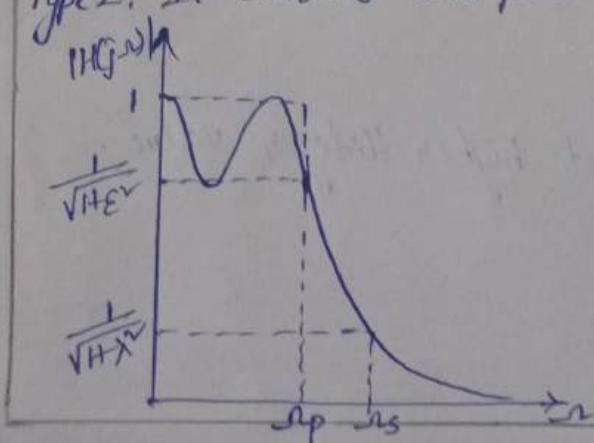
$$\omega_c = \frac{\omega_p}{(10^{0.1A_p - 1})^{1/2N}}, \quad \omega_c = \frac{\omega_s}{(10^{0.1A_s - 1})^{1/2N}}$$

Analog Lowpass Chebyshev filter:

There are two types of Chebyshev filters.

Type 1: It contains only poles so it is called all pole filter.

Type 2: It contains both poles and zeros. So it is called pole-zero filter.



Magnitude response of N^{th} order Type I filter

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2\left(\frac{\omega}{\omega_p}\right)} \quad N=1, 2, 3, \dots$$

ϵ is parameter of the filter related to the ripple in the passband

$C_N(x)$ is N^{th} order Chebyshev polynomial defined as

$$C_N(x) = \cos(N \cos^{-1} x) \quad |x| \leq 1 \quad (\text{passband})$$

$$C_N(x) = \cosh(N \cosh^{-1} x) \quad |x| \geq 1 \quad (\text{stopband})$$

$C_N(x)$ has the following properties

$$C_N(x) = -C_N(-x) \text{ for } N \text{ odd, } C_N(1) = 1 \text{ for all } N$$

$$C_N(x) = C_N(-x) \text{ for } N \text{ even, } C_N(-1) = 1 \text{ for } N \text{ even}$$

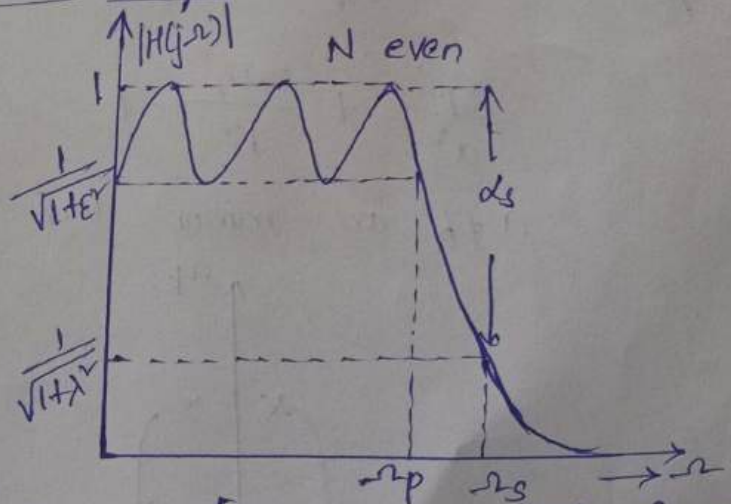
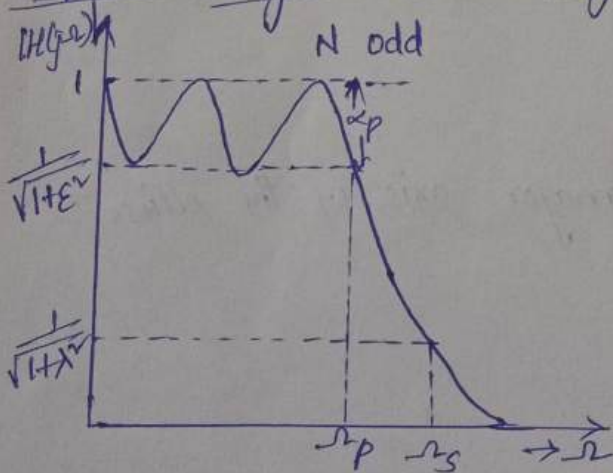
$$C_N(0) = (-1)^{N/2} \text{ for } N \text{ even, } C_N(-1) = 1 \text{ for } N \text{ odd}$$

$$C_N(0) = 0 \text{ for } N \text{ odd}$$

$C_N(x)$ oscillates with equal ripple between ± 1 for $|x| \leq 1$

$C_N(x)$ is monotonically increasing for $|x| > 1 \quad \forall N$

Lowpass Chebyshev filter magnitude response:-



For N odd oscillatory curve starts from unity

For N even it starts from $\frac{1}{\sqrt{1+\epsilon^2}}$

Taking logarithm for equation (1) we get

$$20 \log |H(j\omega)| = 10 \log 1 - 10 \log \left(1 + \epsilon^2 C_N^2\left(\frac{\omega}{\omega_p}\right) \right)$$

At $\omega = \omega_p$ $\alpha_p = 10 \log(1 + \epsilon^2) \Rightarrow \epsilon = (10^{0.1\alpha_p - 1})^{1/2} \therefore C_N(1) = 1$

At $\omega = \omega_s$ $\alpha_s = 10 \log \left\{ 1 + \epsilon^2 C_N^2 \left(\frac{\omega_s}{\omega_p} \right) \right\}$
 $\alpha_s = 10 \log \left[1 + \epsilon^2 \left\{ \cosh(N \cosh^{-1}(\omega_s/\omega_p)) \right\}^2 \right]$

$$\cosh(N \cosh^{-1}(\omega_s/\omega_p)) = \sqrt{\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1}}$$

$$N \cosh^{-1}\left(\frac{\omega_s}{\omega_p}\right) = \cosh^{-1} \sqrt{\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1}}$$

$$N \geq \frac{\cosh^{-1} \sqrt{(10^{0.1\alpha_s} - 1)/(10^{0.1\alpha_p} - 1)}}{\cosh^{-1}(\omega_s/\omega_p)}$$

Substitute A & k

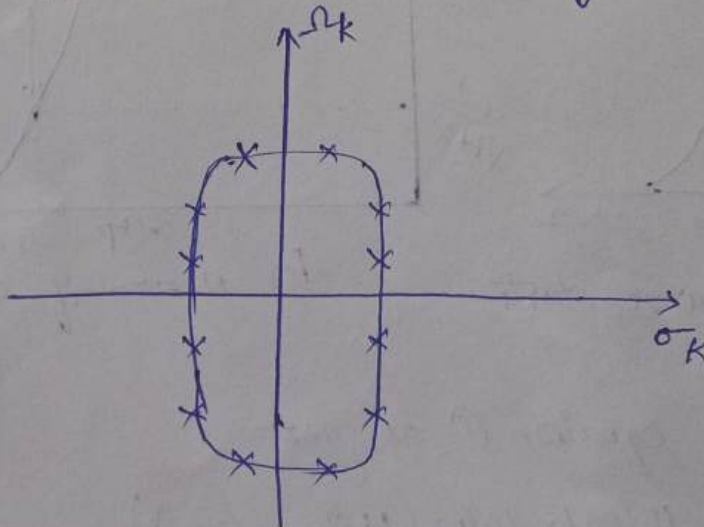
$$N \geq \cosh^{-1}(A) / \cosh^{-1}(1/k)$$

Pole location for Chebyshev filter:-

The poles of Chebyshev transfer function are located on ellipse in the s -plane. Equation for ellipse.

$$\frac{\sigma_k^2}{a^2} + \frac{\omega_k^2}{b^2} = 1$$

a & b are minor and major axis of the ellipse.



P Design a chebyshev filter with a maximum passband attenuation of 2.5dB at $\omega_p = 20$ rad/sec and the stopband attenuation of 30dB at $\omega_s = 50$ rad/sec.

Sol Given $\omega_p = 20$ rad/sec, $\alpha_p = 2.5$ dB, $\omega_s = 50$ rad/sec, $\alpha_s = 30$ dB

$$N = \frac{\cosh^{-1} A}{\cosh^{-1} (1/k)}$$

Type-II chebyshev filter

It consists of both poles and zeros. The magnitude response is equal to

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 \left[\frac{C_N^2(\omega_s/\omega_p)}{C_N^2(\omega_s)} \right]}$$

where $C_N(x)$ is the N^{th} order chebyshev polynomial, ω_s is the stopband frequency and ω_p is the passband frequency. zeroes are located on the imaginary axis at the points

$$s_k = \frac{j\omega_s}{\sin \phi_k} \quad k = 1, 2, 3, \dots, N$$

poles are located at the points (x_k, y_k) where

$$x_k = \frac{\omega_s \sigma_k}{\sigma_k^2 + \omega_k^2} \quad k = 1, 2, \dots, N$$

$$y_k = \frac{\omega_s \sigma_k}{\sigma_k^2 + \omega_k^2} \quad k = 1, 2, \dots, N$$

where $\sigma_k = a \cos \phi_k$

$$k = 1, 2, \dots, N$$

$$\omega_k = b \sin \phi_k \quad k = 1, 2, \dots, N$$

$$\mu = \lambda + \sqrt{1 + \lambda^2}$$

For the given specifications $\epsilon, \lambda, \omega_s$ and ω_p , the order of the filter

$$N = \frac{\cosh^{-1} (1/\epsilon)}{\cosh^{-1} (\frac{\omega_s}{\omega_p})}$$

where $A = 1/\epsilon$

$$N = \frac{\cosh^{-1} (A)}{\cosh^{-1} (1/k)}$$

$$k = \frac{\omega_p}{\omega_s}$$

$$\epsilon = \left[10^{0.1 \alpha_p} - 1 \right]^{1/2} \text{ and } \lambda = \left[10^{0.1 \alpha_s} - 1 \right]^{1/2}$$

Comparison between Butterworth and Chebyshev filter:-

1. The magnitude response of Butterworth filter decreases monotonically as the frequency ω increases from 0 to ∞ , whereas the magnitude response of the Chebyshev filter exhibits ripples in passband or stopband according to type.
2. The transition band is more in Butterworth filter when compared to Chebyshev filter.
3. The poles of Butterworth filter lie on a circle whereas the poles of Chebyshev filter lie on an ellipse.
4. For the same specifications the number of poles in Butterworth is more when compared to Chebyshev filter i.e., order of the Chebyshev filter is less than that of Butterworth.

Comparison between IIR and FIR filters:-

IIR filter	FIR filter
(i) Design is based on all the infinite samples of the impulse response.	(i) Design is based on only a finite number of samples of impulse response.
(ii) The impulse response cannot be directly converted to digital filter transfer function.	(ii) The impulse response can be directly converted to digital filter transfer function.
(iii) The digital filter cannot be directly designed. First an analog filter is to be designed and then it has to be transformed to a digital filter.	(iii) The digital filter can be directly designed.
(iv) The specifications include the desired characteristics for magnitude response.	(iv) The specifications include the desired characteristics for both magnitude and phase response.
(v) Linear phase characteristics cannot be achieved.	(v) Linear phase characteristics can be achieved.

Design of IIR filter from analog filters:-

There are four methods for describing analog filter into digital filter.

- Approximation of derivatives.
- Impulse Invariant Transformation.
- Bilinear Transformation
- Matched z -transformation Technique.

Design of IIR filter By Approximation of Derivatives:-

The analog filter having the rational system function $H(s)$ can also be described by the linear constant coefficient differential equation.

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

In this method of IIR filter design by approximation of derivative an analog filter is converted into a digital filter by approximating the above differential equation into an equivalent difference equation.

The backward difference formula is substituted for the derivative $\frac{dy(t)}{dt}$ at time $t = nT$.

$$\text{Thus } \left. \frac{dy(t)}{dt} \right|_{t=nT} = \frac{y(nT) - y(n-1)T}{T} \quad (\text{or}) \quad \left. \frac{dy(t)}{dt} \right|_{t=nT} = \frac{y(n) - y(n-1)}{T}$$

where T is the sampling interval and $y(n) = y(nT)$.

The system function of an analog differentiator with an output $dy(t)/dt$ is $H(s) = s$, and the digital system which produces the output $[y(n) - y(n-1)]/T$ has the system function $H(z) = [1 - z^{-1}]/T$.

Comparing these two, we can say that the frequency domain equivalent for the relationship

$$\left. \frac{dy(t)}{dt} \right|_{t=nT} = \frac{y(n) - y(n-1)}{T} \text{ is:}$$
$$s = \frac{1 - z^{-1}}{T}$$

Thus, this is the analog domain to digital domain transformation.

Also, the second derivative $\frac{d^2y(t)}{dt^2}$ can be replaced by the second backward difference:

$$\begin{aligned}\left. \frac{d^2y(t)}{dt^2} \right|_{t=nT} &= \left. \frac{d}{dt} \left[\frac{dy(t)}{dt} \right] \right|_{t=nT} \\ &= \frac{[y(nT) - y(nT-T)]/T - [y(nT-T) - y(nT-2T)]/T}{T} \\ &= \frac{y(n) - 2y(n-1) + y(n-2)}{T^2}\end{aligned}$$

The equivalent expression in frequency domain is:

$$\tilde{S} = \frac{1 - 2z^{-1} + z^{-2}}{T^2} \quad (\text{or}) \quad \tilde{S} = \left[\frac{1 - z^{-1}}{T} \right]^2$$

The i^{th} derivative of function $y(t)$ results in the equivalent frequency domain relationship as:

$$\tilde{S}^i = \left[\frac{1 - z^{-1}}{T} \right]^i$$

As a result, the digital filter's system function $H(z)$ can be obtained from the analog filter's system function $H_a(s)$ by the method of approximation of the derivatives as:

$$H(z) = H_a(s) \Big|_{s = \frac{1 - z^{-1}}{T}}$$

The outcomes of the mapping of the z -plane from the s -plane are discussed below.

We have $s = \frac{1 - z^{-1}}{T}$, i.e., $z = \frac{1}{1 - sT}$

Substituting $s = j\omega$ in the expression for z , we have

$$\begin{aligned}z &= \frac{1}{1 - j\omega T} \\ &= \frac{1}{1 + \omega^2 T^2} + j \frac{\omega T}{1 + \omega^2 T^2}\end{aligned}$$

Varying ω from $-\infty$ to ∞ the corresponding locus of points in the z -plane is a circle with radius $\frac{1}{2}$ and with centre at $z = \frac{1}{2}$.

P. Convert the analog low-pass filter specified by $H_a(s) = \frac{2}{s+3}$ into a digital filter making use of the backward difference for the derivative.

Sol. We know that the mapping formula for the backward difference for the derivative is given by

$$s = \frac{1-z^{-1}}{T}$$

For the given analog filter function $H_a(s) = \frac{2}{s+3}$, the corresponding digital filter function is:

$$H(z) = H_a(s) \Big|_{s = \frac{1-z^{-1}}{T}} = \frac{2}{\frac{(1-z^{-1})}{T} + 3}$$

$$= \frac{2T}{1-z^{-1}+3T} = \frac{2}{1-z^{-1}+3} \quad \text{If } T=1s.$$

$$\therefore H(z) = \frac{2}{4-z^{-1}}$$

P. Making use of the backward difference for the derivative, convert the analog filter function given below to a digital filter function.

$$H_a(s) = \frac{4}{s^2+9}$$

Sol. The mapping formula for the backward difference by the derivative is:

$$s = \frac{1-z^{-1}}{T}$$

Therefore, for the given $H_a(s)$, the corresponding digital filter function is:

$$H(z) = H_a(s) \Big|_{s = \frac{1-z^{-1}}{T}} = \frac{4}{\left[\frac{1-z^{-1}}{T}\right]^2 + 9}$$

$$= \frac{4T^2}{1-2z^{-1}+z^{-2}+9T^2}$$

If $T=1s$, then

$$H(z) = \frac{4}{1-2z^{-1}+z^{-2}+9} = \frac{4}{10-2z^{-1}+z^{-2}}$$

Design of IIR filter by Impulse Invariant Transformation:-

In this technique, the desired impulse response of the digital filter is obtained by uniformly sampling the impulse response of the equivalent analog filter.

This is used to convert an analog filter to digital filter. We can convert filter transfer function from analog to digital filter, by applying Impulse Invariance method.

$H_a(s)$ is analog filter $H(z)$ Digital filter
 $H_a(s) \rightarrow H(z)$

Step 1:- Find the analog filter Transfer function $H_a(s)$

Step 2:- Select the sampling period 'T'.

Step 3:- Transfer function of analog filter

$$H_a(s) = \sum_{k=1}^N \frac{C_k}{s - P_k} \rightarrow (1) \quad P_k \rightarrow \text{Poles}$$

Step 4:- Convert Transfer function $H_a(s)$ to $H(z)$

$$H(z) = \sum_{k=1}^N \frac{C_k}{1 - e^{P_k T} z^{-1}} \rightarrow (2)$$

Comparing the above expressions for $H_a(s)$ and $H(z)$, we can say that the impulse invariant transformation is accomplished by the mapping.

$$\frac{1}{s - P_k} \xrightarrow[\text{mapped to}]{\text{is transformed to}} \frac{1}{1 - e^{P_k T} z^{-1}}$$

$s - P_k \xrightarrow{\text{transformed}} 1 - e^{P_k T} z^{-1}$

For high sampling rate:-

$$H(z) = \sum_{k=1}^N \frac{T C_k}{1 - e^{P_k T} z^{-1}}$$

Relation between analog and digital poles

z and s is $z = e^{sT}$

The above mapping shows that the analog pole at $s = p_k$ is mapped into a digital pole at $z = e^{p_k T}$. Therefore, the analog poles and the digital poles are related by the relation.

$$z = e^{sT} \rightarrow \textcircled{1}$$

The general characteristics of the mapping $z = e^{sT}$ can be obtained by substituting $s = \sigma + j\omega$ and expressing the complex variable z in polar form as $z = re^{j\omega} \rightarrow \textcircled{2}$

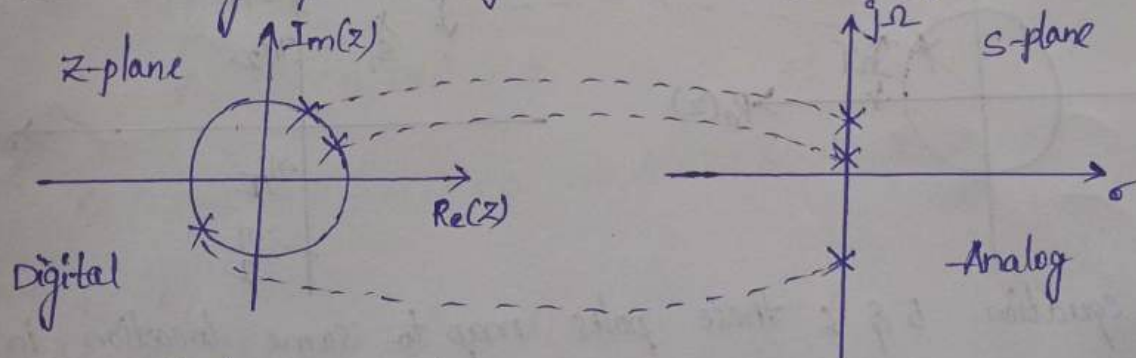
$$re^{j\omega} = e^{(\sigma + j\omega)T} = e^{\sigma T} e^{j\omega T}$$

That means $|z| = r = e^{\sigma T}$

$$\angle z = \omega T = \omega T$$

So the relationship between analog frequency ω and digital frequency ' ω ' is $\omega = \Omega T$ or $\Omega = \frac{\omega}{T}$.

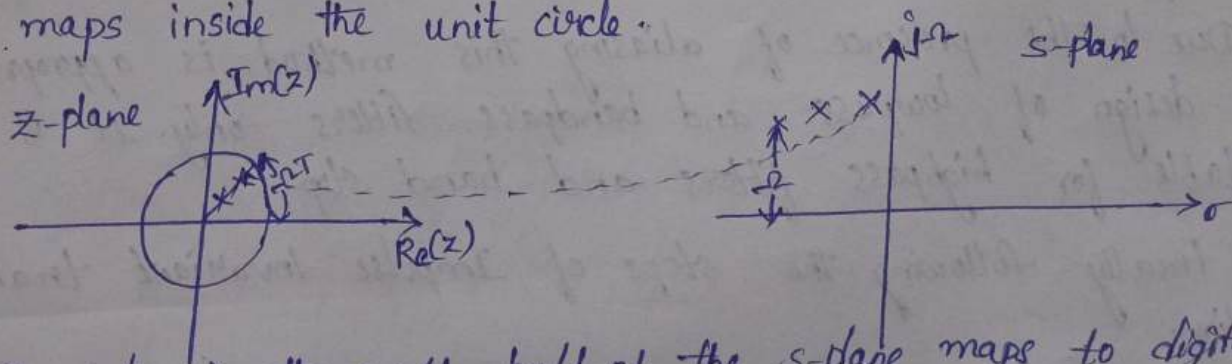
Consider any pole on $j\omega$ axis where $\sigma = 0$



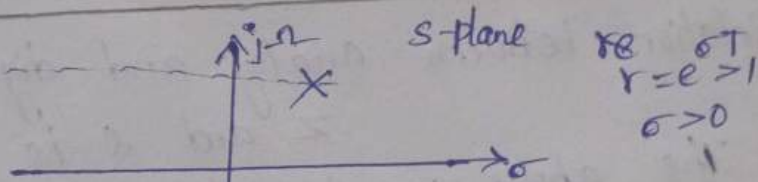
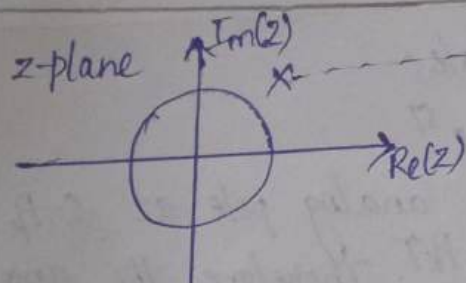
The term $e^{\sigma T}$ has magnitude $e^{\sigma T}$ and angle 0

The term $e^{j\omega T}$ has unity magnitude and angle ωT .

Consider the poles in the left half of s -plane, where $\sigma < 0$. These poles map inside the unit circle.



All the poles in the right half of the s -plane map to digital poles outside the unit circle.



The poles are mapped in many to one mapping where many points in s-plane are mapped to a single point in the z-plane.

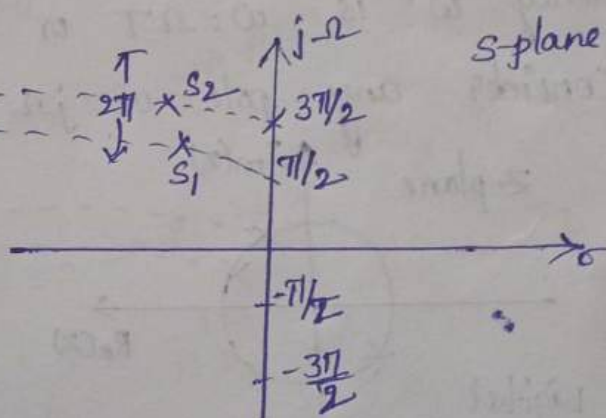
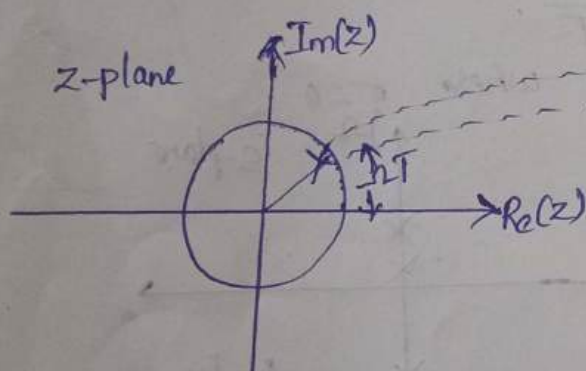
Consider two poles in the s-plane with identical real parts but with Imaginary components differ by $\frac{2\pi}{T}$.

Let the poles are

$$s_1 = \sigma + j\omega, s_2 = \sigma + j\left(\omega + \frac{2\pi}{T}\right)$$

$$z_1 = e^{s_1 T} = e^{\sigma T + j\omega T} = e^{\sigma T} \cdot e^{j\omega T} \rightarrow (5)$$

$$z_2 = e^{s_2 T} = e^{\sigma T + j\left(\omega + \frac{2\pi}{T}\right)T} = e^{\sigma T} \cdot e^{j\omega T} \cdot e^{j2\pi} = z_1$$



from equation 5 & 6 these poles map to same location in z-plane. There are Infinite number of s-plane poles that map to same location in z-plane. This is the disadvantage of impulse invariant mapping.

Due to the presence of aliasing this method is appropriate for design of lowpass and bandpass filters only. It is not suitable for highpass filters and band stop.

Finally following the steps of Impulse Invariant transformation.

Useful impulse invariant transformations:-

$$(i) \frac{s+a}{(s+a)^2+b^2} \rightarrow \frac{1-e^{-aT}(\cos bT)z^{-1}}{1-2e^{-aT}(\cos bT)z^{-1}+e^{-2aT}z^{-2}}$$

$$(ii) \frac{b}{(s+a)^2+b^2} \rightarrow \frac{e^{-aT}(\sin bT)z^{-1}}{1-2e^{-aT}(\cos bT)z^{-1}+e^{-2aT}z^{-2}}$$

P. For the analog transfer function $H_a(s) = \frac{2}{(s+1)(s+3)}$ determine $H(z)$ if a) $T=1s$ and b) $T=0.5s$ using impulse invariant method.

Soln Given, $H_a(s) = \frac{2}{(s+1)(s+3)}$

Using partial fractions, $H_a(s)$ can be expressed as:

$$H_a(s) = \frac{A}{s+1} + \frac{B}{s+3}$$

$$\frac{A(s+3)+B(s+1)}{(s+1)(s+3)} = \frac{2}{(s+1)(s+3)}$$

$$A(s+3)+B(s+1) = 2$$

put $s=-3$

$$A(-3+3)+B(-3+1) = 2 \Rightarrow A(0)+B(-2) = 2 \Rightarrow -2B = 2 \Rightarrow B = -1$$

put $s=-1$

$$A(-1+3)+B(-1+1) = 2 \Rightarrow A(2) = 2 \Rightarrow A = 1$$

$$\therefore H_a(s) = \frac{1}{s+1} - \frac{1}{s+3} = \frac{1}{s-(-1)} - \frac{1}{s-(-3)}$$

By impulse invariant transformation, we know that

$$\frac{C_k}{s-P_k} \xrightarrow{\text{(is transformed to)}} \frac{C_k}{1-e^{P_k T}z^{-1}}$$

system function of digital filter is:

$$H(z) = \frac{1}{1-e^{P_1 T}z^{-1}} - \frac{1}{1-e^{P_2 T}z^{-1}}$$

Here $H_a(s)$ has two poles and $P_1 = -1$, and $P_2 = -3$.

(a) When $T=1s$

$$\begin{aligned}
 H(z) &= \frac{1}{1-e^{-1}z^{-1}} - \frac{1}{1-e^{-3}z^{-1}} \\
 &= \frac{1}{1-0.3678z^{-1}} - \frac{1}{1-0.0497z^{-1}} \\
 &= \frac{(1-0.0497z^{-1}) - (1-0.3678z^{-1})}{(1-0.3678z^{-1})(1-0.0497z^{-1})} \\
 &= \frac{0.3181z^{-1}}{1-0.4175z^{-1}+0.0182z^{-2}}
 \end{aligned}$$

(b) When $T=0.5s$

$$\begin{aligned}
 H(z) &= \frac{1}{1-e^{-0.5}z^{-1}} - \frac{1}{1-e^{-3 \times 0.5}z^{-1}} \\
 &= \frac{1}{1-0.606z^{-1}} - \frac{1}{1-0.223z^{-1}} \\
 &= \frac{(1-0.223z^{-1}) - (1-0.606z^{-1})}{(1-0.606z^{-1})(1-0.223z^{-1})} \\
 &= \frac{0.383z^{-1}}{1-0.829z^{-1}+0.135z^{-2}}
 \end{aligned}$$

P. Convert the analog filter with transfer function $H_a(s) = \frac{s+0.1}{(s+0.1)^2 + 9}$ into a digital filter using the impulse invariant transformation.

Sol. Observe that the given system function of the analog filter is of the standard form $H_a(s) = \frac{s+a}{(s+a)^2 + b^2}$, where we are given $a=0.1$ and $b=3$.

By the impulse invariant transformation, we know that

$$\frac{s+a}{(s+a)^2 + b^2} \xrightarrow{\text{(is transformed to)}} \frac{1-e^{-aT}(\cos bT)z^{-1}}{1-2e^{-aT}(\cos bT)z^{-1}+e^{-2aT}z^{-2}}$$

Therefore, for the given $H_a(s)$, we can write the system function of the digital filter

$$H(z) = \frac{1 - e^{-0.1T}(\cos 3T)z^{-1}}{1 - 2e^{-0.1T}(\cos 3T)z^{-1} + e^{-2(0.1)T}z^{-2}}$$

Assuming $T=1s$, we have

$$\begin{aligned} H(z) &= \frac{1 - e^{-0.1}(\cos 3)z^{-1}}{1 - 2e^{-0.1}(\cos 3)z^{-1} + e^{-0.2}z^{-2}} \\ &= \frac{1 - 0.9048(-0.9899)z^{-1}}{1 - 2(0.9048)(-0.9899)z^{-1} + 0.8187z^{-2}} \end{aligned}$$

$$\therefore H(z) = \frac{1 + 0.8956z^{-1}}{1 + 1.7913z^{-1} + 0.8187z^{-2}}$$

P. The system function of an analog filter is expressed as: $H_a(s) = \frac{2}{s(s+2)}$. Find the corresponding $H(z)$ using the impulse invariant method for a sampling frequency of 4 samples per second.

Sol. Given sampling rate = 4 samples/second.

$$\therefore \text{Sampling period } T = \frac{1}{4} = 0.25s$$

Expressing the given $H_a(s)$ in terms of partial fractions, we have

$$H_a(s) = \frac{2}{s(s+2)} = \frac{1}{s} - \frac{1}{s+2} = \frac{1}{s-(0)} - \frac{1}{s-(-2)}$$

By the impulse invariant transformation, we know that

$$\frac{1}{s - p_k} \xrightarrow{\text{(is transformed to)}} \frac{1}{1 - e^{p_k T} z^{-1}}$$

Here $H_a(s)$ has two poles and $p_1 = 0$, $p_2 = -2$.

Therefore, the system function of the digital filter is:

$$\begin{aligned} H(z) &= \frac{1}{1 - e^{p_1 T} z^{-1}} - \frac{1}{1 - e^{p_2 T} z^{-1}} \\ &= \frac{1}{1 - e^{(0)T} z^{-1}} - \frac{1}{1 - e^{(-2)T} z^{-1}} \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{1-z^{-1}} - \frac{1}{1-e^{-2(0.25)}z^{-1}} \\
 &= \frac{1}{1-z^{-1}} - \frac{1}{1-0.606z^{-1}} \\
 &= \frac{(1-0.606z^{-1}) - (1-z^{-1})}{(1-z^{-1})(1-0.606z^{-1})} \\
 H(z) &= \frac{0.394z^{-1}}{1-1.606z^{-1}+0.606z^{-2}}
 \end{aligned}$$

Design of IIR filter by the Bilinear Transformation method:

Bilinear transformation is a one-to-one mapping from the s -domain to the z -domain. That is, the bilinear transformation is a conformal mapping that transforms the imaginary axis of s -plane into the unit circle in the z -plane only once, thus avoiding aliasing of frequency components.

In this mapping, all points in the left half of s -plane are mapped inside the unit circle in the z -plane, and all points in the right half of s -plane are mapped outside the unit circle in the z -plane.

So the transformation of a stable analog filter results in a stable digital filter. The bilinear transformation can be obtained by using the trapezoidal formula for the numerical integration.

Let the system function of the analog filter be $H_a(s) = \frac{b}{s+a}$

The differential equation describing the above analog filter can be obtained as:

$$H_a(s) = \frac{Y(s)}{X(s)} = \frac{b}{s+a}$$

$$sY(s) + aY(s) = bX(s)$$

Taking inverse Laplace transform on both sides, we get

$$\frac{dy(t)}{dt} + a y(t) = b x(t)$$

Integrating the above equation between the limits $(nT-T)$ and nT , we have

$$\int_{nT-T}^{nT} \frac{dy(t)}{dt} dt + a \int_{nT-T}^{nT} y(t) dt = b \int_{nT-T}^{nT} x(t) dt$$

The trapezoidal rule for numeric integration is expressed as:

$$\int_{nT-T}^{nT} a(t) dt = \frac{T}{2} [a(nT) + a(nT-T)]$$

Therefore, we get

$$y(nT) - y(nT-T) + a \frac{T}{2} y(nT) + a \frac{T}{2} y(nT-T) = b \frac{T}{2} x(nT) + b \frac{T}{2} x(nT-T)$$

Taking z-transform, we get

$$Y(z)[1 - z^{-1}] + a \frac{T}{2} [1 + z^{-1}] Y(z) = b \frac{T}{2} [1 + z^{-1}] X(z)$$

Therefore, the system function of the digital filter is:

$$\frac{Y(z)}{X(z)} = H(z) = \frac{b}{\frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} + a}$$

Comparing this with the analog filter system function $H_a(s)$ we get

$$s = \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] = \frac{2}{T} \left[\frac{z-1}{z+1} \right]$$

Rearranging, we can get

$$z = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s}$$

This is the relation between analog and digital poles in bilinear transformation. So to convert an analog filter function into an equivalent digital filter function,

$$s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} \text{ in } H_a(s)$$

The general characteristic of the mapping $z = e^{sT}$ may be obtained by putting $s = \sigma + j\omega$ and expressing the complex variable z in the polar form as $z = re^{j\omega}$ in the above equation for s .

$$\text{Thus, } s = \frac{2}{T} \left[\frac{z-1}{z+1} \right] = \frac{2}{T} \left[\frac{re^{j\omega} - 1}{re^{j\omega} + 1} \right]$$

$$\text{(or) } s = \frac{2}{T} \frac{(re^{j\omega} - 1)(re^{-j\omega} + 1)}{(re^{j\omega} + 1)(re^{-j\omega} + 1)} = \frac{2}{T} \left[\frac{r^2 - 1}{1 + r^2 + 2r\cos\omega} + j \frac{2r\sin\omega}{1 + r^2 + 2r\cos\omega} \right]$$

Since $s = \sigma + j\omega$, we get

$$\sigma = \frac{2}{T} \left[\frac{r^2 - 1}{1 + r^2 + 2r\cos\omega} \right]$$

$$\text{and } \omega = \frac{2}{T} \left[\frac{2r\sin\omega}{1 + r^2 + 2r\cos\omega} \right]$$

From the above equation for σ , we observe that if $r < 1$ then $\sigma < 0$ and if $r > 1$, then $\sigma > 0$, and if $r = 1$, then $\sigma = 0$.

Hence the left half of the s -plane maps into points inside the unit circle in the z -plane, the right half of the s -plane maps into points outside the unit circle in the z -plane and the imaginary axis of s -plane maps into the unit circle in the z -plane. This transformation results in a stable digital system.

Relation between analog and digital frequencies:

On the imaginary axis of s -plane $\sigma = 0$ and correspondingly in the z -plane $r = 1$.

$$\omega = \frac{2}{T} \left[\frac{2\sin\omega}{1 + 1 + 2\cos\omega} \right] = \frac{2}{T} \left[\frac{\sin\omega}{1 + \cos\omega} \right]$$

$$= \frac{2}{T} \left[\frac{2\sin\frac{\omega}{2}\cos\frac{\omega}{2}}{1 + 2\cos^2\frac{\omega}{2} - 1} \right] = \frac{2}{T} \tan\frac{\omega}{2}$$

$\therefore$ The relation between analog and digital frequencies is,

$$\omega = \frac{2}{T} \tan\frac{\omega}{2}$$

or equivalently, we have $\omega = 2 \tan^{-1} \frac{\Omega T}{2}$.

The above relation between analog and digital frequencies shows that the entire range in Ω is mapped only once into the range $-\pi \leq \omega \leq \pi$.

The mapping is non-linear and the lower frequencies in analog domain are expanded in the digital domain, whereas the higher frequencies are compressed.

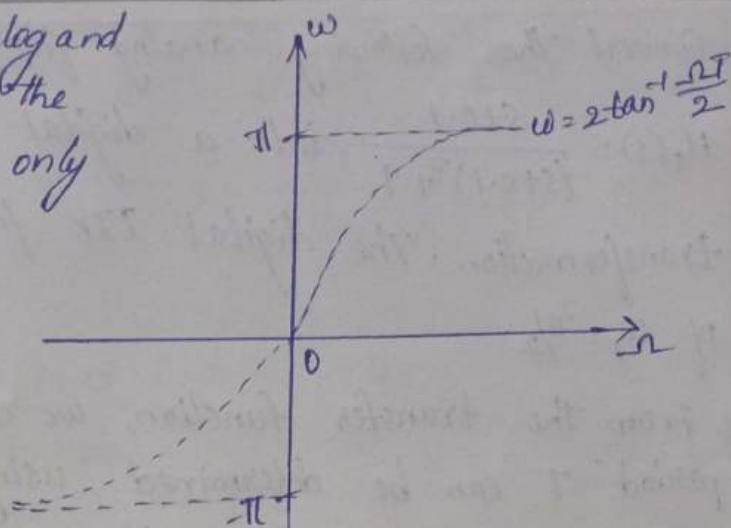


Figure 3.5: Mapping between Ω and ω in bilinear transformation.

This is due to the nonlinearity of the arctangent function and usually known as frequency warping.

Warping Effect:-

The effect of warping on the magnitude response can be explained by considering an analog filter with a number of passbands as shown in figure 3.6.

Ω, ω are the frequency variable in analog and digital filters respectively.

Relation between two ω 's are $\Omega = \frac{2}{T} \tan \frac{\omega}{2}$

$$\therefore \tan \theta = 0$$

$$\tan \frac{\omega}{2} = \frac{\omega}{2}$$

$$\Omega = \frac{2}{T} \frac{\omega}{2} \Rightarrow \Omega = \frac{\omega}{T} \Rightarrow \omega = \Omega T$$

This equation is valid only for low frequencies. For low frequencies the relation between Ω and ω are linear. So, the digital filter have the same amplitude response as the analog filter. For high frequencies the relation between Ω and ω becomes non-linear and distortion is introduced. This is known as warping effect.

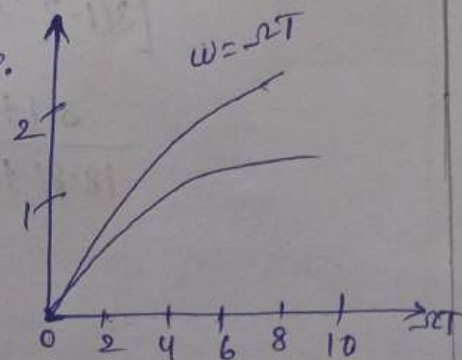


Figure 3.6

P. Convert the following analog filter with transfer function $H_a(s) = \frac{s+0.1}{(s+0.1)^2+9}$ into a digital IIR filter by using bilinear transformation. The digital IIR filter is having a resonant frequency of $\omega_r = \pi/2$.

Sol. From the transfer function, we observe that $\Omega_c = 3$. The sampling period T can be determined using the equation:

$$\Omega_c = \frac{2}{T} \tan \frac{\omega_r}{2}$$

$$\therefore T = \frac{2}{\Omega_c} \tan \frac{\omega_r}{2} = \frac{2}{3} \tan \frac{\pi/2}{2} = 0.666s$$

Using the bilinear transformation, the digital filter system function is:

$$H(z) = H_a(s) \Big|_{s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}} = H_a(s) \Big|_{s = 3 \frac{1-z^{-1}}{1+z^{-1}}}$$

$$H(z) = \frac{s+0.1}{(s+0.1)^2+9} \Big|_{s = 3 \frac{1-z^{-1}}{1+z^{-1}}}$$

$$= \frac{3 \frac{1-z^{-1}}{1+z^{-1}} + 0.1}{\left[3 \frac{1-z^{-1}}{1+z^{-1}} + 0.1\right]^2 + 9}$$

$$= \frac{[3(1-z^{-1}) + 0.1(1+z^{-1})][1+z^{-1}]}{[3(1-z^{-1}) + 0.1(1+z^{-1})]^2 + 9(1+z^{-1})^2}$$

$$= \frac{3.1 + 0.2z^{-1} - 2.9z^{-2}}{18.61 + 0.02z^{-1} + 17.41z^{-2}}$$

Design procedure for low-pass digital Butterworth IIR filter:

Step 1: Derive the specification from the given data.

Step 2: Find the order of filter by using following equation.

$$N \geq \frac{\log \sqrt{\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1}}}{\log\left(\frac{\omega_s}{\omega_p}\right)} \quad (\text{or}) \quad N \geq \frac{\log(\gamma_e)}{\log\left(\frac{\omega_s}{\omega_p}\right)}$$

N is roundoff to the next highest value.

where, α_s = stop band attenuation

α_p = passband attenuation

ω_s = stopband frequency

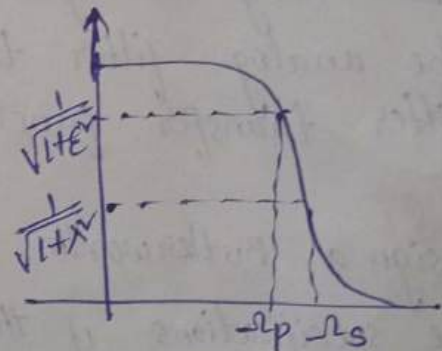
ω_p = passband frequency.

for bilinear transformation

$$\omega_s = \frac{2}{T} \tan \frac{\omega_s}{2}$$

$$\omega_p = \frac{2}{T} \tan \frac{\omega_p}{2}$$

$$\therefore \frac{\omega_s}{\omega_p} = \frac{\tan \omega_p/2}{\tan \omega_s/2}$$



for Impulse transformation,

$$\omega_s = \frac{\omega_s}{T}, \quad \omega_p = \frac{\omega_p}{T} \quad \therefore \frac{\omega_p}{\omega_s} = \frac{\omega_p}{\omega_s}$$

Step 3: Obtain analog filter transfer function $H_a(s)$ of the filter by using the following Table.

Order	Denominator
1	$s+1$
2	$\tilde{s} + \sqrt{2}s + 1$
3	$(s+1)(\tilde{s} + s + 1)$
4	$(\tilde{s} + 0.763s + 1)(\tilde{s} + 1.8471s + 1)$

The numerator is '1'.

Step 4:- Find the cutoff frequency ω_c by using the following equation,

(i) If $\omega_p < \omega_s$
$$\omega_c = \frac{\omega_p}{(10^{0.1A_p} - 1)^{1/2N}} = \frac{\omega_p}{E^{1/N}}$$

(ii) If $\omega_p > \omega_s$

$$\omega_c = \frac{\omega_s}{(10^{0.1A_s} - 1)^{1/2N}} = \frac{\omega_s}{\lambda^{1/N}}$$

Step 5:-

i) For low pass filter design

$$s \rightarrow \frac{s}{\omega_c} \text{ in } H_a(s)$$

ii) For High pass filter design

$$s \rightarrow \frac{\omega_c}{s} \text{ in } H_a(s)$$

Step 6:- Using the given transformation (Bilinear (or) Impulse Invariant) The analog filter transfer function $H_a(s)$ converted into digital filter transfer function $H(z)$.

P. Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired low-pass filter are:

$$0.9 \leq |H(\omega)| \leq 1; 0 \leq \omega \leq \frac{\pi}{2}$$

$$|H(\omega)| \leq 0.2; \frac{3\pi}{4} \leq \omega \leq \pi$$

with $T = 1s$.

Given

$$0.9 \leq |H(\omega)| \leq 1; 0 \leq \omega \leq \frac{\pi}{2}$$

$$|H(\omega)| \leq 0.2; \frac{3\pi}{4} \leq \omega \leq \pi$$

$$\frac{1}{\sqrt{1+\epsilon^2}} = 0.9 \rightarrow \epsilon = 0.484$$

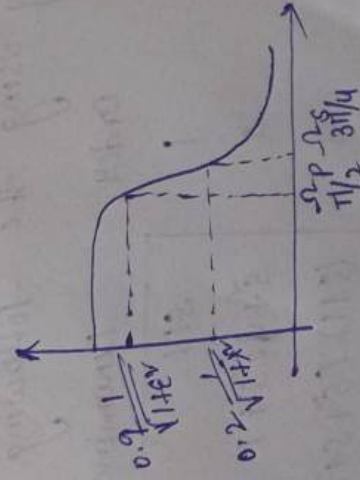
$$\frac{1}{\sqrt{1+\lambda^2}} = 0.2 \rightarrow \lambda = 4.898$$

By using Bilinear Transformation

$$\omega_p = \frac{\pi}{2} \Rightarrow \frac{2}{T} \tan \frac{\pi}{4} = 2$$

$$\omega_s = \frac{3\pi}{4} \Rightarrow \frac{2}{T} \tan \frac{3\pi}{8} = 4.828$$

Step 1:-



Step 2:- $N \geq \frac{\log(Ne)}{\log\left(\frac{\omega_c}{\omega_p}\right)}$

$$= \frac{\log\left(\frac{4.898}{0.484}\right)}{\log\left(\frac{4.828}{2}\right)}$$

$$\geq 2.63$$

N = 3

Step 3:- Analog filter Transfer function (from table)

$$H_a(s) = \frac{1}{(s+1)(s^2+s+1)}$$

Step 4:- Here $\omega_p < \omega_s$

Therefore, the cutoff frequency

$$\omega_c = \frac{\omega_p}{\epsilon^{1/N}} = \frac{2}{(0.484)^{1/3}} = 2.54$$

Step 5:- For low pass filter design

$$s \rightarrow \frac{s}{\omega_c} = \frac{s}{2.54}$$

$$H_a(s) = \frac{1}{\left[\frac{s}{2.54} + 1\right] \left[\left(\frac{s}{2.54}\right)^2 + \frac{s}{2.54} + 1\right]}$$

$$H_a(s) = \frac{1}{\left[\frac{s}{2.54} + 1\right] \left[\frac{s^2}{(2.54)^2} + \frac{s}{2.54} + 1\right]}$$

$$= \frac{1}{\left[\frac{s}{2.54} + 1\right] \left[\frac{s^2}{6.4516} + \frac{s}{2.54} + 1\right]}$$

$$= \frac{16.39}{6.451s + 1} \cdot \frac{1}{(s + 2.54)(s^2 + 2.54s + 6.45)}$$

Step 6:- By using Bilinear transformation $H_a(s)$ is converted into $H(z)$

In this $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$

$$\therefore T = 1$$

$$s = 2 \frac{1-z^{-1}}{1+z^{-1}}$$

Therefore,

$$H(z) = \frac{16.39}{\left[2 \cdot \frac{1-z^{-1}}{1+z^{-1}} + 2.54\right] \left[\left(2 \cdot \frac{1-z^{-1}}{1+z^{-1}}\right)^2 + 2.54 \left(2 \cdot \frac{1-z^{-1}}{1+z^{-1}}\right) + 6.45\right]}$$

$$H(z) = \frac{16.39(1+z^{-1})^3}{(2-2z^{-1}+2.54+2.54z^{-1})(4+4z^{-2}-8z^{-1}+5.08-5.08z^{-2}+6.45+6.45z^{-2}-18.9z^{-1})}$$

$$H(z) = \frac{16.39(1+z^{-1})^3}{(4.54+0.54z^{-1})(15.53+4.9z^{-1}+5.37z^{-1})}$$

$$H(z) = \frac{16.39(1+z^{-1})^3}{70.5+22.24z^{-1}+34.37z^{-2}+8.38z^{-1}+2.64z^{-2}+2.89z^{-3}}$$

$$H(z) = \frac{16.39(1+z^{-1})^3}{70.5+30.62z^{-1}+27.01z^{-2}+2.89z^{-3}}$$

$$H(z) = \frac{0.23(1+z^{-1})^3}{1+0.434z^{-1}+0.38z^{-2}+0.04z^{-3}}$$

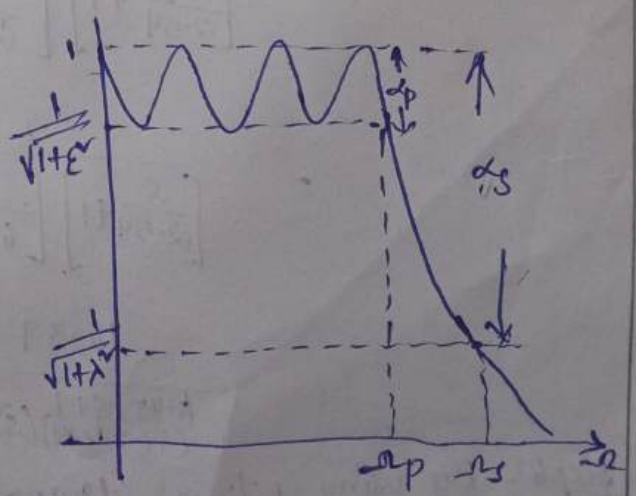
Design of Chebyshev Low pass filter:-

Step 1:- Derive the specifications from the given data.

Step 2:- Find the order of the filter by using the following equations

$$N \geq \frac{\cosh^{-1} \sqrt{\frac{10^{0.1\alpha_s}-1}{10^{0.1\alpha_p}-1}}}{\cosh^{-1} \left(\frac{\omega_s}{\omega_p} \right)}$$

(or) $N \geq \frac{\cosh^{-1} (\sqrt{\lambda_E})}{\cosh^{-1} (\omega_s/\omega_p)}$



Step 3:- Minor & major axis of ellipse a and b.

$$a = \Omega_p \left[\frac{\mu^{1/N} - \mu^{-1/N}}{2} \right]$$

$$b = \Omega_p \left[\frac{\mu^{1/N} + \mu^{-1/N}}{2} \right]$$

$$\mu = \epsilon^{-1} + \sqrt{1 + \epsilon^2}$$

$$\epsilon = (10^{0.1 \alpha_p - 1})^{1/2}$$

Step 4:- poles of chebyshev filter (or) denominator of transfer function.

$$S_k = a \cos \phi_k + j b \sin \phi_k$$

$$\text{where, } \phi_k = \frac{\pi}{2} + \left(\frac{2k-1}{2N} \right) \pi \quad k=1, 2, \dots, N$$

Step 5:- The numerator of the transfer function is evaluated as follow.

Case-i:- If 'N' is odd put $s=0$ in denominator polynomial which is numerator.

Case-ii:- If 'N' is even put $s=0$ in denominator polynomial and divide by $\sqrt{1+\epsilon^2}$ which is numerator.

Step 6:- By using the given transformation convert $H_a(s)$ to $H(z)$.
where, $H(z)$ is transfer function of digital filter.

p. Design a chebyshev IIR digital lowpass filter to satisfy the following constraints. when $T=1$.

$$0.707 \leq |H(\omega)| \leq 1 \quad ; \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(\omega)| \leq 0.1 \quad ; \quad 0.5\pi \leq \omega \leq \pi$$

Sol Step 1:- $\frac{1}{\sqrt{1+\epsilon^2}} = 0.707 \Rightarrow \epsilon = 1$

$$\frac{1}{\sqrt{1+\lambda^2}} = 0.1 \Rightarrow \lambda = 9.95$$

$$\alpha_p = \frac{2}{7} \tan\left(\frac{0.2\pi}{2}\right) = 0.6498$$

$$\alpha_s = \frac{2}{7} \tan\left(\frac{0.5\pi}{2}\right) = 2$$

Step 2:-

$$N \geq \frac{\cosh^{-1}(\sqrt{1/\epsilon})}{\cosh^{-1}\left(\frac{\alpha_s}{\alpha_p}\right)}$$

$$N \geq \frac{\cosh^{-1}\sqrt{\frac{9.95}{1}}}{\cosh^{-1}\left(\frac{2}{0.6498}\right)}$$

$$N \geq \frac{1.8158}{1.7898}$$

$$N \geq 1.04$$

$$\boxed{N=2}$$

Step 3:-

$$\mu = \epsilon^{-1} + \sqrt{1 + \epsilon^2}$$

$$= 1 + \sqrt{2} = 1 + 1.414 = 2.414$$

$$a = 0.6498 \left[\frac{2.414^{1/2} - 2.414^{-1/2}}{2} \right]$$

$$= 0.295$$

$$b = 0.6498 \left[\frac{2.414^{1/2} + 2.414^{-1/2}}{2} \right] = 0.713$$

Step 4:-

$$\phi_k = \frac{\pi}{2} + \left(\frac{2k-1}{2N}\right)\pi \quad k=1, 2$$

$$\phi_1 = 90^\circ + \frac{1}{4} \times \pi = 135^\circ$$

$$\phi_2 = 90^\circ + \frac{3}{4} \times \pi = 225^\circ$$

$$S_k = a \cos \phi_k + j b \sin \phi_k$$

$$S_1 = 0.295 \cos(135^\circ) + j 0.713 \sin(135^\circ) = -0.208 + j 0.504$$

$$S_2 = 0.295 \cos(225^\circ) + j 0.713 \sin(225^\circ) = -0.2085 - j 0.504$$

Denominator:-

$$(s + 0.208 + j 0.504)(s + 0.208 - j 0.504)$$

$$(s + 0.208)^2 - (j 0.504)^2$$

$$s^2 + 0.416s + 0.043 + 0.25$$

$$s^2 + 0.416s + 0.293$$

Step 5:-

$$s = 0 \text{ [}\because N \text{ is even]}$$

$$\text{Numerator} = \frac{0.293}{\sqrt{2}} = 0.207$$

$$H_a(s) = \frac{0.207}{s^2 + 0.416s + 0.293}$$

Step 6:-

$$H(z) = \frac{0.207}{2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 0.416 \times 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.293}$$

$$= \frac{0.207 (1+z^{-1})^2}{4 - 4z^{-2} + 0.832(1-z^{-1})(1+z^{-1}) + 0.293(1+z^{-1})^2}$$

$$H(z) = \frac{0.207(1+z^{-1})^2}{5.125 + 5.461z^{-2} + 7.414z^{-1}}$$

P. Determine the system function $H(z)$ of the lowest order chebyshev IIR filter with following specifications, 3 dB ripple in passband $0 \leq \omega \leq 0.2\pi$, 25dB attenuation in stopband $0.45\pi \leq \omega \leq \pi$ with $T=1$.

Sol Step-1:- $\alpha_p = 3\text{dB}$ $\alpha_s = 25\text{dB}$

$$\omega_p = \frac{2}{T} \tan \frac{\omega_s}{2} = 2 \tan \frac{0.2\pi}{2} = 0.6498$$

$$\omega_s = \frac{2}{T} \tan \frac{\omega_p}{2} = 2 \tan \frac{0.45\pi}{2} = 1.708$$

$$\text{Step 2:- } N \geq \frac{\cosh^{-1} \sqrt{\frac{10^{0.1\alpha_s}}{10^{0.1\alpha_p} - 1}}}{\cosh^{-1} \left(\frac{\omega_s}{\omega_p} \right)}$$

$$N \geq \frac{\cosh^{-1} \sqrt{\frac{10^{0.1(25)} - 1}{10^{0.1(3)} - 1}}}{\cosh^{-1} \left(\frac{1.708}{0.6498} \right)} \geq \frac{3.57}{1.621}$$

$$N \geq 2.202$$

$$\boxed{N=3}$$

Step 3:- $\varepsilon = (10^{0.1 \alpha_p})^{1/2} \Rightarrow \varepsilon = (10^{0.1(3)})^{1/2} \Rightarrow \varepsilon = 1$

$$\mu = \varepsilon^{-1} + \sqrt{1 + \varepsilon^2} \Rightarrow \mu = 1^{-1} + \sqrt{1 + 1^2} \Rightarrow \mu = 1 + \sqrt{2} \Rightarrow \boxed{\mu = 2.414}$$

$$a = \Omega_p \left[\frac{\mu^{1/N} - \mu^{-1/N}}{2} \right] = 0.6498 \left[\frac{2.414^{1/3} - 2.414^{-1/3}}{2} \right] = 0.193$$

$$b = \Omega_p \left[\frac{\mu^{1/N} + \mu^{-1/N}}{2} \right] = 0.6498 \left[\frac{2.414^{1/3} + 2.414^{-1/3}}{2} \right] = 0.678$$

Step 4:-

$$\phi_k = \frac{\pi}{2} + \left(\frac{2k-1}{2N} \right) \pi \quad k=1, 2, 3, \dots, N$$

$$\phi_1 = 90 + \frac{1}{6} \pi = 120$$

$$\phi_2 = 90 + \frac{\pi}{2} = 180$$

$$\phi_3 = 90 + \frac{5}{6} \pi = 240$$

$$S_1 = 0.193 \cos(120) + j0.678 \sin(120) = -0.0965 + j0.587$$

$$S_2 = 0.193 \cos(180) + j0.678 \sin(180) = -0.193 + 0$$

$$S_3 = 0.193 \cos(240) + j0.678 \sin(240) = -0.0965 - j0.58$$

$$(s + 0.0965 + j0.58) (s + 0.0965 - j0.58) (s + 0.19)$$

$$\Rightarrow [(s + 0.0965)^2 - (j0.58)^2] (s + 0.19)$$

$$\Rightarrow [s^2 + 9.31 \times 10^{-3} s + 0.193 s + 0.336] (s + 0.19)$$

$$\Rightarrow (s + 0.19) (s^2 + 0.193 s + 0.345)$$

FIR Digital Filters:- Characteristics of FIR Digital Filters. Design of FIR filters: Using window Techniques, Comparison of IIR & FIR Filters.

Introduction:- A filter is a frequency selective system. Digital filters are classified as finite duration unit impulse response (FIR) filters or infinite duration unit impulse response (IIR) filters, depending on the form of the unit impulse response of the system.

In the FIR system, the impulse response sequence is of finite duration, i.e., it has a finite number of non-zero terms. The IIR system has an infinite number of non-zero terms, i.e., its impulse response sequence is of infinite duration. The response of the FIR filter depends only on the present and past input samples.

The following are the main advantages of FIR filters over IIR filters:

1. FIR filters are always stable.
2. FIR filters with exactly linear phase can easily be designed.
3. FIR filters can be realized in both recursive and non-recursive structures.
4. FIR filters are free of limit cycle oscillations, when implemented on a finite word length digital system.
5. Excellent design methods are available for various kinds of FIR filters.

The disadvantages of FIR filters are as follows:

1. The implementation of narrow transition band FIR filters is very costly, as it requires considerably more arithmetic operations and hardware components such as multipliers, adders and delay elements.
2. Memory requirement and execution time are very high.

Characteristics of FIR filters with linear phase:

The transfer function of a FIR causal filter is given by

$$H(z) = \sum_{n=0}^{N-1} h(n) z^{-n}$$

where $h(n)$ is the impulse response of the filter. The frequency response [fourier transform of $h(n)$] is given by

$$H(\omega) = \sum_{n=0}^{N-1} h(n) e^{-j\omega n}$$

which is periodic in frequency with period 2π , i.e.,

$$H(\omega) = H(\omega + 2k\pi), \quad k=0, 1, 2, \dots$$

Since $H(\omega)$ is complex it can be expressed as

$$H(\omega) = \pm |H(\omega)| e^{j\theta(\omega)}$$

where $|H(\omega)|$ is the magnitude response and $\theta(\omega)$ is the phase response.

We define the phase delay τ_p and group delay τ_g of a filter as:

$$\tau_p = -\frac{\theta(\omega)}{\omega} \quad \text{and} \quad \tau_g = -\frac{d\theta(\omega)}{d\omega}$$

For FIR filters with linear phase, we can define

$$\theta(\omega) = -\alpha\omega \quad -\pi \leq \omega \leq \pi$$

where α is constant phase delay in samples.

$$\tau_g = -\frac{d\theta(\omega)}{d\omega} = -\frac{d}{d\omega}(-\alpha\omega) = \alpha \quad \text{and} \quad \tau_p = \frac{-\theta(\omega)}{\omega} = \frac{\alpha\omega}{\omega} = \alpha$$

i.e., $\tau_p = \tau_g = \alpha$ which means that α is independent of frequency.

We have

$$\sum_{n=0}^{N-1} h(n) e^{-j\omega n} = \pm |H(\omega)| e^{j\theta(\omega)}$$

$$\text{i.e., } \sum_{n=0}^{N-1} h(n) [\cos \omega n - j \sin \omega n] = \pm |H(\omega)| [\cos \theta(\omega) + j \sin \theta(\omega)]$$

This gives us

$$\sum_{n=0}^{N-1} h(n) \cos \omega n = \pm |H(\omega)| \cos \theta(\omega)$$

$$\text{and } -\sum_{n=0}^{N-1} h(n) \sin \omega n = \pm |H(\omega)| \sin \theta(\omega)$$

Therefore,

$$\frac{\sum_{n=0}^{N-1} h(n) \sin \omega n}{\sum_{n=0}^{N-1} h(n) \cos \omega n} = \frac{\sin \theta(\omega)}{\cos \theta(\omega)} = \frac{\sin \alpha\omega}{\cos \alpha\omega}$$

i.e., $\sum_{n=0}^{N-1} h(n) [\sin \omega n \cos \alpha \omega - \cos \omega n \sin \alpha \omega] = 0$

i.e., $\sum_{n=0}^{N-1} h(n) \sin (\alpha - n) \omega = 0$

This will be zero when

$h(n) = h(N-1-n)$ and $\alpha = \frac{N-1}{2}$, for $0 \leq n \leq N-1$

This shows that FIR filters will have constant phase and group delays when the impulse response is symmetrical about

$\alpha = \frac{(N-1)}{2}$

The impulse response satisfying the symmetry condition $h(n) = h(N-1-n)$ for odd and even values of N is shown in figure 4.1. When $N=9$, the centre of symmetry of the sequence occurs at the fourth sample and when $N=8$, the filter delay is $3\frac{1}{2}$ samples.

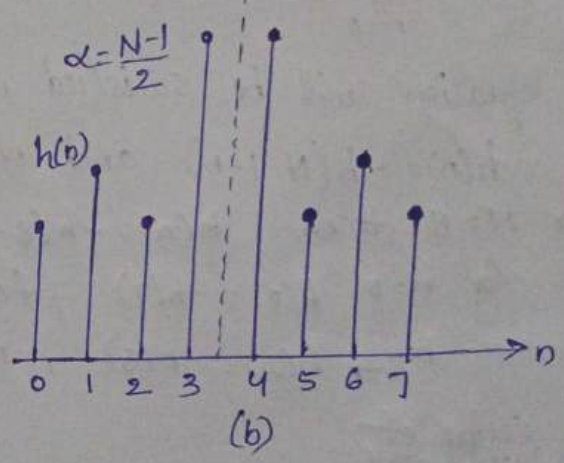
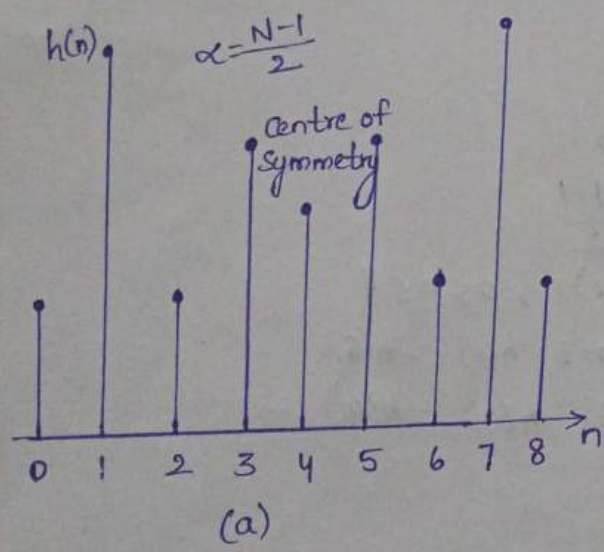


Figure 4.1 Impulse response sequence of symmetrical sequences for (a) N odd (b) N even.

If only constant group delay is required and not the phase delay, we can write

$\theta(\omega) = \beta - \alpha \omega$

Now, we have

$H(\omega) = \pm |H(\omega)| e^{j(\beta - \alpha \omega)}$

i.e., $\sum_{n=0}^{N-1} h(n) e^{-j\omega n} = \pm |H(\omega)| e^{j(\beta - \alpha \omega)}$

i.e., $\sum_{n=0}^{N-1} h(n) [\cos \omega n - j \sin \omega n] = \pm |H(\omega)| [\cos(\beta - \alpha\omega) + j \sin(\beta - \alpha\omega)]$

This gives $\sum_{n=0}^{N-1} h(n) \cos \omega n = \pm |H(\omega)| \cos(\beta - \alpha\omega)$

and $-\sum_{n=0}^{N-1} h(n) \sin \omega n = \pm |H(\omega)| \sin(\beta - \alpha\omega)$

$$\therefore \frac{\sum_{n=0}^{N-1} h(n) \sin \omega n}{-\sum_{n=0}^{N-1} h(n) \cos \omega n} = \frac{\sin(\beta - \alpha\omega)}{\cos(\beta - \alpha\omega)}$$

Cross multiplying and rearranging, we get

$$\sum_{n=0}^{N-1} h(n) [\cos \omega n \sin(\beta - \alpha\omega) + \sin \omega n \cos(\beta - \alpha\omega)] = 0$$

i.e., $\sum_{n=0}^{N-1} h(n) \sin(\beta - (\alpha - n)\omega) = 0$

If $\beta = \frac{\pi}{2}$, the above equation can be written as:

$$\sum_{n=0}^{N-1} h(n) \cos(\alpha - n)\omega = 0$$

This equation will be satisfied when

$$h(n) = -h(N-1-n) \text{ and } \alpha = \frac{N-1}{2}$$

for $N=8$ then $h(n) = -h(8-n-1)$, $0 \leq n \leq 7$

for $n=0$ $h(0) = -h(7)$, for $n=1$ $h(1) = -h(6)$

$n=2$ $h(2) = -h(5)$, $n=3$ $h(3) = -h(4)$

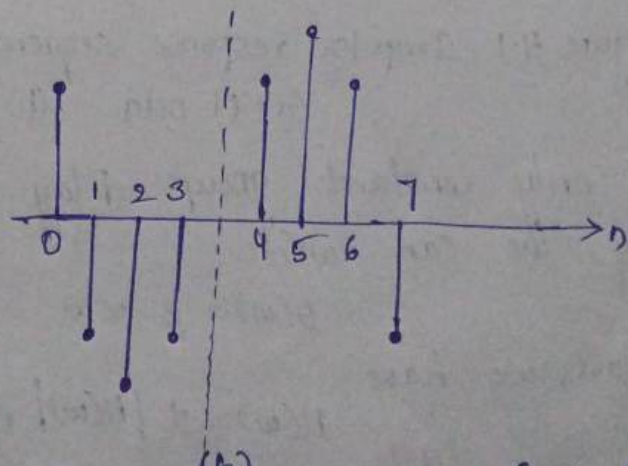
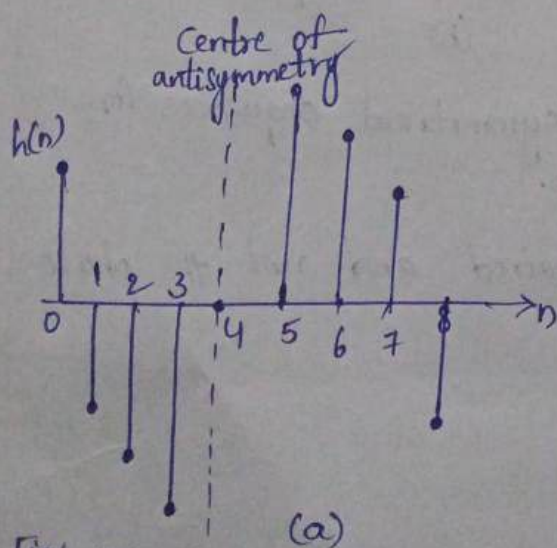


Fig: 4.2 Impulse response sequence of antisymmetric sequence for (a) N odd (b) N even

(5)

This shows that FIR filters have constant delay τ_g and not constant phase delay when the impulse response is antisymmetric about $\alpha = (N-1)/2$.

The impulse response satisfying the antisymmetry condition is shown in figure 4.2. When $N=9$, the centre of antisymmetry occurs at fourth sample and when $N=8$, the centre of antisymmetry occurs at $3\frac{1}{2}$ samples. From figure 4.2, we find that $h[(N-1)/2] = 0$ for antisymmetric odd sequence.

Ex:- The length of an FIR filter is 7. If this filter has a linear phase, show that the equation $\sum_{n=0}^{N-1} h(n) \sin(\alpha - n)\omega = 0$ is satisfied.
 sol: The length of the filter is 7. Therefore, for linear phase,

$$\alpha = \frac{N-1}{2} = \frac{7-1}{2} = 3$$

The condition for symmetry when N is odd, is $h(n) = h(N-1-n)$.
 Therefore, the filter coefficients are $h(0) = h(6)$, $h(1) = h(5)$, $h(2) = h(4)$ and $h(3)$.

Therefore,

$$\begin{aligned} \sum_{n=0}^{N-1} h(n) \sin(\alpha - n)\omega &= \sum_{n=0}^6 h(n) \sin(3 - n)\omega \\ &= h(0) \sin 3\omega + h(1) \sin 2\omega + h(2) \sin \omega + h(3) \sin 0 + \\ &\quad h(4) \sin(-\omega) + h(5) \sin(-2\omega) + h(6) \sin(-3\omega) \\ &= h(0) \sin 3\omega + h(1) \sin 2\omega + h(2) \sin \omega + h(3) \sin 0 + \\ &\quad h(2) \sin \omega - h(1) \sin 2\omega - h(0) \sin 3\omega \\ &= 0 \end{aligned}$$

Hence, the equation $\sum_{n=0}^{N-1} h(n) \sin(\alpha - n)\omega = 0$ is satisfied.

Design techniques for Linear Phase FIR filters:-

There are four well known methods of design techniques for Linear phase FIR filters.

They are

- 1) Fourier Series method
- 2) Window method
- 3) Frequency sampling method
- 4) Optimal frequency design method.

WINDOW METHOD

There are several types of windows some of them are

- i) Rectangular window
- ii) Triangular window (Bartlett)
- iii) Raised Cosine window
 - a) Hamming window
 - b) Hanning window
- iv) Blackman window
- v) Kaiser window

i) Rectangular Window :-

The N -point rectangular window has a window function (weighting function) is defined as

$$w_R(n) = \begin{cases} 1, & -\frac{(N-1)}{2} \leq n \leq \frac{(N-1)}{2} \\ 0, & \text{elsewhere} \end{cases}$$

(or)

$$w_R(n) = \begin{cases} 1, & 0 \leq n \leq (N-1) \\ 0, & \text{elsewhere} \end{cases}$$

The spectrum (frequency response) of rectangular window $w_R(\omega)$ is given by the Fourier transform of $w_R(n)$.

∴ The frequency response of rectangular window is

$$F[W_R(n)] = W_R(\omega) = \sum_{n=0}^{N-1} W_R(n) e^{-j\omega n}$$

$$= \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{-j\omega n}$$

Let, $l = \frac{N-1}{2} + n \Rightarrow n = l - \frac{N-1}{2}$

Lower limit, $n = -\left(\frac{N-1}{2}\right) \Rightarrow l = 0$

Upper limit, $n = \frac{N-1}{2} \Rightarrow l = N-1$

$$= \sum_{l=0}^{N-1} e^{-j\omega \left(l - \frac{N-1}{2}\right)}$$

$$= \sum_{n=0}^{N-1} e^{-j\omega \left(n - \frac{N-1}{2}\right)}$$

$$= \sum_{n=0}^{N-1} e^{-j\omega n} \cdot e^{+j\omega \left(\frac{N-1}{2}\right)}$$

$$= e^{j\omega \left(\frac{N-1}{2}\right)} \sum_{n=0}^{N-1} e^{-j\omega n}$$

$$= e^{j\omega \left(\frac{N-1}{2}\right)} \sum_{n=0}^{N-1} (e^{-j\omega})^n$$

$$= e^{j\omega \left(\frac{N-1}{2}\right)} \left[\frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}} \right]$$

$$= e^{j\omega \left(\frac{N-1}{2}\right)} \frac{e^{-j\omega \frac{N}{2}} \left[e^{j\omega \frac{N}{2}} - e^{-j\omega \frac{N}{2}} \right]}{e^{-j\omega/2} \left[e^{j\omega/2} - e^{-j\omega/2} \right]}$$

$$= e^{j\omega \left(\frac{N-1}{2}\right)} \cdot e^{-j\omega \left(\frac{N}{2}\right)} \cdot \frac{e^{-j\omega/2} \cdot 2j \sin\left(\frac{\omega N}{2}\right)}{e^{-j\omega/2} \cdot 2j \sin\left(\frac{\omega}{2}\right)}$$

$$= e^{j\omega \left[\frac{N-1}{2} - \frac{N}{2} + \frac{1}{2}\right]} \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} = e^0 \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} \therefore F[W_R(n)] = \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}$$

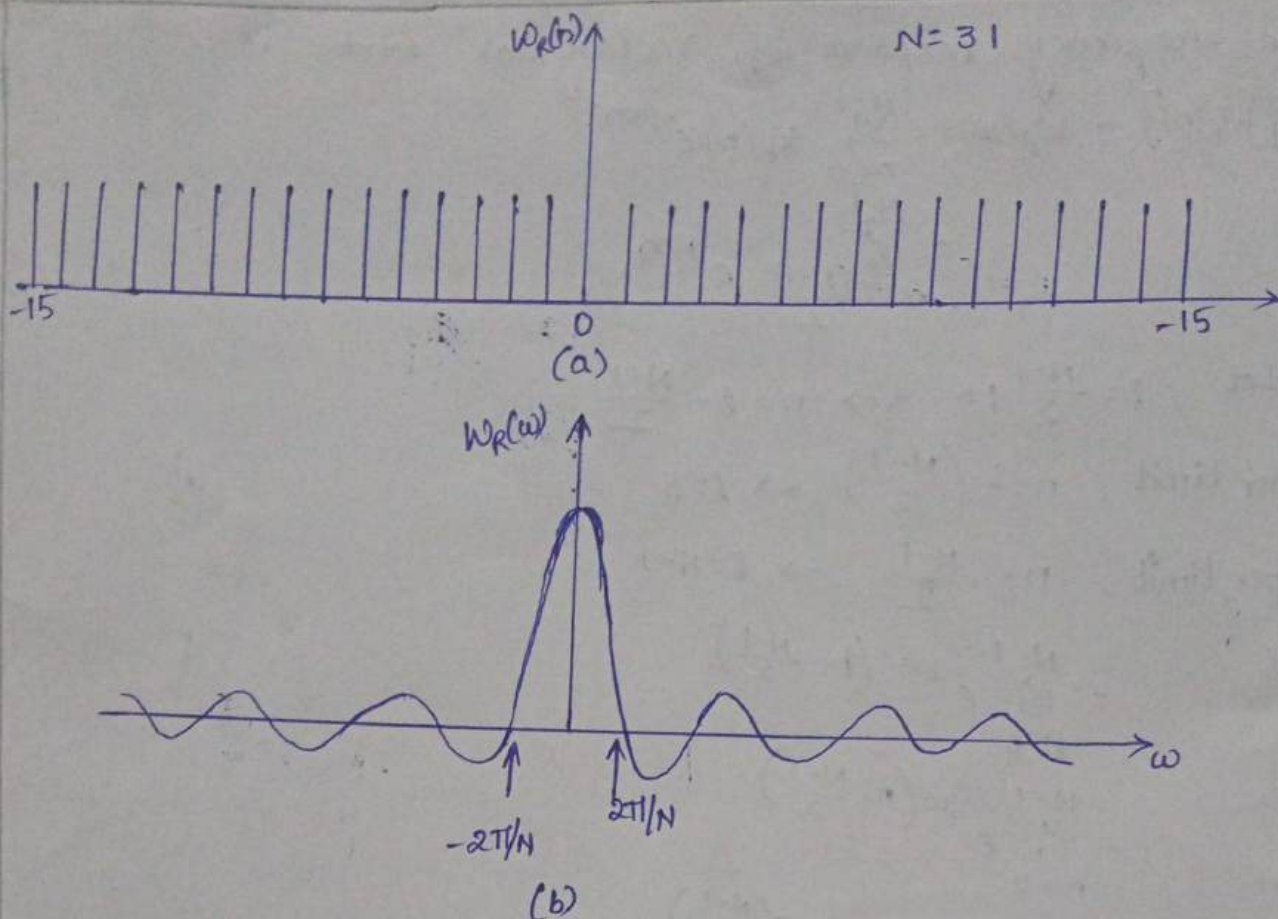


Figure: (a) Rectangular window sequence, (b) magnitude response of rectangular window.

(ii) Triangular or Bartlett Window:-

The N-point triangular window has a window function (or) weighting function is defined as

$$w_T(n) = \begin{cases} 1 - \frac{2|n|}{N-1}, & \text{for } -\left(\frac{N-1}{2}\right) \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{otherwise} \end{cases}$$

$$(or) \quad w_T(n) = \begin{cases} 1 - \frac{2|n - (N-1)/2|}{N-1}, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

iii) Raised Cosine Window:-

The N-point raised cosine window has a window function (or) weighting function is defined as

$$w_R(n) = \begin{cases} \alpha + (1-\alpha) \cos\left(\frac{2\pi n}{N-1}\right) & ; -\left(\frac{N-1}{2}\right) \leq n \leq \frac{N-1}{2} \\ 0 & ; \text{otherwise} \end{cases}$$

For Hamming window, $\alpha = 0.54$
 for Hanning window, $\alpha = 0.5$

a) Hanning Window:-

The Hanning window function is given by

$$w_{Hn}(n) = \begin{cases} 0.5 + 0.5 \cos\left(\frac{2\pi n}{N-1}\right), & \text{for } -\left(\frac{N-1}{2}\right) \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{otherwise} \end{cases}$$

$$\text{or } w_{Hn}(n) = \begin{cases} 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right), & \text{for } 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

b) Hamming Window:-

The Hamming window function is given by

$$w_H(n) = \begin{cases} 0.54 + 0.46 \cos\left(\frac{2\pi n}{N-1}\right), & \text{for } -\left(\frac{N-1}{2}\right) \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{otherwise} \end{cases}$$

$$\text{or } w_H(n) = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

Design of FIR filter using window Techniques

Procedure to design Linear Phase FIR filter using Windows:-

Let $H_d(\omega)$ is desired frequency response and $h_d(n)$ is desired sample or impulse response

$$h_d(n) \xleftrightarrow{\text{F.T}} H_d(\omega) \text{ then,}$$

$$H_d(\omega) = \sum_{n=-\infty}^{\infty} h_d(n) e^{-j\omega n} \rightarrow \text{①}$$

$$\text{and } h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \rightarrow \text{②}$$

Generally eqn ② has an infinite duration, since we are designing FIR filter to make $h_d(n)$ finite, this is multiplied

by window sequence with finite duration 'N'.

Therefore, $h(n) = h_d(n) w_R(n) \rightarrow (3)$

where, $w_R(n) = 1$, $0 \leq n \leq N-1$

$= 0$, otherwise

$\rightarrow (4)$

Substitute (4) in (3) then

$$h(n) = h_d(n) \quad 0 \leq n \leq N-1$$

Generally $h(n)$ can be written as

$$h(n) = h_d(n) \cdot w(n)$$

Similarly,

$$H(\omega) = H_d(\omega) * w(\omega)$$

P. Design the symmetric FIR low pass filter when $H_d(\omega) = e^{-j\omega\tau}$ for $|\omega| \leq \omega_c$ with $N=7$, $\omega_c = 1$ rad/sec. Use rectangular window.

sol

$$h(n) = h_d(n) \times w(n)$$

$$H_d(\omega) = e^{-j\omega\tau}$$

$$|\omega| \leq \omega_c$$

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$h_d(n) = \frac{1}{2\pi} \int_{-1}^1 [e^{-j\omega\tau}] e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-1}^1 e^{j\omega(n-\tau)} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\tau)}}{j(n-\tau)} \right]_{-1}^1$$

$$= \frac{1}{2\pi} \left[\frac{e^{j(n-\tau)} - e^{-j(n-\tau)}}{j(n-\tau)} \right]$$

$$= \frac{1}{\pi(n-\tau)} \left[\frac{e^{j(n-\tau)} - e^{-j(n-\tau)}}{2j} \right]$$

$$h_d(n) = \frac{\sin(n-\gamma)}{\pi(n-\gamma)} \quad \text{for } n \neq \gamma$$

$\Rightarrow$ for $n = \gamma$

$$h_d(n) = \frac{1}{2\pi} \int_{-1}^1 e^{j\omega(n-\gamma)} d\omega$$

$$= \frac{1}{2\pi} \int_{-1}^1 1 d\omega = \frac{1}{2\pi} [\omega]_{-1}^1 = \frac{1}{2\pi} [1+1] = \frac{1}{\pi} \quad \text{for } n = \gamma$$

$$\therefore h_d(n) = \frac{\sin(n-\gamma)}{\pi(n-\gamma)} \quad \text{for } n \neq \gamma$$

$$= \frac{1}{\pi} \quad \text{for } n = \gamma$$

$$\Rightarrow h(n) = h(N-n-1)$$

$$h(n) = h_d(n) \cdot w(n)$$

$$= h_d(N-n-1) \cdot w(n)$$

$$h_d(n) = h_d(N-n-1)$$

$$\Rightarrow \frac{\sin(n-\gamma)}{\pi(n-\gamma)} = \frac{\sin(N-n-1-\gamma)}{\pi(N-n-1-\gamma)}$$

$$\frac{-\sin(n-\gamma)}{-\pi(n-\gamma)} = \frac{\sin(N-n-1-\gamma)}{\pi(N-n-1-\gamma)}$$

$$\frac{\sin(-(n-\gamma))}{-\pi(n-\gamma)} = \frac{\sin(N-n-1-\gamma)}{\pi(N-n-1-\gamma)}$$

$$-n+\gamma = N-n-1-\gamma$$

$$\gamma = \frac{N-1}{2}$$

$$\therefore h_d(n) = \frac{\sin\left[n - \left(\frac{N-1}{2}\right)\right]}{\pi\left[n - \left(\frac{N-1}{2}\right)\right]}, \quad \text{for } n \neq \frac{N-1}{2}$$

$$= \frac{1}{\pi}, \quad \text{for } n = \frac{N-1}{2}$$

$$h_d(n) = \frac{\sin(n-3)}{\pi(n-3)} \quad \text{for } n \neq 3$$

$$= \frac{1}{\pi} \quad \text{for } n = 3$$

$$\gamma = \frac{N-1}{2} = \frac{7-1}{2} = \frac{6}{2} = 3$$

$$\therefore \gamma = 3$$

$$N = 7 \quad n = 0 \text{ to } 6$$

$$h(0) = h_d(0) w_R(0) = 0.01497 \times 1 = 0.01497$$

$$h(1) = h_d(1) w_R(1) = 0.1447$$

$$h(2) = h_d(2) w_R(2) = 0.2678$$

$$h(3) = \frac{1}{\pi} = 0.318$$

$$h(4) = 0.2672$$

$$h(5) = 0.1447$$

$$h(6) = 0.01497$$

$$\therefore h(n) = \{0.01497, 0.1447, 0.2678, 0.318, 0.2672, 0.1447, 0.01497\}$$

P. Design a low pass filter is to be design with the following desired frequency response

$$H_d(e^{j\omega}) = e^{-2j\omega} \quad \text{for } -\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4}$$

$$= 0 \quad \text{for } \frac{\pi}{4} \leq \omega \leq \pi$$

Determine the filter coefficient if the window function is defined as $w(n) = 1$ for $0 \leq n \leq 4$
 $= 0$ otherwise also determine the frequency response. $H(e^{j\omega})$ of the desired filter.

Sol Given $H_d(e^{j\omega}) = e^{-2j\omega}$ for $-\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4}$
 $= 0$ for $\frac{\pi}{4} \leq \omega \leq \pi$

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) \cdot e^{j\omega n} d\omega$$

$$\begin{aligned}
 &= \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{-j2\omega} \cdot e^{j\omega n} d\omega \\
 &= \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{j\omega(n-2)} d\omega \\
 &= \frac{1}{2\pi} \left[\frac{e^{j\omega(n-2)}}{j(n-2)} \right]_{-\pi/4}^{\pi/4} \\
 &= \frac{1}{2\pi} \left[\frac{e^{j(n-2)\pi/4} - e^{-j(n-2)\pi/4}}{j(n-2)} \right] \\
 &= \frac{1}{\pi(n-2)} \left[\frac{e^{j(n-2)\pi/4} - e^{-j(n-2)\pi/4}}{2j} \right]
 \end{aligned}$$

$$h_d(n) = \frac{1}{\pi(n-2)} \sin(n-2) \frac{\pi}{4} \quad n \neq 2$$

$$\begin{aligned}
 h_d(n) &= \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{j\omega(n-2)} d\omega \quad n=2 \\
 &= \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} 1 d\omega \\
 &= \frac{1}{2\pi} \left[\frac{\pi}{4} + \frac{\pi}{4} \right] = \frac{1}{2\pi} \left[\frac{\pi}{2} \right] = \frac{1}{4}
 \end{aligned}$$

$$h_d(n) = \begin{cases} \frac{1}{\pi(n-2)} \sin(n-2) \frac{\pi}{4} & \text{for } n \neq 2 \\ \frac{1}{4} & \text{for } n = 2 \end{cases}$$

$$h(n) = h_d(n) \cdot w(n) \quad [\because w(n) = 1]$$

$$h(n) = h_d(n)$$

$$h(0) = \frac{1}{2\pi}$$

$$h(3) = \frac{1}{\sqrt{2}\pi}$$

$$h(1) = \frac{1}{\sqrt{2}\pi}$$

$$h(4) = \frac{1}{2\pi}$$

$$h(2) = \frac{1}{4}$$

$$h(n) = \left\{ \frac{1}{2\pi}, \frac{1}{\sqrt{2}\pi}, \frac{1}{4}, \frac{1}{\sqrt{2}\pi}, \frac{1}{2\pi} \right\}$$

$$H(z) = \sum_{n=0}^4 h(n) z^{-n} = h(0) + h(1) z^{-1} + h(2) z^{-2} + h(3) z^{-3} + h(4) z^{-4}$$

$$= \frac{1}{2\pi} + \frac{1}{\sqrt{2}\pi} z^{-1} + \frac{1}{4} z^{-2} + \frac{1}{\sqrt{2}\pi} z^{-3} + \frac{1}{2\pi} z^{-4} = z^{-2} \left[0.25 + \frac{1}{\sqrt{2}\pi} (z+z^{-1}) + \frac{1}{2\pi} (z^2+z^{-2}) \right]$$

$$H(e^{j\omega}) = H(\omega) = \sum_{n=0}^4 h(n) e^{-j\omega n}$$

$$= h(0) + h(1) e^{-j\omega} + h(2) e^{-2j\omega} + h(3) e^{-3j\omega} + h(4) e^{-4j\omega}$$

$$\Rightarrow \frac{1}{2\pi} + \frac{1}{\sqrt{2}\pi} e^{-j\omega} + \frac{1}{4} e^{-2j\omega} + \frac{1}{\sqrt{2}\pi} e^{-3j\omega} + \frac{1}{2\pi} e^{-4j\omega}$$

$$\Rightarrow e^{-2j\omega} \left[\frac{1}{2\pi} e^{2j\omega} + \frac{1}{\sqrt{2}\pi} e^{j\omega} + \frac{1}{4} + \frac{1}{\sqrt{2}\pi} e^{-j\omega} + \frac{1}{2\pi} e^{-2j\omega} \right]$$

$$\Rightarrow e^{-2j\omega} \left[\frac{1}{2\pi} (e^{2j\omega} + e^{-2j\omega}) + \frac{1}{\sqrt{2}\pi} (e^{j\omega} + e^{-j\omega}) + \frac{1}{4} \right]$$

$$\therefore H(e^{j\omega}) = e^{-2j\omega} \left[0.25 + \frac{\sqrt{2}}{\pi} \cos \omega + \frac{1}{\pi} \cos 2\omega \right]$$

P. A filter is to be design with the following desired frequency response $H_d(e^{j\omega}) = \begin{cases} 0 & -\pi/2 \leq \omega \leq \pi/2 \\ e^{-j2\omega} & \pi/2 \leq |\omega| \leq \pi \end{cases}$. Determine the coefficients $h(n)$ if the window function is $w(n) = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & \text{otherwise} \end{cases}$ also

find the frequency response $H(e^{j\omega})$ of the designed filter.

$$\text{Sol} \quad H_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$H_d(n) = \frac{1}{2\pi} \left[\int_{-\pi}^{-\pi/2} e^{-2j\omega} \cdot e^{j\omega n} d\omega + \int_{\pi/2}^{\pi} e^{-2j\omega} \cdot e^{j\omega n} d\omega \right]$$

$$H_d(n) = \frac{1}{2\pi} \left[\int_{-\pi}^{-\pi/2} e^{j\omega(n-2)} d\omega + \int_{\pi/2}^{\pi} e^{j\omega(n-2)} d\omega \right]$$

$$H_d(n) = \frac{1}{2\pi} \left[\left[\frac{e^{j\omega(n-2)}}{j(n-2)} \right]_{-\pi}^{-\pi/2} + \left[\frac{e^{j\omega(n-2)}}{j(n-2)} \right]_{\pi/2}^{\pi} \right]$$

$$H_d(n) = \frac{1}{2\pi} \left[\frac{e^{-j(n-2)\pi/2} - e^{-j(n-2)\pi}}{j(n-2)} + \frac{e^{j(n-2)\pi} - e^{j(n-2)\pi/2}}{j(n-2)} \right]$$

$$H_d(n) = \frac{1}{2\pi j(n-2)} \left[\frac{e^{-j(n-2)\pi/2} - e^{-j(n-2)\pi}}{-e} + \frac{e^{j(n-2)\pi} - e^{j(n-2)\pi/2}}{-e} \right]$$

$$= \frac{1}{\pi(n-2)} \left\{ \left[\frac{e^{j(n-2)\pi} - e^{-j(n-2)\pi}}{2j} \right] - \left[\frac{e^{j(n-2)\pi/2} - e^{-j(n-2)\pi/2}}{2j} \right] \right\}$$

$$= \frac{1}{\pi(n-2)} \left[\sin(n-2)\pi - \sin(n-2)\pi/2 \right] \quad \text{for } n \neq 2$$

$$h_d(n) = \frac{1}{2\pi} \left[\left(-\pi/2 + \pi \right) + \left(\pi - \pi/2 \right) \right] \quad \text{for } n=2$$

$$= \frac{1}{2}$$

$$\therefore h_d(n) = \frac{1}{\pi(n-2)} \left[\sin(n-2)\pi - \sin(n-2)\pi/2 \right] \quad \text{for } n \neq 2$$

$$= \frac{1}{2} \quad \text{for } n=2$$

$$h(n) = h_d(n) \cdot w(n)$$

$$h(n) = h_d(n) \quad \text{for } 0 \leq n \leq 4$$

$$h(0) = \frac{-1}{2\pi} [0-0] = 0$$

$$h(1) = \frac{1}{-\pi} [0+1] = -\frac{1}{\pi}, \quad h(2) = \frac{1}{2}$$

$$h(3) = \frac{1}{\pi} [0-1] = -\frac{1}{\pi}, \quad h(4) = \frac{1}{2\pi} [0-0] = 0$$

$$h(n) = \left\{ 0, -\frac{1}{\pi}, \frac{1}{2}, -\frac{1}{\pi}, 0 \right\}$$

$$H(z) = \sum_{n=0}^4 h(n) z^{-n} = h(0) + h(1) z^{-1} + h(2) z^{-2} + h(3) z^{-3} + h(4) z^{-4}$$

$$= \frac{-1}{\pi} z^{-1} + \frac{1}{2} z^{-2} - \frac{1}{\pi} z^{-3}$$

$$H(e^{j\omega}) = H(\omega) = \sum_{n=0}^4 h(n) e^{-j\omega n}$$

$$= h(0) + h(1) e^{-j\omega} + h(2) e^{-j2\omega} + h(3) e^{-j3\omega} + h(4) e^{-j4\omega}$$

$$= \frac{-1}{\pi} e^{-j\omega} + \frac{1}{2} e^{-j2\omega} - \frac{1}{\pi} e^{-j3\omega}$$

$$= \frac{-1}{\pi} [e^{-j\omega} + e^{-j3\omega}] + \frac{1}{2} e^{-j2\omega}$$

$$= \frac{-1}{\pi} [e^{-j2\omega} (e^{j\omega} + e^{-j\omega})] + \frac{1}{2} e^{-j2\omega}$$

$$= e^{-j2\omega} \left[\frac{1}{2} - \frac{1}{\pi} [e^{j\omega} + e^{-j\omega}] \right]$$

$$= e^{-j2\omega} \left[\frac{1}{2} - \frac{2}{\pi} \cos \omega \right]$$

$$= e^{-j2\omega} [0.5 - 0.636 \cos \omega]$$

P) Design a filter with $H_d(\omega) = \begin{cases} e^{-j2\omega} & -\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4} \\ 0 & \frac{\pi}{4} \leq \omega \leq \pi \end{cases}$ using a

hamming window with $N=7$

The desired frequency response,

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} H_d(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} [e^{-j2\omega}] e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{j\omega(n-2)} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega(n-2)}}{j(n-2)} \right]_{-\pi/4}^{\pi/4}$$

$$= \frac{1}{2\pi} \left[\frac{e^{j(n-2)\pi/4} - e^{-j(n-2)\pi/4}}{j(n-2)} \right]$$

$$= \frac{1}{\pi(n-2)} \left[\frac{e^{j(n-2)\pi/4} - e^{-j(n-2)\pi/4}}{2j} \right]$$

$$= \frac{1}{\pi(n-2)} \sin(n-2)\frac{\pi}{4} \quad n \neq 2$$

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} 1 d\omega = \frac{1}{2\pi} [\omega]_{-\pi/4}^{\pi/4} = \frac{1}{2\pi} \left[\frac{\pi}{4} + \frac{\pi}{4} \right] = \frac{1}{2\pi} \times \frac{\pi}{2} = \frac{1}{4}$$

$$h_d(n) = \begin{cases} \frac{\sin(n-2)\pi/4}{\pi(n-2)} & \text{for } n \neq 2 \\ \frac{1}{4} & \text{for } n=2 \end{cases}$$

$$h_d(0) = \frac{\sin \pi(0-3)/4}{\pi(0-3)} = \frac{0.707}{3\pi}, \quad h_d(1) = \frac{\sin \pi(1-3)/4}{\pi(1-3)} = \frac{1}{2\pi}$$

$$h_d(2) = \frac{\sin \pi(2-3)/4}{\pi(2-3)} = \frac{0.707}{\pi}, \quad h_d(4) = \frac{\sin \pi(4-3)/4}{\pi(4-3)} = \frac{0.707}{\pi}$$

$$h_d(5) = \frac{\sin \pi(5-3)/4}{\pi(5-3)} = \frac{1}{2\pi}, \quad h_d(6) = \frac{\sin \pi(6-3)/4}{\pi(6-3)} = \frac{0.707}{3\pi}$$

The Hamming window function of a causal filter is,

$$w(n) = \begin{cases} 0.54 - 0.46 \cos \frac{2\pi n}{N-1}, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

Therefore, with $N=7$

$$w(0) = 0.54 - 0.46 \cos 0 = 0.08, \quad w(4) = 0.54 - 0.46 \cos \frac{2\pi \times 4}{7-1} = 0.08$$

$$w(1) = 0.54 - 0.46 \cos \frac{2\pi \times 1}{7-1} = 0.31, \quad w(5) = 0.54 - 0.46 \cos \frac{2\pi \times 5}{7-1} = 0.31$$

$$w(2) = 0.54 - 0.46 \cos \frac{2\pi \times 2}{7-1} = 0.77, \quad w(6) = 0.54 - 0.46 \cos \frac{2\pi \times 6}{7-1} = 0.08$$

$$w(3) = 0.54 - 0.46 \cos \frac{2\pi \times 3}{7-1} = 1$$

The filter coefficients of the resultant filter are,

$$h(n) = h_d(n) w(n), \quad n=0, 1, 2, 3, 4, 5, 6$$

Therefore,

$$h(0) = h_d(0) w(0) = \frac{0.707}{3\pi} \times 0.08 = 0.006$$

$$h(1) = h_d(1) w(1) = \frac{1}{2\pi} \times 0.31 = 0.049$$

$$h(2) = h_d(2) w(2) = \frac{0.707}{\pi} \times 0.77 = 0.173$$

$$h(3) = h_d(3) w(3) = \frac{1}{4} \times 1 = \frac{1}{4} = 0.25$$

$$h(4) = h_d(4) w(4) = \frac{0.707}{\pi} \times 0.77 = 0.173$$

$$h(5) = h_d(5) w(5) = \frac{1}{2\pi} \times 0.31 = 0.049$$

$$h(6) = h_d(6) w(6) = \frac{0.707}{3\pi} \times 0.08 = 0.006$$

The frequency response of a causal filter is given by

$$H(\omega) = \sum_{n=0}^6 h(n) e^{-j\omega n}$$

$$= h(0) + h(1) e^{-j\omega} + h(2) e^{-j2\omega} + h(3) e^{-j3\omega} + h(4) e^{-j4\omega} + h(5) e^{-j5\omega} + h(6) e^{-j6\omega}$$

$$= e^{-j3\omega} \left\{ h(3) + [h(0) e^{j3\omega} + h(6) e^{-j3\omega}] + [h(1) e^{j2\omega} + h(5) e^{-j2\omega}] + [h(2) e^{j\omega} + h(4) e^{-j\omega}] \right\}$$

$$= e^{-j3\omega} [h(3) + 2h(0) \cos 3\omega + 2h(1) \cos 2\omega + 2h(2) \cos \omega]$$

$$= e^{-j3\omega} [0.25 + 0.012 \cos 3\omega + 0.098 \cos 2\omega + 0.346 \cos \omega]$$

The transfer function of the digital FIR low-pass filter is?

$$H(z) = \sum_{n=0}^{N-1} h(n) z^{-n}$$

$$= h(0) + h(1) z^{-1} + h(2) z^{-2} + h(3) z^{-3} + h(4) z^{-4} + h(5) z^{-5} + h(6) z^{-6}$$

$$= z^{-3} [h(3) + h(2) (z^{-1} + z) + h(1) (z^{-2} + z^2) + h(0) (z^{-3} + z^3)]$$

$$= z^{-3} [0.25 + 0.173 [z + z^{-1}] + 0.049 [z^2 + z^{-2}] + 0.006 [z^3 + z^{-3}]]$$

$$= 0.006 + 0.049 z^{-1} + 0.173 z^{-2} + 0.25 z^{-3} + 0.173 z^{-4} + 0.049 z^{-5} + \underline{\underline{0.006 z^{-6}}}$$

* The system that use single sample rate from A/D & D/A converters are known as single rate systems. The discrete time system processes data at more than one sampling rate are known as multirate system.

Ex: In digital audio the different sampling rates are used. They are 30 kHz for broadcasting, 44.1 kHz for compact disc and 48 kHz for audio tape.

Different sampling rates can be obtained using an upsampler and down-sampler.

The basic operations in multirate processing is decimation ($\downarrow$) and interpolation ($\uparrow$).

Decimation is for reducing a sample rate and interpolation is for increasing the sampling rate.

applications of multirate dsp

- * High quality data acquisition and storage system
- * audio signal processing

- * video signal processing
- * speech processing
- * Advantages multirate DSP
 - * Computational requirements are less
 - * Storage for filter coefficients are less
 - * filter order required for multi rate applications is low.

* Sampling theorem

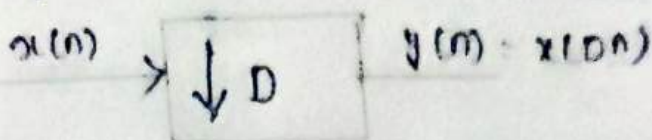
- * Sampling theorem states that a band limited signal $x(t)$ which has ^{NO} ~~more~~ spectral components higher than f_m Hz. It can be completely reconstructed from its samples when samples $f_s \geq 2f_m$
- * Therefore the sampling rate $f_s = 2f_m$

Down Sampling

Reducing the sampling rate of a discrete time signal is known as down sampling. In this process the sampling rate can be decreased by an integer factor 'D' by keeping every D samples and removing D-1 in between samples.

- * mathematically down sampling can be expressed as $y(n) = x(Dn)$

- * D is also called decimation-factor
- * Graphically - the down sampling is shown below



Ex: consider a discrete time signal

$$x(n) = \{1, 2, 3, 1, 2, 3, 1, 2, 3\}$$

The down sampling signals for $D=2$, $D=3$ and $D=4$ is shown below

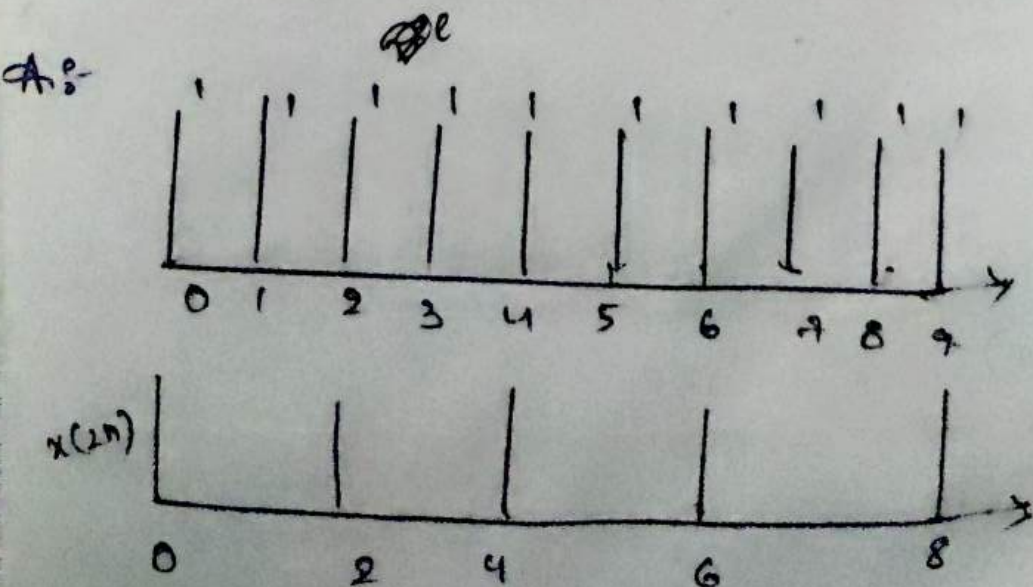
$$x(2n) = \{1, 3, 1, 3, 1, 3\}$$

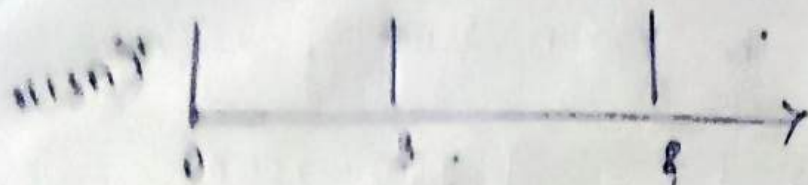
$$x(3n) = \{1, 1, 1\}$$

$$x(4n) = \{1, 2, 3\}$$

- * Consider a signal $x(n) = u(n)$ for $n \geq 0$

~~2.40~~ obtain a signal with decimation factor 2 and 3
 $= 0$ for elsewhere



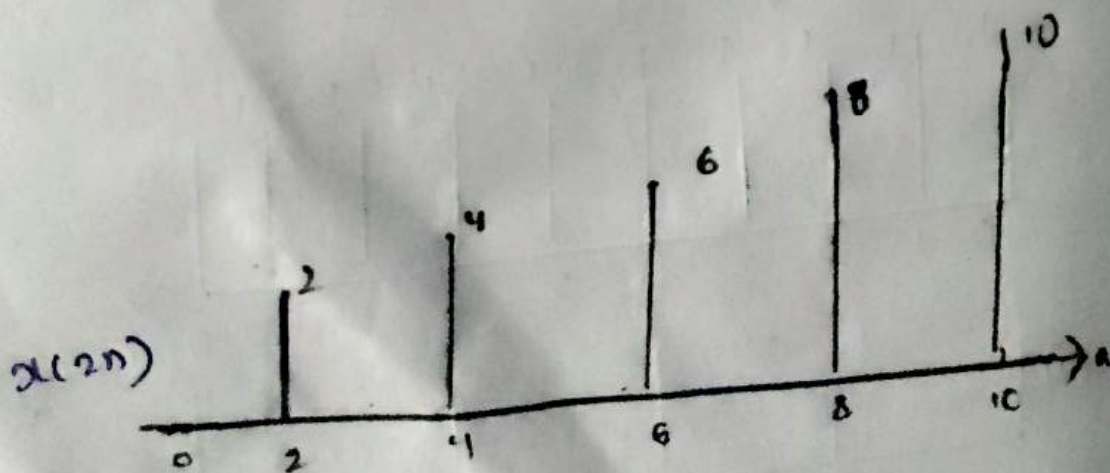
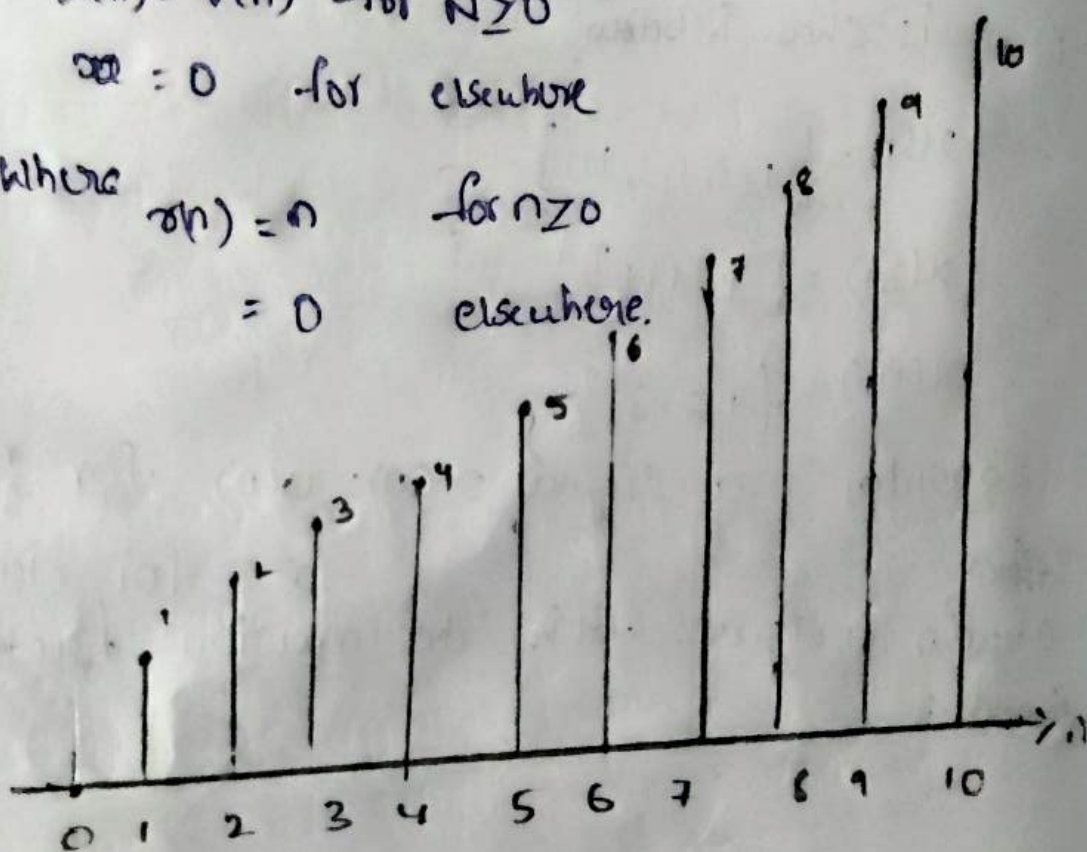


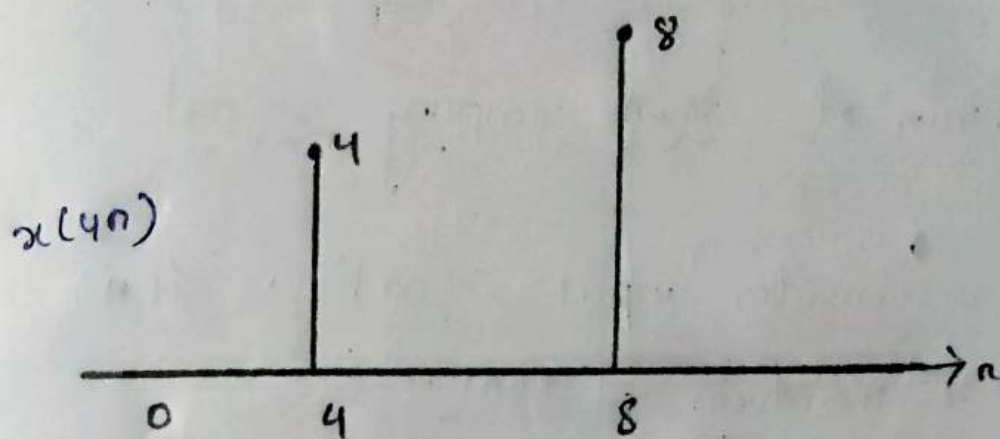
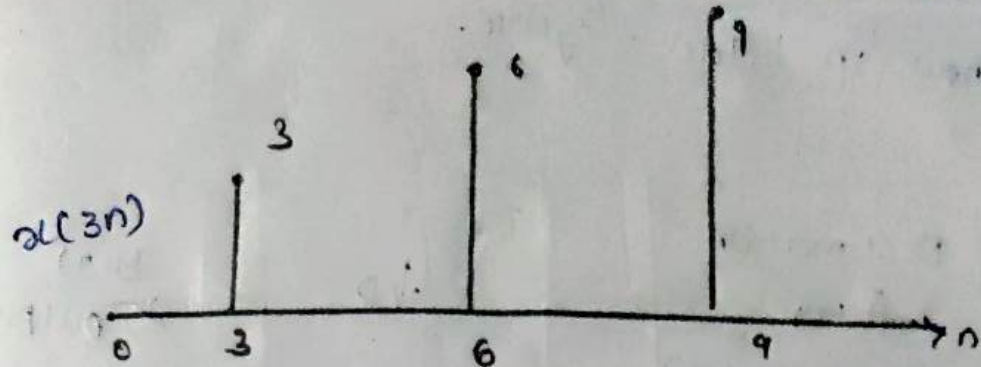
* consider a ramp signal Sketch the desimated version of - for signal for the factor 13, 4

Ans: $x(n) = n$ for $n \geq 0$

$x(n) = 0$ for elsewhere

where $x(n) = n$ for $n \geq 0$
 $= 0$ elsewhere.



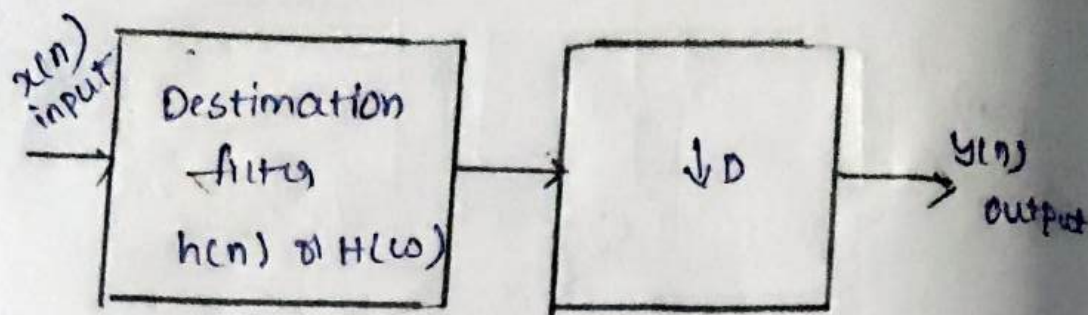


Decimation:-

if the signal $x(n)$ is not band limited then there will be overlapping of spectra at the output of down sampler. This overlapping of spectra is called aliasing which is undesirable. This aliasing problem can be eliminated by band limiting the input signal by inserting a low pass filter called anti alias filter before the down sampler.

* The anti alias filter and down sampler together is called decimator. The block diagram of the decimator

is show in below figure.



* spectrum of a down sampling signal

* Let us consider input signal is $x(n)$

The z -transform of $x(n)$ is

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \rightarrow (1)$$

The down sample signal $y(n)$, is obtain by multiplying the input sequence $x(n)$ with an impulse train $p(n)$ with a period 'D' and then leaving out $d-1$ samples

The impulse train is given by

$$p(n) = \begin{cases} 1 & \text{for } n=0, \pm D, \pm 2D, \pm 3D, \dots \\ 0 & \text{for otherwise} \end{cases} \rightarrow (2)$$

The discrete Fourier series of $p(n)$ is

$$p(n) = \frac{1}{D} \sum_{k=0}^{D-1} e^{j2\pi kn/D} \rightarrow (3)$$

multiple the sequence $x(n)$ with $p(n)$ is given by

$$x'(n) = p(n)x(n) \rightarrow (4)$$

$$\left. \begin{aligned} x'(n) &= x(n) \text{ for } n=0, \pm D, \pm 2D, \dots \\ &= 0 \text{ for otherwise} \end{aligned} \right\} \rightarrow (5)$$

if we ~~take~~ ^{leave} the $D-1$ samples between each sample we get $y(n) = x'(nD)$

$$= x(nD)p(nD)$$

$$= x(nD) \rightarrow (6)$$

The z -transform of output is given by

$$Y(z) = \sum_{n=-\infty}^{\infty} y(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x'(nD) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) z^{-n/D} \rightarrow (7)$$

$$= \sum_{n=-\infty}^{\infty} [x(n)p(n)] z^{-n/D}$$

$$= \sum_{n=-\infty}^{\infty} \left[x(n) \frac{1}{D} \sum_{k=0}^{D-1} e^{j2\pi kn/D} \right] z^{-n/D}$$

$$= \frac{1}{D} \sum_{k=0}^{D-1} \sum_{n=-\infty}^{\infty} [x(n) \cdot e^{j2\pi kn/D}] z^{-n/D}$$

$$= \frac{1}{D} \sum_{k=0}^{D-1} \sum_{n=-\infty}^{\infty} x(n) [e^{-j2\pi kn/D} z^{1/D}]^{-n}$$

$$= \frac{1}{D} \sum_{k=0}^{D-1} x \left(e^{-j2\pi k/D} z^{1/D} \right)$$

$z = e^{-j\omega}$

$$= \frac{1}{D} \sum_{k=0}^{D-1} x \left(e^{-j2\pi k/D} \cdot e^{j\omega/D} \right)$$

$$= \frac{1}{D} \sum_{k=0}^{D-1} x \left(e^{-j(2\pi k - \omega)/D} \right)$$

$$y(z) = \frac{1}{D} \sum_{k=0}^{D-1} x \left(\frac{2\pi k - \omega}{D} \right) \rightarrow \textcircled{8}$$

$$y(e^{j\omega}) = \frac{1}{D} \sum_{k=0}^{D-1} x \left[\frac{e^{j(\omega - 2\pi k)}}{D} \right]$$

* consider discrete signal $x(n) = \{1, -1, 2, -2\}$
 determine the interpolated version of the signal for a) $T=2$ b) $T=3$ c) $T=4$

a) $x(n) = \{1, -1, 2, -2\}$

$x(0) = 1$ for $T=2$

$x(1) = -1$

$x(2) = 2$

$x(3) = -2$

$n=0 \quad x(n/2) = x(0) = 1$

$n=1 \quad x(0.5) = 0$

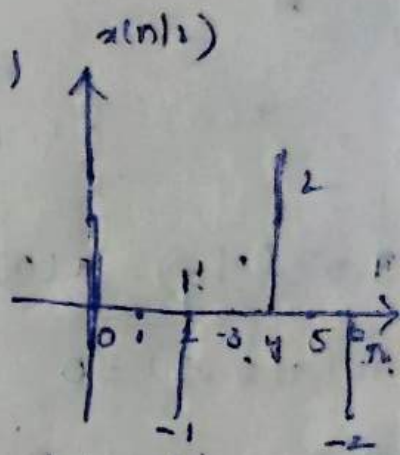
$n=2 \quad x(1) = -1$

$n=3 \quad x(1.5) = 0$

$n=4 \quad x(2) = 2$

$n=5 \quad x(2.5) = 0$

$n=6 \quad x(3) = -2$



$\therefore x(n/2) = \{1, 0, -1, 0, 2, 0, -2\}$

$T=3$

$x(n) = \{1, -1, 2, -2\}$

$n=0 \quad x(n/3) \quad x(0) = 1$

$n=1 \quad x(n/3) \quad (0.33) = 0$

$n=2 \quad , x(0.66) = 0$

$n=3 \quad , x(1) = -1$

$n=4 \quad x(1.33) = 0$

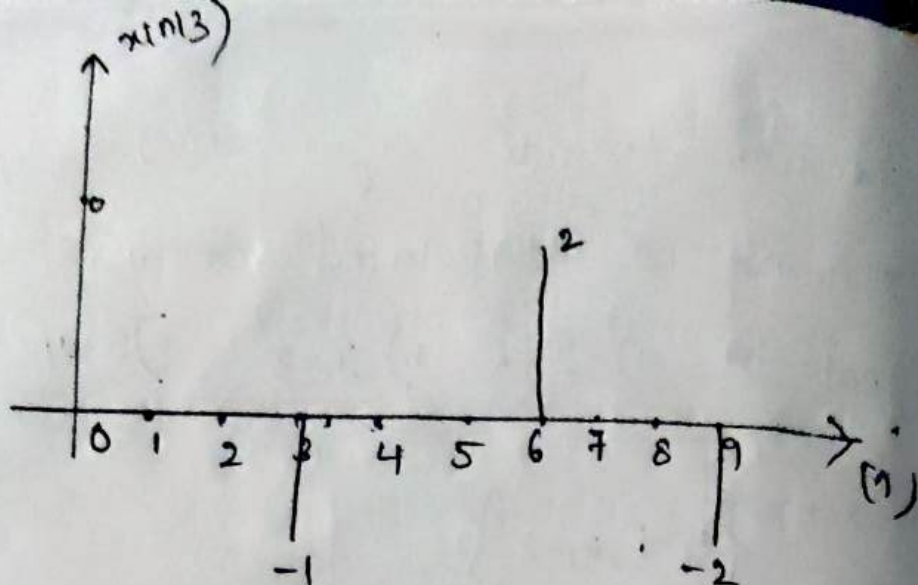
$n=5 \quad , x(1.66) = 0$

$n=6 \quad , x(2) = 2$

$n=7 \quad , x(2.33) = 0$

$n=8 \quad , x(2.66) = 0$

$n=9 \quad , x(3) = -2$



$$\underline{\underline{P=84}}$$

$$x(n) = \{1, -1, 2, -2\}$$

$$n=0, x(n/4) = x(0) = 1$$

$$n=1, x(0.25) = 0$$

$$n=2, x(0.5) = 0$$

$$n=3, x(0.75) = 0$$

$$n=4, x(1) = -1$$

$$n=5, x(1.25) = 0$$

$$n=6, x(1.5) = 0$$

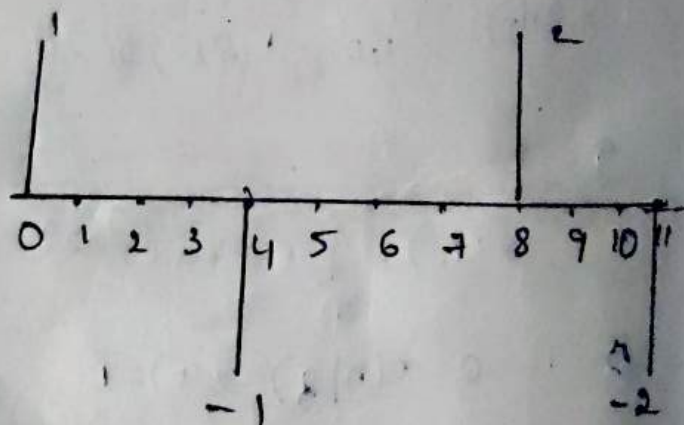
$$n=7, x(2) = 2$$

$$n=8, x(2.25) = 0$$

$$n=9, x(2.5) = 0$$

$$n=10, x(2.75) = 0$$

$$n=11, x(3) = -2$$



*** Spectrum of Up Sampled ***
old paper

* The spectrum of original signal of
The original signal of up sampler is
 $y(z) = x(z^I)$

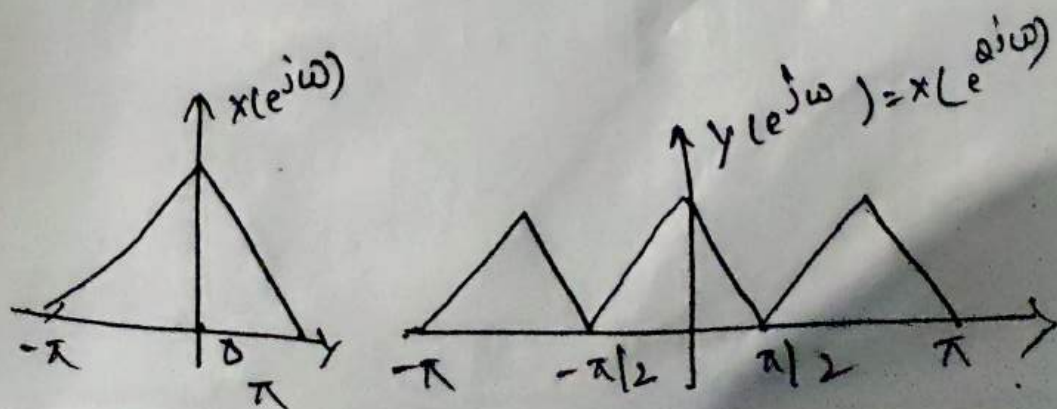
for stable system let $z = e^{j\omega}$

$$\text{Then } Y(e^{j\omega}) = X(e^{j\omega I})$$

* The term $X(e^{j\omega I})$ is the compressed version of $X(e^{j\omega})$ by a factor 'I'

If the frequency response of $X(e^{j\omega})$ is periodic with a period 2π $X(e^{j\omega I})$ will repeat 'I' times in a period of 2π in the spectrum of upsampled signal

let The spectrum of original signals and the spectrum of upsampled signal for $I=2$ is shown in below figure.

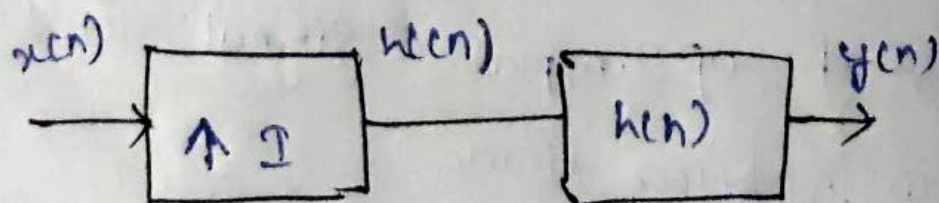


Anti Imaging Filter :

* The frequency spectrum of upsampling signal of $y(e^{j\omega})$ with a factor 'I' contains additional images of the Input Spectrum. These additional images are due to the addition of samples between successive values of $x(n)$.

Since we are not interested in image spectra, a low pass filter with a cutoff frequency $\omega_c = \pi/I$ can be used after upsample. This filter is known as anti image filter.

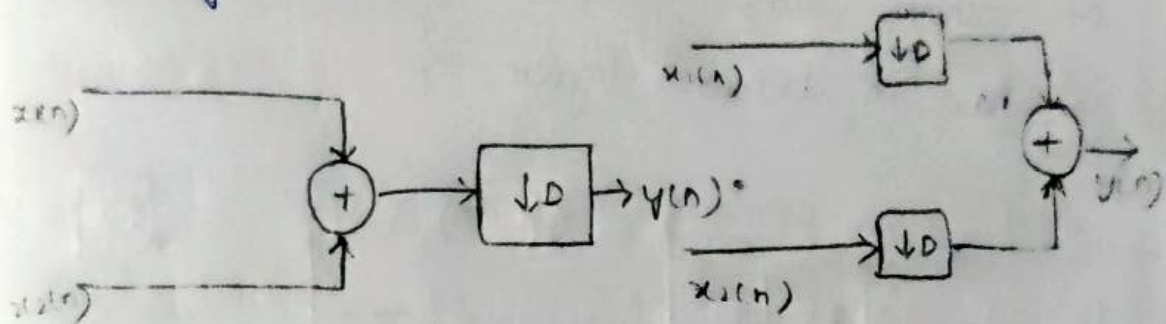
* The combination of upsampler and anti imaging filter is known as an interpolator. It is shown in the below fig.



Identity

* The scaling of discrete time signals and their addition at the nodes are independent of the sampling rate.

* The first Identity is indicated in the following figures.

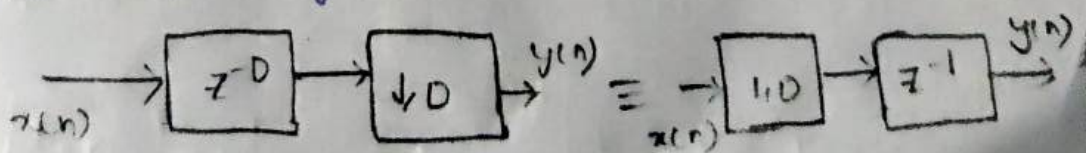


Second Identity

* A delay of 'D' sample periods before down sampling is same as adding of one sampling period after the down

sampler.

The below diagram shows in the Second Identity



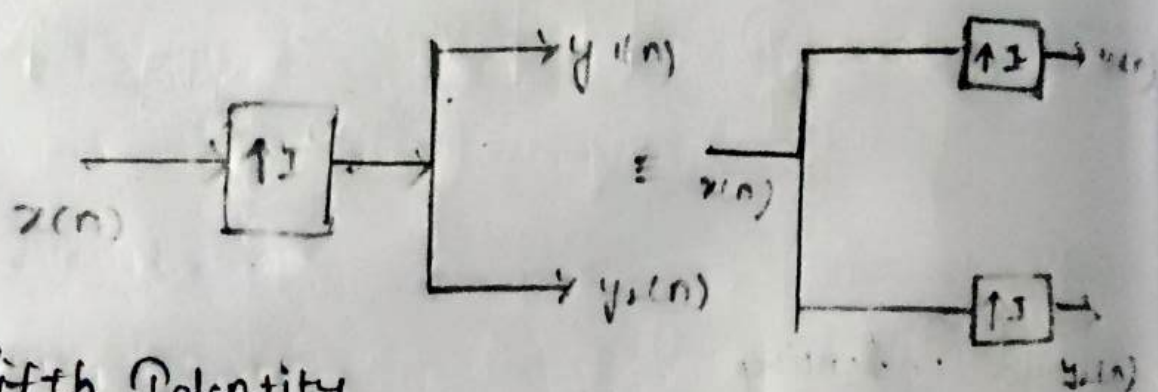
Third Identity

* Third Identity is shown in below figure



Fourth Identity

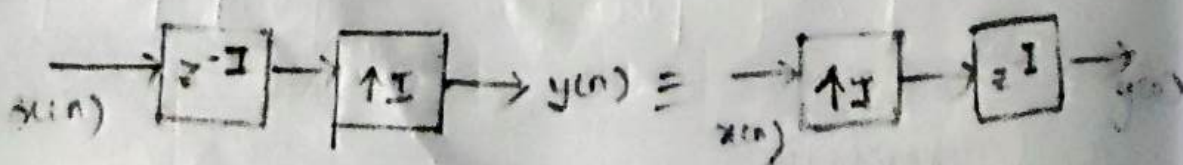
The scaling of discrete time signals and their addition at the nodes, are independent of their sampling rate. The fourth identity is shown in the below figure.



Fifth Identity

A delay of one sample before upsampling leads to a delay of 2 sample periods after upsampling.

* Fifth Identity is shown in the below figure.



Sixth Identity

Sixth Identity is shown in the below figure.



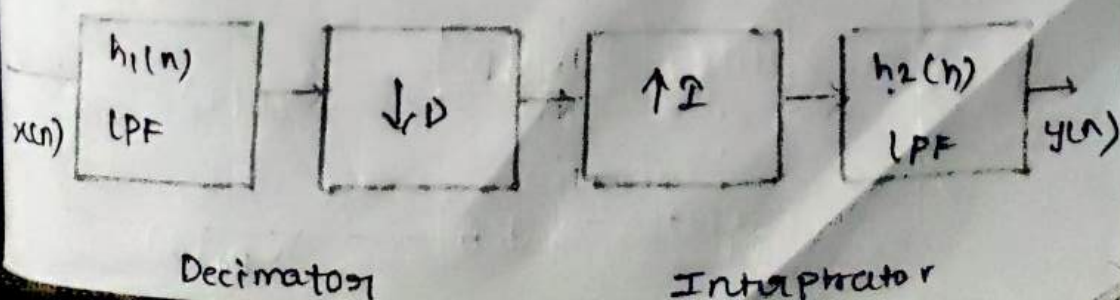
Applications of multirate dsp:

- * The Applications of multirate dsp are
- * Implementation of narrow band lowpass filter
- * filter banks

Implementation of narrow band low pass filter

A narrow band lowpass filter is characterized by narrow pass band and narrow transition band. It requires very large number of coefficients. To overcome this multi-rate approach is used to implement a narrow band low pass filter.

* A narrow band lowpass filter using multirate approach is shown in below figure



- * The filter $h_1(n)$ and $h_2(n)$ in the above figure are low pass filter
- * The Input signal $x(n)$ is passed first through a low pass filter
- * Then the frequency of $x(n)$ is reduced by a factor D . This signal again raised by factor D again a signal is passed through a low pass filter
- * To meet a desired specifications $H_1(n)$ & $H_2(n)$ should be identical

Filter banks :-

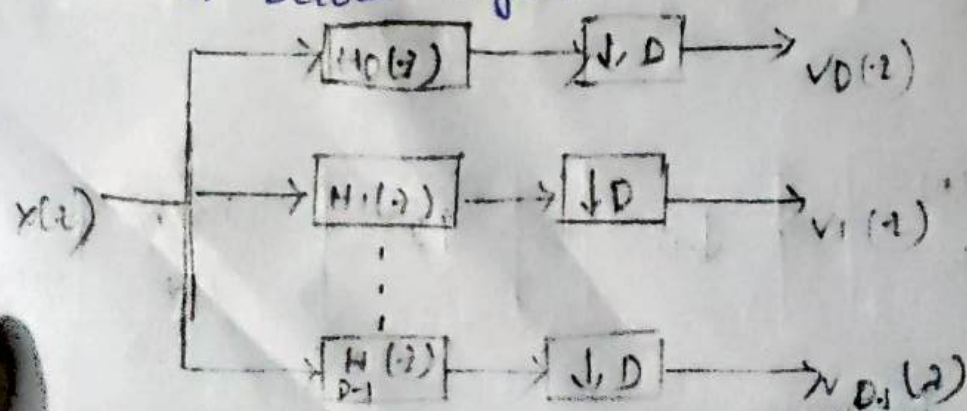
- * Filter banks are usually classified into two types

* analysis filterbanks

* synthesis filterbanks

Analysis filterbanks

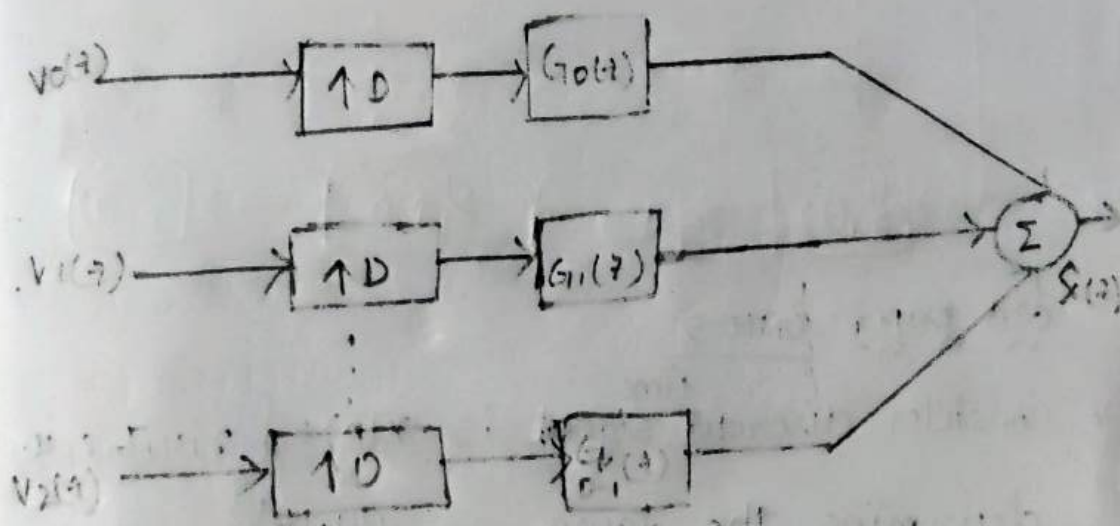
The D -channel analysis filter bank is shown in below figure



It consist of D -sub-filters all the sub-filters are equally space in frequency and each have the same bandwidth the spectrum of input signal lies in the range $0 \leq \omega \leq \pi$ The sampling rate can be reduced by a factor D

Synthesis filter bank

The D -channel synthesis filter bank is show in below figure.



It is dual of analysis filter bands in this each $V_D(z)$ is fed to up-sampling. These signals are applied to filters $G_0(z)$ and finally added to get the output signal $\hat{x}(z)$. The filters $G_0(z), G_{D-1}(z)$ have the same characteristics as the analysis filter $H_0(z), H_{D-1}(z)$.

Subband Coding - filter bank

The combining analysis - filter bank and the synthesis - filter bank we can obtain a D-channel Sub band coding - filter bank is show in below figure.

